Quantum Theory
As A Mechanism Tree

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This book reorganizes quantum theory away from historical article names and toward the recurring construction detected across the quantum page archive: context-selected Hilbert space, state carrier, generator, spectral question, probability/readout, compatibility limits, boundary realization, many-mode extension, and protocols.

Contents

How To Use The Mechanism Tree

What The Tree Is

This book is a derivation map for quantum topics. Each named topic is treated as a contribution to one prediction-making mechanism. The base object is a carrier or context jointly typed with an operator apparatus. Closure, readout, and ordered protocol attach when the topic requires them. Boundaries, fields, detectors, and encodings specify realizations. The page title is therefore not the root of the explanation; the root is the dependency that the page supplies.

> Mechanism-first reading: identify the operation a named topic performs in the quantum construction.
>
> **Constructor structure:** typed carrier–operator identity $\oplus$ closure/readout/protocol completion $\to$ physical realization. Derivations may add roles, project consequences, or rewrite laterally.

Completion DAG And Source Constructor

The mechanism tree uses two graph layers. The completion DAG points from records with fewer constructor roles to records with more roles. It is a partial order of formal specification, not physical time and not a compulsory order of exposition. The source constructor follows adjacent equations inside papers and records completion, projection, and lateral rewrite. This distinction matters because a complete theory is rarely written in one equation. The DAG measures how much of the role contract is explicit. The source constructor shows how authors distribute that contract across neighboring equations, often expanding a relation and then projecting it onto a term, limit, observable, or consequence.

$$\begin{aligned} \text{identity}: &\quad \mathcal{I}_Q = ((\mathcal{H}_C, \mathcal{D}_C), \mathcal{O}_C) \\ \text{completion}: &\quad \mathcal{F}_Q = (\mathcal{K}_C, \mathcal{E}_C, \mathcal{P}_C) \\ \text{source moves}: &\quad +\,\text{role}, \quad \text{projection}, \quad \text{lateral rewrite}. \end{aligned}$$

Standard quantum mechanics already contains Hilbert spaces, operators, spectra, Born probabilities, and commutators. The corpus result concerns their organization: carrier and operator form the reusable typed identity, completion is attached locally, and source derivations move in both directions through completion rank.

Why Hilbert Space Is Central

Hilbert space supplies the carrier on which states, operator domains, inner products, spectra, and probability measures are defined. A position or configuration space instead labels one possible representation of that carrier. The two spaces coincide neither conceptually nor generally. For a spinless particle on a configuration space Q with measure μ, a pure state is a ray $[\psi]$ in

$\mathcal{H} = L^2(Q, d\mu)$. The function $\psi(x) = \langle x|\psi\rangle$ is a representative of that ray in the generalized position basis; $x \in Q$ labels a configuration, not the representation itself. Functions equal almost everywhere define the same vector, and vectors related by a global phase define the same pure state. For $Q = \mathbb{R}^3$ with Lebesgue measure this becomes $L^2(\mathbb{R}^3, d^3x)$, but spin gives $L^2(Q, d\mu) \otimes \mathbb{C}^{2s+1}$, identical particles require symmetric or antisymmetric subspaces, and mixed states are density operators rather than rays.

$$
\begin{aligned}
[\psi] \in \mathbb{P}(\mathcal{H}), \quad |\psi\rangle \sim e^{i\alpha}|\psi\rangle &\qquad \text{pure state ray} \\
\psi(x) = \langle x|\psi\rangle, \quad \Pr(X \in \Delta) = \int_\Delta |\psi(x)|^2 d\mu(x) &\qquad \text{position representation} \\
\rho \geq 0, \quad \operatorname{Tr}\rho = 1, \quad \Pr_A(\Delta) = \operatorname{Tr}(\rho E_A(\Delta)) &\qquad \text{general state and readout} \\
A = A^\dagger, \quad A = \int_{\sigma(A)} \lambda \, dE_A(\lambda) &\qquad \text{operator-to-spectrum conversion} \\
\rho_t = U(t)\rho U(t)^\dagger, \quad U^\dagger U = I &\qquad \text{identity-preserving evolution.}
\end{aligned}
$$

The carrier $(\mathcal{H}_C, \mathcal{D}_C)$ and its operator apparatus must be typed together. States inhabit the carrier, generators act on specified domains, observables supply spectral measures, and Born probabilities refer to those measures. A choice of carrier constrains the legal questions but does not select an outcome.

The Compact Representation

The compact representation is the minimal formal record needed to say what a quantum mechanism is doing. It is a compressed version of standard quantum mechanics:

> **Compact tree:** carrier/context $\leftrightarrow$ operator apparatus, with closure, readout, and protocol attached as required.
>
> Realization changes boundaries, fields, detectors, encodings, and scaling limits without forcing a new universal construction order.

- **Context** means the experimental or mathematical setting in which a quantum question is posed: preparation, basis, boundary, gauge, detector, domain, or representation.
- **Hilbert-space carrier and domain** means the Hilbert space, operator domain, and constraints selected by that context. Examples include normalization, positivity of a density operator, boundary conditions, gauge constraints, and domain conditions for an unbounded operator.
- **Generator/evolution** means the Hamiltonian, unitary map, quantum channel, action, or other lawful map that transports the state before readout.
- **Observable spectrum** means the allowed answer set of a measurable question, represented by a self-adjoint operator, projection-valued measure, POVM, or channel readout.
- **Probability readout** means the Born or trace rule that turns the state and spectral channels into probabilities for observed outcomes.
- **Compatibility constraint** means a restriction on which questions can be jointly resolved, usually expressed by commutators, uncertainty relations, contextuality tests, conservation laws, or admissibility checks.

- **Boundary/protocol realization** means the physical or engineered implementation: a potential well, scattering boundary, cavity, field mode, detector, circuit, channel, or measurement protocol.

$$
\begin{aligned}
\mathfrak{M}_Q &= (C, \mathcal{H}_C, \mathcal{D}_C, \rho, H_C, A_C, E_{A_C}, \mathrm{Pr}, \mathcal{K}, \mathcal{R}) \\
C &: \text{ context, preparation, basis, boundary, gauge, or detector arrangement} \\
\mathcal{H}_C, \mathcal{D}_C &: \text{ admissible state space and operator domain} \\
\rho_t &= U_C(t)\rho U_C(t)^\dagger, \qquad U_C(t) = \exp(-iH_C t/\hbar) \\
A_C &= \int_{\sigma(A_C)} \lambda \, dE_{A_C}(\lambda) \\
\mathrm{Pr}(\Delta \mid C, \rho, A) &= \mathrm{Tr}(\rho_t E_{A_C}(\Delta)) \\
\mathcal{K} &: \text{compatibility tests such as commutators, constraints, and admissibility checks} \\
\mathcal{R} &: \text{realization layer: boundary, field mode, detector, circuit, or scaling limit.}
\end{aligned}
$$

This representation explains the tree. Context selects the Hilbert-space carrier and operator domain. The wave function and density matrix are states on that carrier. The Hamiltonian and unitary operator contribute the generator. Observables and projection-valued measures contribute the spectral question. The Born rule contributes the probability readout. Commutators, uncertainty, Bell-type pages, and EPR-type pages contribute compatibility tests. Tunnelling, particle-in-a-box, scattering, fields, gauge theory, circuits, and channels contribute realization or protocol layers.

Recipe For Reading A Page

1. **Locate the page in the tree.** The branch tells which dependency of $\mathfrak{M}_Q$ the page mainly specifies.
2. **Fill the compact tuple.** Identify C, $\mathcal{H}_C$, ρ, H_C or the relevant map, A_C, the spectral measure or readout, and any compatibility condition.
3. **Read the topic page in full.** Every retained topic remains in the book. Native equations and source links show how its local construction differs from the shared branch structure.
4. **Read anomalies as diagnostics.** An anomaly is a junction where the topic uses several roles at once; the diagnosis is decomposition, not rejection.
5. **Use transfer carefully.** A mechanism can be transferred only when the state carrier, operator role, readout, and compatibility test survive the move; field-specific nouns may change.

What Was Found In This Run

The current export contains 146 quantum pages. Of these, 145 have topic-specific equation skeletons or explicit constructor overrides, and 1 are expanded from the compact constructor as core-derived mechanisms. The sparse-attention profile is strongly operator/spectral: 145 pages exceed the operator/spectrum threshold, with mean signal 0.328. State evolution is also broad (145 pages), while boundary/context (54), incompatibility (59), and explicit protocol (32) are more selective. The practical result is a compact reading of quantum theory as operator-to-spectrum conversion under context and compatibility constraints. This is the main structural finding of the rewrite. Quantum theory can be introduced as a constructor in which particles, waves, fields, detectors, boundaries, circuits, and interpretations are roles attached to a smaller state-operator-readout spine. The portable unit is the legal transformation from context and state to spectral readout.

Branch Map

Role	Use in the derivation	Pages
Hilbert-space context: admissible carrier and basis	A quantum calculation first fixes the Hilbert space, operator domain, basis, preparation context, representation, gauge, or boundary condition. This is not the measured answer; it is the legal carrier on which states, transformations, observables, and probabilities can be defined.	7
State carrier inside Hilbert space	A state is the probability-bearing element of the selected Hilbert space or its density-operator state space. Wave functions, density matrices, superpositions, and coherent states are different representations of this predictive carrier.	24
Generator: lawful change before readout	Hamiltonians, unitary maps, equations of motion, and path weights describe the lawful transport of the state before a question is resolved.	22
Spectral question: what can be asked	An observable is a permitted question whose operator form determines the possible numerical answers.	10
Readout rule: how answers become probabilities	Measurement connects a state and an observable to recorded frequencies. Projection, POVMs, Born weights, and collapse language are alternative ways of presenting this state-to-spectrum readout step.	10
Compatibility limit: what cannot be jointly sharp	The non-classical part of the theory appears when two otherwise legal questions do not compose into one common sharp question. Commutators, uncertainty relations, contextuality, Bell tests, and entanglement live here.	6
Boundary realization: how effects appear	Many named quantum effects are boundary realizations of the same construction. A potential, barrier, box, cavity, detector, or medium changes the allowed spectral channels without changing the basic prediction problem.	12
Many-mode extension: fields, particles, and scaling	Quantum field theory, gauge theory, renormalization, photons, fermions, and related topics extend the same state-operator-spectrum logic to variable particle number, local fields, and scale-dependent descriptions.	21
Protocol layer: engineered transformations	Quantum computing, channels, circuits, algorithms, networks, sensors, and error correction turn the same formal machinery into controlled sequences of operations.	18

Role	Use in the derivation	Pages
Annotations: history, interpretations, and popular frames	Some pages help readers navigate the subject but do not form steps in the mechanism. They are kept as annotations so books, historical figures, interpretations, and popular frames do not distort the constructive tree.	16

Preamble: Findings From The Rewrite

Core Insight

The rewrite reveals a compact dependency structure beneath the usual quantum topic list. The new DAG and source constructor distinguish the identity of a mechanism from the local moves used to derive it:

> **Constructor structure:** carrier/context $\leftrightarrow$ operator apparatus; closure, readout, and protocol attach as completion modes; boundaries, fields, and devices supply realizations. Source derivations move by completion, projection, and lateral rewrite.

Here, context means the preparation, basis, boundary, gauge, domain, detector, or representation in which a question is posed. It selects the Hilbert space and operator domain allowed by that context, together with normalization, positivity, boundary, gauge, or domain constraints. This carrier is central: it defines which states are legal, how amplitudes and norms are compared, which operators may act, and how spectral projectors can be read probabilistically. The generator is the Hamiltonian, unitary map, channel, or action that carries the state before readout. The observable spectrum is the allowed answer set of the question being asked. The probability readout is the Born or trace-rule assignment to those answers. The compatibility constraint records which otherwise valid questions cannot be resolved together, for example through a non-zero commutator, uncertainty relation, contextuality test, conservation law, or admissibility condition. Boundary and protocol realization specify how the same construction is embodied as a potential well, scattering boundary, cavity, field mode, detector, circuit, or measurement sequence.

Main Finding And Why It Matters

The rewrite reorganizes the quantum corpus by the sequence needed to make a prediction: select a domain or Hilbert space, assign a state, specify a generator or measurement map, resolve the relevant spectrum, assign probabilities, and check compatibility and realization conditions. Named topics then enter as positions in that sequence. Particles, waves, measurement, tunnelling, entanglement, collapse, fields, and interpretations are not treated as equal roots; each supplies a state, map, readout, compatibility limit, boundary condition, protocol, or annotation.

This gives a test for comparing pages. A page is technically specified when it states what state is carried, what operator or map acts, what spectrum or readout is produced, what compatibility constraint applies, and what changes when the physical realization changes. Two topics can therefore differ in vocabulary while preserving the same state-operator-readout structure.

This organization also changes how quantum models can be compared. The comparison is no longer based on topic names alone; it asks whether the Hilbert space or domain, operator class, spectral readout, and compatibility condition remain well defined after a change of representation or realization. Mixed-role pages then become specific decomposition problems rather than isolated puzzles.

Transition Sparse-Attention Result

A second sparse-attention pass compared the article ordering with the rewritten derivation ordering. The comparison asks which formal roles become visible when the same quantum topics are sorted by Hilbert-space context, state, generator, spectrum, probability rule, compatibility condition, and physical realization rather than by article title.

The transition run covered 146 pages. Of these, 145 currently have topic-specific equation skeletons or explicit page overrides, while 1 are expanded from the common quantum formalism. The distinction is evidential, not structural: both page types are read through the same context-state-generator-spectrum-readout sequence.

The transition signal is asymmetric: the rewrite adds formal roles more strongly than it adds object names. It exposes state, operator, readout, compatibility, boundary, and protocol roles that are implicit in the article ordering.

Route role	Mean signal	Interpretation
operator-to-spectrum readout	0.3284	formal role preserved in the rewrite
state evolution	0.2318	formal role preserved in the rewrite
normalization or admissibility	0.1651	formal role preserved in the rewrite
preparation, basis, or boundary context	0.0947	formal role preserved in the rewrite
non-commuting compatibility limits	0.0865	formal role preserved in the rewrite
controlled update protocol	0.0756	formal role preserved in the rewrite

New Structural Information

1. **The rewrite converts a noun-indexed encyclopedia into a derivation graph.** The new object is not a better summary of each topic; it is an ordering relation: which topic plays context, state, generator, observable, readout, compatibility, boundary, field, protocol, or annotation.

2. **The dominant stable role is operator-to-spectrum readout, not object naming.** The rewrite makes explicit that many named quantum topics become different ways of asking a legal spectral question of a state.

3. **Particles become stable role-realizations inside field/mode/readout machinery.** The particle pages are not discarded; they are relocated as field/mode/statistics/readout constructions. This is a more precise statement than 'particles are not fundamental'.

4. **Interpretations mostly act on state, probability, and update semantics.** QBism, relational quantum mechanics, collapse language, and popular frames can be kept without letting them become false roots of the derivation tree.

5. **Boundary pages are realization gates: they change allowed spectra without changing the core prediction problem.** Tunnelling, particle-in-a-box, scattering, cavities, and spectral lines become one family: boundary-shaped spectra.

6. **Protocol pages compose state preparation, maps, and readouts.** Quantum computing is reorganized as controlled composition of states, operators, readouts, and error constraints rather than a separate ontology of qubits.

7. **Anomalies identify where several formal roles coincide.** EPR, the measurement problem, quantum gravity, quantum biology, and related pages combine several formal questions in one topic: state preparation, compatibility, boundary or environmental coupling, protocol order, and readout.

Highest-Attention Transition Pages

- **Quantum logic** (generators, constructed): dominant roles after rewriting are measurement, context, state. Formal reading: This page sits between generators and annotations. The ambiguity marks a place where two formal roles meet and require separate treatment before the page supports a derivation.

- **Quantum biology** (generators, constructed): dominant roles after rewriting are context, observable, measurement. Formal reading: Quantum biology is an open-system transfer problem. The anomaly is that the biological environment is not background noise only; it is part of the boundary that may preserve, destroy, or select coherence. The unresolved formal fields are the state carrier, environmental coupling, coherence or transport observable, and the classical control that would remove the quantum contribution.

- **Schrödinger's cat** (states, constructed): dominant roles after rewriting are measurement, context, state. Formal reading: Schrödinger's cat is a macroscopic readout protocol, not a spectrum-first topic. It couples microscopic unitary evolution to a macroscopic boundary and forces the reader to separate three steps: coherent state transport, decoherence or apparatus coupling, and the rule by which one record is selected or conditioned.

- **Measurement problem** (measurement, constructed): dominant roles after rewriting are measurement, context, state. Formal reading: The measurement problem is a readout junction. It sits where unitary state transport, detector context, probability assignment, and state update meet. The formal decomposition is pre-measurement evolution, apparatus or environment coupling, POVM or projection readout, and the rule used to condition the state after the record.

- **Delayed-choice quantum eraser** (measurement, constructed): dominant roles after rewriting are measurement, context, state. Formal reading: The delayed-choice eraser is a protocol-order stress test. Its mechanism is the arrangement of which-path information, later measurement choice, and conditional correlation readout. The anomaly is not retrocausality by default; it is that the relevant statistics are defined only after the full measurement protocol is specified.

- **Introduction to quantum mechanics** (annotations, constructed): dominant roles after rewriting are measurement, context, state. Formal reading: An introductory page is anomalous because pedagogy compresses the whole formal sequence into one narrative. It mixes states, operators, spectra, measurement, examples, and interpretations. Its technical content separates into the individual branches before supporting specific derivations.

- **Einstein–Podolsky–Rosen paradox** (incompatibility, constructed): dominant roles after rewriting are measurement, context, state. Formal reading: EPR is a compatibility test, not a page about a peculiar object. Its mechanism is a bipartite state, separated measurement contexts, and a correlation readout that cannot be reduced to pre-existing local values. The formal starting point is the joint state and the allowed local observables; the question is which correlation constraints fail.

- **Quantum nonlocality** (incompatibility, constructed): dominant roles after rewriting are measurement, context, state. Formal reading: This incompatibility page mixes operator-to-spectrum readout, state evolution, controlled update protocol. It identifies which otherwise legal questions fail to share a single sharp representation, and what experiment or inequality exposes that failure.

Consequences For The Quantum Presentation

The domain or Hilbert space precedes the state; the state precedes the generator or measurement map; the spectrum and probability rule precede interpretation. This order keeps particles, protocols, and interpretations downstream of the formal construction they use.

The 145 topic-specific pages already supply local equation skeletons. The 1 core-derived pages mark topics where the book can state the expected quantum ingredients, but still needs a page-native state, operator or map, spectrum or readout, compatibility condition, and realization.

Two topics can be compared when they preserve the same state-operator-readout pattern, even if their historical names differ. A valid comparison should keep track of the Hilbert space or domain, operator class, spectrum, probability rule, and compatibility condition.

Pages such as EPR, the measurement problem, quantum gravity, and quantum biology combine compatibility, boundary, protocol, and transport roles. They require decomposition into separate formal questions before they can be presented as derivations.

Interpretive pages change the status assigned to state vectors, probabilities, updates, observers, or recorded outcomes. They do not by themselves replace the Hamiltonian, operator algebra, spectral decomposition, or Born-rule assignment.

The resulting presentation is a derivation tree rather than a topic list. Topic-specific pages show local equations. Core-derived pages state which quantum ingredients must be supplied before the page supports a full derivation.

Main Findings

1. **A compact order is enough to reconstruct most pages.** The recurring order is Hilbert space or domain, state, generator, observable, spectral resolution, probability rule, and compatibility condition. The book uses this order to present the standard formalism before moving to particles, fields, protocols, and interpretations.

2. **Observables and spectra carry the largest recurrent signal.** 145 of 147 pages exceed the operator/spectrum threshold, with mean signal 0.328. This is why self-adjoint operators, projectors, eigenvalues, and spectral measures appear early in the reconstruction.

3. **State evolution is widespread but requires a later question.** 145 pages carry state-evolution or transport signal. In the quantum reading, a Hamiltonian, unitary map, semigroup, channel, or path integral evolves a state before an observable defines the recorded distribution.

4. **Domains and boundary conditions specify admissible realizations.** 54 pages emphasize preparation, boundary, or domain context. Tunnelling, cavities, scattering,

and particle-in-a-box are grouped as cases where the domain or potential changes the allowed spectrum or transition amplitude.

5. **Incompatibility marks failure of joint spectral resolution.** 59 pages carry explicit incompatibility signal. Entanglement, Bell phenomena, commutators, and uncertainty enter through observables that cannot be assigned one common sharp classical readout under the stated assumptions.

6. **Protocols are ordered implementations of the same formal steps.** Only 32 pages exceed the protocol threshold. Quantum computing and information pages are therefore placed after the state, map, measurement, and compatibility layers that their circuits or channels compose.

7. **Topic names map onto formal roles.** Hilbert-space context: admissible carrier and basis: 7, State carrier inside Hilbert space: 24, Generator: lawful change before readout: 22, Spectral question: what can be asked: 10, Readout rule: how answers become probabilities: 10, Compatibility limit: what cannot be jointly sharp: 6, Boundary realization: how effects appear: 12, Many-mode extension: fields, particles, and scaling: 21, Protocol layer: engineered transformations: 18, Annotations: history, interpretations, and popular frames: 16. This gives a formal table of contents for the book: the branch of a page is determined by the role its equations or witnesses emphasize.

8. **Particles enter through representations, statistics, and readout-stable excitations.** Particle pages resolve into field modes, spectra, occupation rules, statistics, and detector-stable excitations. Particle identity is treated as a role inside the state-operator-readout formalism, not as the first organizing variable of the book.

9. **String theory and AdS/CFT are field/geometry correspondence cases.** String theory is placed with many-mode spectral constructions. AdS/CFT is placed closer to geometry because boundary data, correlators, and reconstruction maps carry more of its formal content in the current evidence profile.

10. **Black-hole topics require a dedicated source set.** The sparse-attention analysis places black-hole physics near boundary, information, and closure questions. A reliable black-hole chapter needs a targeted run over horizons, Hawking radiation, entropy, holography, and information-loss source equations.

11. **Interpretations modify state, probability, update, or observer language.** Interpretive pages are retained as annotations on the formal machinery. They are not used to replace the Hamiltonian, operator algebra, spectral decomposition, or Born-rule probability assignment.

12. **Geometry supplies domains, boundary data, and reconstruction maps.** Geometry enters when the carrier space, metric, boundary, gauge representation, or reconstruction map determines which operators and readouts are admissible. In the current evidence profile, operator and spectral roles are more portable than geometric presentation.

What Changed Relative To Wikipedia

A conventional encyclopedia organizes quantum theory by names and historical articles. The source-equation rewrite organizes the same material by the operations needed to construct a prediction. The page named measurement no longer becomes the origin of the theory; it becomes the readout junction. The page named particle becomes a realization. The page named tunnelling becomes a boundary spectral channel. The page named quantum field theory becomes a scaling and many-mode extension. The tree therefore gives a new order of explanation rather than a new interpretation of consciousness, observation, or ontology.

Unusual Findings And Anomalies

The most useful anomalies are not errors. They are pages that sit between branches or violate the dominant operator/spectrum pattern. These pages mark conceptual junctions where a theory-builder should look for missing assumptions, hidden boundary conditions, or possible transfers to other fields.

The anomaly labels describe a page's role in the mechanism tree, not the literal physical object named by the page. For example, a page can be structurally anomalous because context, protocol, or compatibility carries the explanation before spectra are read out.

- **pre-spectral admissibility**: another construction step fixes the admissible question before eigenvalues become meaningful.
- **context participates in the law**: preparation, apparatus, context, representation, or boundary conditions are part of the mechanism rather than surrounding detail.
- **admissibility meets incompatibility**: the page joins the rules that make a quantum state legal with the rules that restrict jointly resolvable questions.
- **operation order matters**: the order of operations matters; the topic cannot be reduced to a static state, operator, and spectrum.
- **several mechanisms meet**: several construction steps meet in one topic, so the page is a junction rather than a clean branch leaf.
- **branch interface**: the topic belongs at an interface between two explanatory roles and should be read as a bridge before being assigned to one branch.

- **Schrödinger's cat**. Primary reading: State carrier inside Hilbert space; neighboring reading: Annotations: history, interpretations, and popular frames. Schrödinger's cat is a macroscopic readout protocol, not a spectrum-first topic. It couples microscopic unitary evolution to a macroscopic boundary and forces the reader to separate three steps: coherent state transport, decoherence or apparatus coupling, and the rule by which one record is selected or conditioned.
- **Einstein–Podolsky–Rosen paradox**. Primary reading: Compatibility limit: what cannot be jointly sharp; neighboring reading: Annotations: history, interpretations, and popular frames. EPR is a compatibility test, not a page about a peculiar object. Its mechanism is a bipartite state, separated measurement contexts, and a correlation readout that cannot be reduced to pre-existing local values. The formal starting point is the joint state and the allowed local observables; the question is which correlation constraints fail.
- **Quantum biology**. Primary reading: Generator: lawful change before readout; neighboring reading: Hilbert-space context: admissible carrier and basis. Quantum biology is an open-system transfer problem. The anomaly is that the biological environment is not background noise only; it is part of the boundary that may preserve, destroy, or select coherence. The unresolved formal fields are the state carrier, environmental coupling, coherence or transport observable, and the classical control that would remove the quantum contribution.

- **Introduction to quantum mechanics**. Primary reading: Annotations: history, interpretations, and popular frames; neighboring reading: State carrier inside Hilbert space. An introductory page is anomalous because pedagogy compresses the whole formal sequence into one narrative. It mixes states, operators, spectra, measurement, examples, and interpretations. Its technical content separates into the individual branches before supporting specific derivations.

- **Measurement problem**. Primary reading: Readout rule: how answers become probabilities; neighboring reading: Generator: lawful change before readout. The measurement problem is a readout junction. It sits where unitary state transport, detector context, probability assignment, and state update meet. The formal decomposition is premeasurement evolution, apparatus or environment coupling, POVM or projection readout, and the rule used to condition the state after the record.

- **Quantum gravity**. Primary reading: Many-mode extension: fields, particles, and scaling; neighboring reading: Generator: lawful change before readout. Quantum gravity is a field/boundary junction. It asks whether geometry itself becomes part of the quantum state carrier or remains a realization layer for an operator theory. The missing formal objects are explicit: a state of geometry, a constraint or evolution operator, a boundary or semiclassical readout, and a test showing which geometric quantities survive quantization.

- **Scattering**. Primary reading: Boundary realization: how effects appear; neighboring reading: Generator: lawful change before readout. Scattering is a boundary-to-spectrum mechanism. The relevant object is the map from asymptotic in-states to out-states, together with the interaction region, boundary or asymptotic channels, S-matrix or cross-section readout, and conservation constraints.

- **Quantum state**. Primary reading: State carrier inside Hilbert space; neighboring reading: Generator: lawful change before readout. Quantum state is the carrier, not the final prediction. It is anomalous because it precedes several downstream roles: admissibility, evolution, observable choice, and probability readout. The unresolved distinction is whether the carrier is a vector, density operator, field state, or register, and which transformations preserve its legality.

Interpretation Of The Anomalies

The sparse-attention interpretation is aggregated from role labels assigned by the mechanism pass. It summarizes recurring junction types in the mechanism tree rather than presenting a list of physical anomalies.

- **several mechanisms meet**. several construction steps meet in one topic, so the page is a junction rather than a clean branch leaf. Mechanism implication: one topic name covers several formal roles, so the page is a junction rather than a single branch leaf.

- **pre-spectral admissibility**. another construction step fixes the admissible question before eigenvalues become meaningful. Mechanism implication: the eigenvalue problem is downstream of another role that first defines the admissible question.

- **operation order matters**. the order of operations matters; the topic cannot be reduced to a static state, operator, and spectrum. Mechanism implication: the order of preparation, transformation, and readout is part of the mechanism.

- **context participates in the law**. preparation, apparatus, context, representation, or boundary conditions are part of the mechanism rather than surrounding detail. Mechanism implication: preparation, apparatus, domain, detector, or boundary conditions participate in the observed law.
- **admissibility meets incompatibility**. the page joins the rules that make a quantum state legal with the rules that restrict jointly resolvable questions. Mechanism implication: admissibility conditions and jointly resolvable questions meet in the same topic.
- **branch interface**. the topic belongs at an interface between two explanatory roles and should be read as a bridge before being assigned to one branch. Mechanism implication: the topic lies at an interface between explanatory roles and should be read as a bridge.

Constructor-Detected Research Leads

The useful anomalies in this book are constructor junctions: places where a topic requires several formal roles at once. Some are established foundational problems; others are under-formulated mechanism questions that can guide derivation and experiment.

- **Measurement as a four-map junction**. The established measurement problem becomes formally sharper when split into pre-measurement evolution, apparatus or environment coupling, POVM or projection readout, and post-record state conditioning. The useful signal is the missing local map between probability readout and state update.
- **Geometry as a possible state carrier**. Quantum gravity is a field/geometry interface problem. The constructor reading isolates the specific unresolved choice: whether geometry remains a substrate on which operators act, or becomes part of the quantum state carrier itself. The corresponding test is which geometric quantities survive as admissible observables after quantization.
- **Boundary-selected spectra across different phenomena**. Scattering, tunnelling, cavities, particle-in-a-box systems, spectral lines, and optical boundary problems share a boundary-to-spectrum construction. The research lead is to treat them as one transferable mechanism family rather than as separate examples tied to different physical nouns.
- **Environment as an active selector in quantum biology**. In quantum biology, the useful constructor signal is that the environment can be part of the mechanism: it may preserve, destroy, or select coherence. A useful derivation must name the carrier, coupling, coherence or transport observable, and classical control.
- **Hamiltonian dual role**. The Hamiltonian is both a generator of time evolution and an observable with an energy spectrum. This is standard formalism. The constructor lead is to keep those roles distinct when deriving mechanisms that use energy conservation, spectral measurement, and unitary transport in the same page.
- **Quantum simulation as a target-carrier-validation triangle**. A simulator is an engineered realization of another Hamiltonian, channel, or field theory. The anomaly is practical: the physical carrier, encoded target, and validation observable are distinct roles. A simulator claim is incomplete until all three are specified.

Relation To Lagrangian, Transfer, And Stability Layers

The later validation chapter explains how the global evidence layers check this tree. Stability tests ask which part of a formalism remains the same after a rewrite. Typed transfer and bridge-family layers ask which constructor roles can move between contexts. The Lagrangian navigation layer asks whether a proposed move through the atlas is a low-resistance continuation, a strained bridge, or a void-boundary target. In ordinary quantum language: the invariant is mostly operator/structural role, while geometry, boundary, and representation are where the role becomes embodied.

Practical Consequence

A reader, teacher, or AI constructor should not start from named quantum objects. The constructive order is: define the admissible context, define the predictive carrier, define lawful change, define the legal question, resolve its spectrum, assign probabilities, test incompatibility, then dress the construction in the relevant physical realization. That is the main structural result of the rewrite.

Part I

The Mechanism Tree

Chapter 1

Root Construction

Given a context-selected Hilbert space, an admissible state, and a permitted observable, quantum theory determines which numerical outcomes can occur, assigns probabilities to those outcomes, and marks which questions cannot be made simultaneously sharp.

1.1 Re-Derivation Path

1. **Type the identity.** Specify the Hilbert-space carrier, operator domain, state class, and operator apparatus together. Neither carrier nor operator is complete without the other.
2. **Attach completion.** Add normalization, symmetry, compatibility, spectral readout, probability assignment, or an ordered protocol where the problem requires it.
3. **Choose a realization.** State how boundaries, fields, scaling limits, detectors, gauges, or encodings embody the typed identity.
4. **Follow the derivation moves.** Track when a source equation adds a role, projects a relation onto a consequence, or rewrites the same role in another representation.

$$
\begin{aligned}
B &\longmapsto \rho_B \\
\rho_t &= U_t \rho_B U_t^\dagger \\
O &= \sum_i \lambda_i P_i \\
p_i &= \mathrm{Tr}(P_i \rho_t) \\
[O_1, O_2] &\neq 0.
\end{aligned}
$$

Chapter 2

Sparse Attention Summary

Operation	Mean signal	Pages above 0.10
State evolution	0.232	145
Normalization and admissibility	0.165	136
Observables and spectra	0.328	145
Preparation and boundary context	0.095	54
Incompatible questions	0.086	59
Controlled update protocol	0.076	32

The dominant recurrent signal is observables and spectra. Evolution, context, incompatibility, closure, and protocol are branch-forming operations. This is why the book begins with admissible context, state carrier, generator, and legal question rather than with particles or measurement folklore.

Chapter 3

Compact Operator Formulation

The compact form of nonrelativistic quantum theory is a typed carrier-operator identity with closure and readout attached where a prediction requires them.

This chapter is the formal spine of the book. It does not impose one temporal sequence on every derivation. Each later page is placed by asking which part of the typed identity it specifies, which completion condition it adds, and which realization it changes.

3.1 The Minimal Constructor

$$
\begin{aligned}
C &\longmapsto (\mathcal{H}_C, \mathcal{D}_C) \\
\rho \in \mathcal{S}(\mathcal{H}_C), &\qquad \rho \geq 0, \quad \operatorname{Tr}\rho = 1 \\
\rho_t = U_C(t)\rho U_C(t)^\dagger, &\qquad U_C(t) = \exp(-iH_C t/\hbar) \\
A_C &= \int_{\sigma(A_C)} \lambda \, dE_{A_C}(\lambda) \\
\Pr(\Delta \mid C, \rho, A) &= \operatorname{Tr}(\rho_t \, E_{A_C}(\Delta)) \\
[A_C, B_C] \neq 0 &\quad \Rightarrow \quad \text{no generic common sharp spectral refinement.}
\end{aligned}
$$

3.2 Interpretation Of The Symbols

- C is the context: preparation, basis, boundary condition, gauge choice, detector arrangement, domain, or representation.
- $\mathcal{H}_C$ is the admissible state space selected by that context.
- $\mathcal{D}_C$ is the domain or admissibility condition for the generator and observables.
- ρ is the predictive state carrier. It may be a state vector, density operator, field state, or register state.
- H_C is the generator of lawful change before readout.
- A_C is the question being asked. Its spectral measure E_{A_C} supplies the outcome channels.
- $\Pr(\Delta \mid C, \rho, A)$ is the Born probability of an outcome set Δ.
- A non-zero commutator marks a compatibility limit: two legal questions may not share one sharp readout basis.

3.3 Identity, Completion, And Realization

The carrier and operator apparatus are jointly typed: an operator has a domain on a carrier, while the carrier determines which operators and states are admissible. Closure, readout, and protocol are attached completion modes. A boundary, field representation, detector, or circuit is a realization of this structure rather than a later universal stage.

$$
\begin{aligned}
&\mathcal{I}_Q = \big((\mathcal{H}_C, \mathcal{D}_C),\, \mathcal{O}_C\big) && \text{typed identity} \\
&\mathcal{F}_Q = \big(\mathcal{K}_C,\, \mathcal{E}_C,\, \mathcal{P}_C\big) && \text{closure, readout, protocol} \\
&\mathcal{R}_Q : \mathcal{I}_Q \oplus \mathcal{F}_Q \longrightarrow \text{boundary, field, device, or encoding.}
\end{aligned}
$$

A calculation can add a missing role, project a complete relation onto one consequence, or rewrite the same role in another representation. These are derivation moves on the constructor, not new physical postulates.

3.4 Why This Is More Compact

The usual presentation introduces particles, waves, measurement, operators, interpretations, and fields as separate conceptual blocks. The compact constructor shows that many of these are changes of role rather than separate roots. A particle is a stable state/readout role. A barrier is a context that changes the operator domain. A field is a many-mode extension of the state space. A quantum circuit is a protocolized composition of maps. An interpretation changes the meaning of state, probability, or update, but usually leaves the constructor intact.

Chapter 4

Dependencies Exposed By The Constructor

The corpus supports a typed carrier-operator identity, an attached completion fibre, and a reversible derivation calculus. The completion DAG records a partial order of formal specification; the source constructor records how equations are actually developed inside papers.

4.1 How The Structure Was Obtained

The analysis combines 14,523,959 retained equation morphisms with a source-local graph of 14,354,227 equations from 2,151,957 paper groups. The factorization was selected on 100,000 sampled rows and then checked against the full assignment.

1. Each equation record was separated into carrier or context evidence, operator-apparatus evidence, and completion evidence for closure, readout, or ordered protocol.
2. Candidate grammars were compared by a description-length objective. The selected model retains carrier and operator as primitive factors and attaches completion as a fibre.
3. Mechanism edges were oriented only when the destination contained more constructor roles. This orientation defines completion rank, not physical time.
4. Equations adjacent in the same source were classified as role completion, role projection, or lateral rewrite. These edges restore the local order omitted by the corpus-wide DAG.

Measured quantity	Value	Consequence
Selected grammar description score	0.309	carrier–operator identity with attached completion
Three-factor alternative score	0.400	completion is not promoted to a third primitive space
Mechanism edges increasing completion rank	24.8%	the DAG is a partial order, not the whole graph
Coupled carrier–operator edges remaining lateral	99.99%	similarity usually preserves rank
Complete source constructor frames	14.2%	mechanisms are distributed across neighboring equations

Measured quantity	Value	Consequence
Completion–projection balance	0.992	source derivation is approximately bidirectional

4.2 Static Identity And Derivation Dynamics

Two structures must therefore be kept separate. The static identity says which operator apparatus is defined on which carrier and domain. The derivation dynamics says how a paper adds assumptions or readout, projects a relation onto a consequence, or changes representation without changing completion rank.

$$\begin{aligned}
\text{identity:}\quad & \mathcal{I} = ((\mathcal{H}, \mathcal{D}), \mathcal{O}),\\
\text{completion:}\quad & \mathcal{F} = (\mathcal{K}, \mathcal{E}, \mathcal{P}),\\
\text{derivation moves:}\quad & x \xrightarrow{+\,\text{role}} y, \qquad x \xrightarrow{\text{projection}} y, \qquad x \xleftrightarrow{\text{rewrite}} y.
\end{aligned}$$

The source graph contains 3,761,468 completion moves, 3,703,292 projections, and 4,737,510 lateral rewrites. The near balance between the first two classes explains why a derivation cannot be represented faithfully as a one-way funnel.

4.3 Dependencies Made Explicit

Hamiltonian. Generator and observable are distinct roles of the same operator. Time evolution and energy readout share domain conditions but are not the same operation.

Measurement. Probability readout and post-measurement state update are separate maps. Treating them as one step hides the measurement junction.

Wave function. A wave function is a coordinate representation of a state. Basis change alters the function while preserving the carrier-level state and its probabilities.

Entanglement. The factorization of the carrier determines which observables count as local. Subsystem structure therefore enters the definition of the readout, not only the state.

Gauge theory. Representational redundancy must be quotiented or constrained before a physical readout is assigned. Closure is part of admissibility rather than a later correction.

Quantum gravity. Geometry may be a realization domain or part of the quantum carrier. The two cases require different operator domains and different notions of observable.

Path integrals. Integral and differential descriptions are alternative realizations of generator dynamics. Their equivalence depends on measure, boundary, and regularization data.

Quantum simulation. Physical carrier, encoded target, and validation observable are three distinct roles. Agreement of a programmed Hamiltonian alone does not complete the simulation claim.

4.4 How This Chapter Organizes The Book

The branch chapters remain complete. They are grouped by the role each topic contributes to the typed identity, its completion, or its realization. Every topic page is retained; the constructor supplies cross-references between pages that share a formal dependency but use different physical vocabulary.

Chapter 5

Rewiring Quantum Concepts

A useful connection between two topics must specify the mathematical object that is retained, the part of the construction that is changed, and an independent calculation that can fail. Similar terminology or nearby evidence profiles are not sufficient.

5.1 Four Legal Moves

Representation change. Move between descriptions while preserving amplitudes, expectation values, or correlators on their common domain.

Carrier refactorization. Change the subsystem, algebra, boundary, or encoding structure; locality and entanglement must then be recomputed.

Completion attachment. Add a constraint, readout, or protocol to an otherwise incomplete carrier–operator pair.

Dual-role split. Separate two operations performed by one mathematical object, such as generation of motion and spectral readout by a Hamiltonian.

These moves do not assert that the connected theories are physically equivalent. They identify a controlled question: which predictions survive when one part of a quantum construction is replaced? The comparisons below were nominated by overlap among the equation-derived route profiles of the connected pages. That corpus signal is used only for selection; the displayed invariant and failure test determine whether a proposed connection survives mathematical scrutiny.

5.2 One operator, two roles: generator and observable

Connected topics: Hamiltonian Quantum Mechanics, Observable, Spectral Theory

Read the same operator through its exponential for transport and through its spectral measure for energy readout **Retained object or prediction:** the self-adjoint operator and its domain

$$U(t) = e^{-iHt/\hbar}$$
$$H = \int_{\sigma(H)} E \, dP_H(E)$$

Failure test: Check domain and self-adjointness once, then verify separately that the generated dynamics is unitary and that the spectral measure reproduces energy statistics.

5.3 Schrodinger, Heisenberg, and path-integral representations

Connected topics: Schrodinger Picture, Heisenberg Picture, Path Integral Formulation

Move time dependence between states and operators, or replace the propagator by an action-weighted integral **Retained object or prediction:** transition amplitudes and expectation values

$$A_H(t) = U(t)^\dagger A_S U(t)$$
$$\langle q_f | U(t_f - t_i) | q_i \rangle = \int \mathcal{D}q\, e^{iS[q]/\hbar}$$

Failure test: Specify domains, measure, boundary conditions, and regularization; the formulations are connected only where they yield the same amplitudes or correlators.

5.4 Carrier factorization defines locality

Connected topics: Quantum Entanglement, Quantum Information, Quantum Field Theory

Change the tensor-product or algebraic subsystem decomposition and track which observables remain local **Retained object or prediction:** the global state and algebraic predictions

$$\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$$
$$\rho_A = \mathrm{Tr}_B\, \rho_{AB}$$

Failure test: State the subsystem algebra or tensor factorization explicitly; entanglement and locality claims are not invariant under arbitrary refactorization.

5.5 Constraints define physical readout

Connected topics: Gauge Theory, Quantum Gravity, Measurement In Quantum Mechanics

Attach gauge or diffeomorphism closure before assigning outcome effects to physical observables **Retained object or prediction:** predictions on the physical state space

$$\widehat{\mathcal{C}}_a |\psi_{\mathrm{phys}}\rangle = 0$$
$$[\widehat{O}_{\mathrm{phys}}, \widehat{\mathcal{C}}_a] |\psi_{\mathrm{phys}}\rangle = 0$$

Failure test: Verify that candidate effects descend to the constrained or quotient state space; gauge-dependent quantities cannot be promoted directly to physical records.

5.6 Boundary conditions are spectral control variables

Connected topics: Particle In A Box, Quantum Tunnelling, Scattering, Spectral Theory

Change domain, boundary conditions, or asymptotic channels and follow the induced spectrum or transmission map **Retained object or prediction:** the differential operator family and probability conservation

$$H_D = -\frac{\hbar^2}{2m}\Delta_D + V$$
$$S : \mathcal{H}_{\mathrm{in}} \to \mathcal{H}_{\mathrm{out}}$$

Failure test: A claimed connection must identify the operator domain and conserved flux; similar-looking wave equations with different domains need not have comparable spectra.

5.7 Measurement, channels, and error correction share an instrument calculus

Connected topics: Quantum Channel, Measurement In Quantum Mechanics, Quantum Error Correction

Retain or discard the classical outcome of a quantum instrument, then condition a recovery channel on that outcome **Retained object or prediction:** complete positivity and total probability

$$\mathcal{I}_i(\rho) = \sum_\alpha K_{i\alpha}\rho K_{i\alpha}^\dagger$$
$$\mathcal{E} = \sum_i \mathcal{I}_i$$
$$\mathcal{R}_i \circ \mathcal{I}_i$$

Failure test: Check complete positivity, normalization, and recovery fidelity with a reference system; state-update rules alone are insufficient.

5.8 Duality and simulation require an intertwining map

Connected topics: Quantum Simulator, Ads Cft Correspondence, Quantum Error Correction

Encode one carrier in another and require the encoding to intertwine the relevant dynamics and readouts **Retained object or prediction:** a selected operator algebra and its correlators

$$VH_{\text{target}} \simeq H_{\text{carrier}}V$$
$$VO_{\text{target}} \simeq O_{\text{carrier}}V$$

Failure test: Validate more than state overlap: compare a generating set of observables or correlators and report the approximation regime and error bounds.

5.9 What The Rewiring Adds

The resulting organization is neither a chronology nor a vocabulary tree. It exposes dependencies that are scattered across conventional chapters: domains connect dynamics to spectroscopy; factorization connects entanglement to locality; constraints connect gauge redundancy to observable construction; and instruments connect measurement to channels and error correction. Each connection can be used to design a calculation, but none bypasses its domain, normalization, or approximation conditions.

Chapter 6

Mechanism Validation Layers

The mechanism tree is evaluated by three independent evidence layers: role-current stability asks what survives a rewrite, Gromov-Wasserstein (GW) neighborhoods measure relational bridge candidates, and the learned Lagrangian asks whether a proposed move is an easy continuation, a strained bridge, or a void-boundary design target.

6.1 Role-Current Layer: What Stays Stable

The role-current layer measures stability of formal roles under rewrite and translation. When a theory is rewritten, translated, or moved through the corpus, the test asks which part of the formal apparatus keeps its role. For a mechanism tree, this is the identity test: two formulations belong together when the same role contract survives the move.

- The run promoted ? strict currents and ? near currents. These mark role-preserving directions in the learned representation.

6.2 Structural Bridge Layer: Candidate Translations

The Gromov-Wasserstein layer compares relational neighborhoods. It contributes a structural bridge score: two formulations are stronger bridge candidates when their local neighborhoods preserve comparable relations among roles. In the current constructor, this layer is combined with typed transfer edges, bridge-family candidates, source-local transitions, and source-equation validation.

- The run scanned ? Gromov-Wasserstein candidates, hardened ? bridge records, and retained ? artifact-validated method bridges.
- GW nominates relationally compatible bridge candidates; typed transfer edges, directed source transitions, source equations, and residual checks rank them for use.

6.3 Lagrangian Layer: Which Moves Are Easy Or Strained

The learned Lagrangian is the navigation layer of the atlas. It is a representation-space action surrogate over source-equation fingerprints, separate from the physical action of a quantum system. Given a construction-role state, it asks which next states are cheap continuations, which require a strained bridge, and which absences sit on a meaningful void front. In this sense it finds roads through the formal grammar: low-action corridors where a formal role can be continued, translated, or tested. In the current quantum tree this layer supplies the global road-map constraint. A full page-coordinate export will allow direct page-level action scoring.

- For the quantum tree, the Lagrangian is therefore a construction constraint rather than a leaf selector. It tells how open construction steps should be interpreted in the atlas, but the present export cannot yet say that a particular Wikipedia page lies on a particular low-action road. That requires full fingerprints or witness transition vectors for the page.

6.4 How These Layers Change The Quantum Tree

A quantum concept is not placed by name alone. It is placed by the role it plays in the mechanism, checked against source equation witnesses, and interpreted through stability and transfer layers. If a page preserves structure and spectral role, it belongs near the spine. If it changes the admissible domain, it belongs in boundary realization. If it introduces many modes or scale flow, it belongs in the field/scaling extension. If it describes a sequence of controlled operations, it belongs in the protocol layer. This is the human-readable reason for the tree: it unfolds quantum theory as a construction, then asks which parts survive when notation, representation, geometry, or field changes.

Chapter 7

Sparse-Attention Result

Quantum theory, in this sparse-attention reading, is a constructor that turns admissible state descriptions into spectra and probabilities under compatibility and boundary constraints.

Particles, strings, fields, measurements, and interpretations collapse into roles in the same constructor rather than separate root categories.

7.1 Minimal Constructor

The sparse-attention run returns the same constructor as a role sequence. Rendered in standard quantum notation, the sequence is:

$$\begin{aligned}
B &\longmapsto (\mathcal{H}_B, \mathcal{D}_B) \\
\rho_B(t) &= U_B(t)\rho_B(0)U_B(t)^\dagger \\
O_B &= \sum_i \lambda_i P_i \qquad \text{or, more generally,} \qquad O_B = \int_{\sigma(O_B)} \lambda \, dE_{O_B}(\lambda) \\
p_i &= \mathrm{Tr}(P_i \rho_B(t)) , \qquad \Pr(\Delta) = \mathrm{Tr}(\rho_B(t) E_{O_B}(\Delta)) \\
[O_1, O_2] \neq 0 \quad &\Rightarrow \quad \text{no generic common sharp readout} \\
\mathcal{R} :&\ \text{boundary, field, detector, circuit, or scaling realization.}
\end{aligned}$$

- B denotes the context: preparation, basis, boundary, domain, gauge, or detector arrangement.
- $(\mathcal{H}_B, \mathcal{D}_B)$ is the admissible state space and operator domain selected by that context.
- $\rho_B(t)$ is the propagated state carrier.
- O_B, P_i, and E_{O_B} express the observable question and its spectral readout channels.
- The probabilities p_i or $\Pr(\Delta)$ are Born-rule readouts of the state against those channels.
- Non-commuting observables define compatibility limits; realization layers specify how the same role is embodied.

7.2 Route Statistics

Role	Mean signal	Pages > 0.10
state evolution	0.232	146
admissibility and closure	0.165	137
observable spectra	0.327	146
preparation and boundary context	0.094	54
incompatible questions	0.086	59
controlled update protocol	0.075	32

The strongest recurrent signal is observable spectra. In the current export it is active above threshold on nearly every page, while protocol/update is much sparser. This is the quantitative reason the tree begins with state, generator, observable, spectrum, and readout before it introduces algorithms, experiments, or interpretations.

7.3 Constructor Regularities

The regularities below are the strongest reusable patterns in the sparse-attention export. They are public reading rules for the quantum book, not internal diagnostics or a fixed ontology. Title-token rules and other lexical artifacts are omitted here because they are useful for auditing but too weak for the main exposition.

7.3.1 Operator-to-spectrum as the readout spine

Reading. The stable readout of a quantum construction is usually spectral. A state becomes experimentally or mathematically predictive when an allowed operator question exposes eigenvalues, projectors, or a spectral measure.

Reading consequence. Introduce observables, self-adjointness, spectral projectors, and the Born rule as the common readout interface before presenting particles, fields, or protocols as separate subjects.

7.3.2 State transport as the carrier of change

Reading. Quantum topics repeatedly require a lawful carrier of change: a state vector, density operator, field mode, or channel is propagated before it is read.

Reading consequence. Derive state evolution as the map between preparation and later readout, using unitary, semigroup, or channel language depending on whether the system is closed, open, or operational.

7.3.3 Admissibility as the legal spine

Reading. Closure and admissibility form the legal spine of the theory: states must be normalizable or positive, maps must preserve the allowed state set, and operators must have a domain on which the question is well-defined.

Reading consequence. Every derivation should state what makes the state, operator, map, or probability assignment legal before it discusses physical interpretation.

7.3.4 Incompatibility as a selective limit

Reading. Incompatibility is selective. It becomes central when two questions, bases, or transformations cannot be made sharp at the same time.

Reading consequence. Use commutators, non-common eigenbases, Bell-type constraints, or contextuality tests only where the page is actually about jointly unavailable readouts.

7.3.5 Context as the active boundary of the question

Reading. Context is sometimes part of the law. Preparation, boundary conditions, detector arrangement, basis choice, or representation can change which state and operator are admissible.

Reading consequence. When context is active, write the context before the state: specify the Hilbert space, domain, boundary, apparatus, or channel that makes the later spectral question meaningful.

7.3.6 Protocol as ordered implementation

Reading. Protocols are a sparse but real layer. They become central when an ordered sequence of operations, measurements, controls, or updates determines what can be inferred.

Reading consequence. Place quantum algorithms, circuits, finite automata, Bell tests, and delayed-choice experiments in this layer: their content depends on operation order, not only on a static Hamiltonian.

7.3.7 Junction pages require decomposition

Reading. Some pages are junctions rather than leaves. EPR, wave function, delayed-choice eraser, entanglement, quantum biology, and similar topics join state, evolution, readout, boundary, protocol, and compatibility in one place.

Reading consequence. Such pages should be read as junctions of several formal roles: state carrier, transformation, readout, context, and compatibility condition. Their position marks where a single topic name covers more than one construction step.

7.3.8 Geometry enters as realization

Reading. Geometry enters mainly through realization-heavy branches. In this export, geometric language most often specifies boundary, field, context, or reconstruction layers rather than replacing the operator-spectral core.

Reading consequence. Introduce geometry when the theory needs a domain, boundary, metric, representation, field realization, or reconstruction map. Keep the operator and admissibility structure visible underneath.

7.4 Topic-Specific Readings

- **Particle**: In this reading, a particle is not the root unit. It is a stable field/mode/readout realization of the state-operator-spectrum constructor. Photon, electron, fermion, and boson do not share one identical route profile; 'particle' decomposes into spectral, boundary, closure, and statistics roles. Relevant pages: Boson, Fermion, Photon, Fock space, Creation and annihilation operators, Electron.
- **Black-hole direction**: Black holes do not appear as a primary constructor node in this export. They enter as a boundary/geometry stress test through quantum gravity, string, AdS/CFT, and horizon-like witnesses. This suggests a useful rerun: black-hole topics

should be processed explicitly to test whether horizon physics is a boundary readout, an information-closure problem, or an geometry-free operator-kernel gap. Relevant pages: Quantum gravity, AdS/CFT correspondence, Quantum geometry, Loop quantum gravity, Electron, Fermion.

- **String-theory direction**: String theory appears as a many-mode field/scale extension with strong spectral content and geometry/boundary pressure, not as a separate noun ontology. The string is best read as a carrier of spectra under constraints; AdS/CFT is the sharper geometry-translation page because its boundary signal is higher. Relevant pages: String theory, AdS/CFT correspondence, Renormalization, Quantum mechanics, Quantum geometry, Loop quantum gravity.

- **Measurement and interpretations**: Measurement is a readout/probability layer, and interpretations modify the meaning of that layer rather than the constructor equations. QBism and relational pages can have strong operator/spectrum evidence while still being annotations, because they narrate the readout layer instead of adding new dynamics. Relevant pages: QBism, Wave function collapse, POVM, Interpretations of quantum mechanics, Born rule, Quantum mechanics.

- **Geometry**: Geometry behaves as realization and reconstruction: it makes the operator construction legal on domains, boundaries, gauges, or holographic duals, but it is not the strongest preserved current. Quantum gravity, AdS/CFT, and spin foam cluster near field/boundary roles; this matches the current source-equation stability checks. Relevant pages: Spin foam, Loop quantum gravity, Quantum geometry, AdS/CFT correspondence, Electron, Fermion.

- **Void leads**: The best discovery leads are missing gates: contexts where a state, operator, boundary, or compatibility relation is implied but not closed by evidence. The black-hole/holography area and geometry-free operator kernels are natural tests for this, They require explicit topic reruns and typed equations. Relevant pages: Quantum mechanics, Quantum geometry, Fock space, Photon, Electron, Fermion.

These statements are role hypotheses produced by deterministic sparse attention over the quantum page archive and existing source-equation findings. They are not final physics claims. The next test is to rerun the same analysis on explicit black-hole, horizon, Hawking-radiation, entropy, holography, and information-loss topics with typed equation witnesses.

Chapter 8

Anomalies And Leads

8.1 Structural Anomalies

The labels below describe why a page is structurally interesting in the mechanism tree. They do not describe the literal object named by the page.

- **pre-spectral admissibility**: another construction step fixes the admissible question before eigenvalues become meaningful.
- **context participates in the law**: preparation, apparatus, context, representation, or boundary conditions are part of the mechanism rather than surrounding detail.
- **admissibility meets incompatibility**: the page joins the rules that make a quantum state legal with the rules that restrict jointly resolvable questions.
- **operation order matters**: the order of operations matters; the topic cannot be reduced to a static state, operator, and spectrum.
- **several mechanisms meet**: several construction steps meet in one topic, so the page is a junction rather than a clean branch leaf.
- **branch interface**: the topic belongs at an interface between two explanatory roles and should be read as a bridge before being assigned to one branch.

- **Schrödinger's cat**. Schrödinger's cat is a macroscopic readout protocol, not a spectrum-first topic. It couples microscopic unitary evolution to a macroscopic boundary and forces the reader to separate three steps: coherent state transport, decoherence or apparatus coupling, and the rule by which one record is selected or conditioned.
- **Einstein–Podolsky–Rosen paradox**. EPR is a compatibility test, not a page about a peculiar object. Its mechanism is a bipartite state, separated measurement contexts, and a correlation readout that cannot be reduced to pre-existing local values. The formal starting point is the joint state and the allowed local observables; the question is which correlation constraints fail.
- **Quantum biology**. Quantum biology is an open-system transfer problem. The anomaly is that the biological environment is not background noise only; it is part of the boundary that may preserve, destroy, or select coherence. The unresolved formal fields are the state carrier, environmental coupling, coherence or transport observable, and the classical control that would remove the quantum contribution.

- **Introduction to quantum mechanics**. An introductory page is anomalous because pedagogy compresses the whole formal sequence into one narrative. It mixes states, operators, spectra, measurement, examples, and interpretations. Its technical content separates into the individual branches before supporting specific derivations.

- **Measurement problem**. The measurement problem is a readout junction. It sits where unitary state transport, detector context, probability assignment, and state update meet. The formal decomposition is pre-measurement evolution, apparatus or environment coupling, POVM or projection readout, and the rule used to condition the state after the record.

- **Quantum gravity**. Quantum gravity is a field/boundary junction. It asks whether geometry itself becomes part of the quantum state carrier or remains a realization layer for an operator theory. The missing formal objects are explicit: a state of geometry, a constraint or evolution operator, a boundary or semiclassical readout, and a test showing which geometric quantities survive quantization.

- **Scattering**. Scattering is a boundary-to-spectrum mechanism. The relevant object is the map from asymptotic in-states to out-states, together with the interaction region, boundary or asymptotic channels, S-matrix or cross-section readout, and conservation constraints.

- **Quantum state**. Quantum state is the carrier, not the final prediction. It is anomalous because it precedes several downstream roles: admissibility, evolution, observable choice, and probability readout. The unresolved distinction is whether the carrier is a vector, density operator, field state, or register, and which transformations preserve its legality.

- **Wave–particle duality**. Wave-particle duality is a representation/readout switch. The same carrier is interrogated through incompatible experimental contexts, so the observed pattern changes from interference-like to count-like. Its compact form is context selection plus readout channel.

- **Quantum entanglement**. Entanglement is a tensor-factorization and correlation constraint. The anomaly is that the state is not reducible to independently readable subsystem states, while the readout is still local and spectral. The required distinction is between joint state, subsystem observables, and correlation test.

- **Fermi–Dirac statistics**. Fermi-Dirac statistics is an admissibility rule for many-particle states. The central mechanism is antisymmetry and occupation restriction, not an ordinary eigenvalue list. The formal content is anticommutation, exclusion, occupation numbers, and the thermodynamic readout derived from that constrained state space.

- **Hamiltonian (quantum mechanics)**. The Hamiltonian has two roles at once: it generates time evolution and, when treated as an observable, supplies an energy spectrum. That double role explains the anomaly. A clean page must separate domain/self-adjointness, unitary transport, conserved energy, and spectral readout.

- **Wave function**. The wave function is a representation of the state carrier, not a material wave by itself. Its anomaly is that it stores amplitude, phase, normalization, basis choice, and probability potential in one object. The formal decomposition separates representation, admissibility, evolution, and Born readout.

- **Delayed-choice quantum eraser**. The delayed-choice eraser is a protocol-order stress test. Its mechanism is the arrangement of which-path information, later measurement choice, and conditional correlation readout. The anomaly is not retrocausality by default;

it is that the relevant statistics are defined only after the full measurement protocol is specified.

- **Relativistic quantum mechanics**. Relativistic quantum mechanics is a compatibility junction between quantum state evolution and spacetime symmetry. Its construction preserves relativistic covariance, defines the correct state carrier, and explains how spin, energy, and causality constraints enter the operator algebra.

- **Quantum electrodynamics**. Quantum electrodynamics is a field-interaction construction. Its anomaly comes from combining gauge admissibility, charged matter states, photon modes, perturbative transport, and scattering/readout. The relevant formal objects are field operators, gauge constraints, interaction terms, and observable amplitudes.

- **Quantum field theory**. This field-level page mixes operator-to-spectrum readout, state evolution, normalization or admissibility. It is a many-mode or geometric realization problem: state sector or field algebra, constraints, and observable readout determine the field content.

- **Bell's theorem**. This incompatibility page mixes operator-to-spectrum readout, state evolution, normalization or admissibility. It identifies which otherwise legal questions fail to share a single sharp representation, and what experiment or inequality exposes that failure.

8.2 Research Leads

- Search for systems where a state-like carrier and a legal-question operator exist, but the incompatibility relation has not been tested. Those systems are candidates for quantum-like contextual behavior without importing quantum ontology.

- Treat tunnelling, particle-in-a-box, cavity optics, and spectral lines as one family of boundary-shaped spectra. This suggests looking for overlooked boundary controls in systems currently described only by bulk evolution.

- Quantum computing should be read as an engineering layer over the state-operator-readout constructor. New protocols should be searched by composing lawful quantum questions and controlled maps.

- Pages that are branch-ambiguous are useful: they often mark junctions where two constructions meet, such as field theory joining transport, incompatibility, and boundary context.

- Historical, interpretive, and object-name pages should be demoted to annotations. The conceptual spine is context, state, generator, spectral question, probability, compatibility, realization.

Part II

Branch Chapters

Chapter 9

Hilbert-space context: admissible carrier and basis

A quantum calculation first fixes the Hilbert space, operator domain, basis, preparation context, representation, gauge, or boundary condition. This is not the measured answer; it is the legal carrier on which states, transformations, observables, and probabilities can be defined.

9.1 Why This Branch Exists

This branch is the first step because quantum mechanics is not defined on raw objects. It begins by specifying the space, basis, representation, or admissibility condition in which states and questions make sense.

9.2 Page Map

Page
Mathematical formulation of quantum mechanics
Hilbert space
Transformation theory (quantum mechanics)
Quantum differential calculus
Quantum complexity theory
Quantum cellular automaton
Relativistic quantum mechanics

9.3 Mechanism Pages

Each page below is read as a specialization of the compact quantum constructor. Topic-specific pages use explicit equations or overrides; core-derived pages use the branch-level equation form associated with their mechanism role.

9.4 Mathematical formulation of quantum mechanics

Central Claim

a mathematical formulation of quantum mechanics modifies the interpretation of the probability/readout layer while preserving the formal quantum dynamics.

Formal Role

a mathematical formulation of quantum mechanics acts on the readout layer of the quantum constructor. The formal ingredients remain the state assignment, the operator or measurement being applied, and the Born-rule map from projectors to probabilities. What changes is the status assigned to those ingredients: for this topic, the state or probability is treated through the agent, measurement context, or interpretive stance attached to the formalism. The page should therefore be read as a statement about the interpretation of state, probability, update, or recorded outcome while the Hamiltonian, spectral resolution, and commutator structure remain the formal reference layer. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its mathematical presentation emphasizes local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Mathematical formulation of quantum mechanics contributes a topic-native constructor role to the quantum construction.
- **Placement**. This page is read first as a context-setting move: it fixes the arena in which states, domains, and questions are legal.
- **Carrier or domain**. State terms: quantum state and wave function. Context/domain terms: basis.
- **Operator or map**. Operator terms: matrix, Hamiltonian, or unitary. Protocol or update terms: Path integral.
- **Admissibility**. Compatibility or closure terms: uncertainty and complementarity. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: spectrum. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$

$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$

$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$

$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.06537
- arXiv:1604.05385
- arXiv:0912.2823
- arXiv:1612.00682
- arXiv:quant-ph0205159

9.5 Hilbert space

Central Claim

Hilbert space is the admissible state carrier of quantum theory: it supplies the space in which states, operators, bases, spectra, and probabilities become legally defined.

Formal Role

Hilbert space is not physical space and not a geometric background in this book. It is the legal carrier of quantum identity. A state is a vector or density operator on it; the inner product gives amplitudes and norms; observables are self-adjoint operators on it; spectral projectors define possible answers; and unitary evolution preserves norm and probability. Hilbert space is therefore central because it binds state, probability, operator spectrum, and identity preservation into one formal carrier. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its mathematical presentation emphasizes local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Hilbert space contributes a state-carrier role to the quantum construction.
- **Placement**. This page is read first as a context-setting move: it fixes the arena in which states, domains, and questions are legal.
- **Carrier or domain**. A complex Hilbert space, or a density-operator state space built on it.
- **Operator or map**. Self-adjoint observables, unitary maps, spectral projectors, and domain-restricted generators defined on the carrier.
- **Admissibility**. Inner-product structure, normalization, positivity for density states, and operator-domain conditions make states and observables legal.
- **Readout**. Born probabilities, spectral projectors, expectation values, and preserved norms.
- **Check**. Changing basis or representation should preserve probabilities and expectation values when the change is unitary.

Topic Equations

Standard constructor skeleton: normalized states, density states, spectral resolution, Born readout, and unitary identity preservation.

$$
\begin{aligned}
|\psi\rangle \in \mathcal{H}, \qquad \langle\psi|\psi\rangle = 1 \\
\rho \in \mathcal{S}(\mathcal{H}), \qquad \rho \geq 0, \quad \operatorname{Tr}\rho = 1 \\
A = A^\dagger, \qquad A = \int_{\sigma(A)} \lambda \, dE_A(\lambda) \\
\Pr(\Delta \mid \rho, A) = \operatorname{Tr}(\rho E_A(\Delta)) \\
\rho_t = U(t)\rho U(t)^\dagger, \qquad U^\dagger U = I
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:1612.00682
- arXiv:2308.15676
- arXiv:2108.07838
- arXiv:0809.5271

9.6 Transformation theory (quantum mechanics)

Central Claim

A transformation theory (quantum mechanics) can be read as a quantum construction: the potential, domain, initial condition, or boundary condition fixes the admissible state space; the Hamiltonian, whose exponential gives unitary time evolution defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a transformation theory (quantum mechanics) is described by a wave function or density operator defined on the Hilbert space allowed by the system's domain. The physical question is represented by the Hamiltonian, whose exponential gives unitary time evolution; the experimental or mathematical setting is the potential, domain, initial condition, or boundary condition. The observable content is obtained from the eigenvalues and eigenfunctions of the relevant observable. In the local terminology of this topic, the same construction appears through quantum state or wave function, Hamiltonian or observable operator, and eigenvalue or energy level. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in operator-to-spectrum readout, state evolution,

Quantum Mechanism Frame

- **Role**. Transformation theory (quantum mechanics) contributes a topic-native constructor role to the quantum construction.
- **Placement**. This page is read first as a context-setting move: it fixes the arena in which states, domains, and questions are legal.
- **Carrier or domain**. State terms: quantum state. Context/domain terms: preparation condition, measurement setup, or potential or domain.
- **Operator or map**. Operator terms: Hamiltonian, observable operator, or generator. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$\begin{aligned} B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)} \\ \rho_t = U_t \rho_B U_t^\dagger \quad &\text{(unitary evolution from preparation to readout)} \\ O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)} \\ [O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)} \end{aligned}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1801.03283
- arXiv:1708.03640
- arXiv:1612.00682
- arXiv:quant-ph0205159
- arXiv:1506.05598

9.7 Quantum differential calculus

Central Claim

A quantum differential calculus can be read as a quantum construction: the measurement basis and experimental arrangement fixes the admissible state space; the self-adjoint observable being asked of that state defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a quantum differential calculus is described by a prepared quantum state before the measurement. The physical question is represented by the self-adjoint observable being asked of that state; the experimental or mathematical setting is the measurement basis and experimental arrangement. The observable content is obtained from the observable's spectral projectors and the Born probabilities assigned to them. In the local terminology of this topic, the same construction appears through quantum state or wave function, Hamiltonian or observable operator, and eigenvalue or energy level. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its mathematical presentation

Quantum Mechanism Frame

- **Role**. Quantum differential calculus contributes a topic-native constructor role to the quantum construction.
- **Placement**. This page is read first as a context-setting move: it fixes the arena in which states, domains, and questions are legal.

- **Carrier or domain**. State terms: quantum state, wave function, or density operator. Context/domain terms: preparation condition, measurement setup, or potential or domain.
- **Operator or map**. Operator terms: Hamiltonian, observable operator, or generator. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$
\begin{aligned}
B \longmapsto \rho_B \quad & \text{(context specifies an admissible state)} \\
\rho_t = U_t \rho_B U_t^\dagger \quad & \text{(unitary evolution from preparation to readout)} \\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad & \text{(spectral probability measure)} \\
[O_1, O_2] \neq 0 \quad & \text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:0912.2823

- arXiv:1604.06537

- arXiv:1612.00682

- arXiv:1801.03283

- arXiv:quant-ph0205159

9.8 Quantum complexity theory

Central Claim

A quantum complexity theory can be read as a quantum construction: the chosen basis, pulse sequence, or measurement axis fixes the admissible state space; a Hamiltonian or unitary matrix rotating that state between preparation and measurement defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a quantum complexity theory is described by a two-dimensional Hilbert space, usually written as a qubit state or a density matrix. The physical question is represented by a Hamiltonian or unitary matrix rotating that state between preparation and measurement; the experimental or mathematical setting is the chosen basis, pulse sequence, or measurement axis. The observable content is obtained from projectors onto the two eigenstates of the measured observable. In the local terminology of this topic, the same construction appears through state vector or wave function, Hamiltonian or observable operator, and eigenvalue or energy level. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in preparation, basis, or boundary context, state evolution,

Quantum Mechanism Frame

- **Role**. Quantum complexity theory contributes a topic-native constructor role to the quantum construction.

- **Placement**. This page is read first as a context-setting move: it fixes the arena in which states, domains, and questions are legal.

- **Carrier or domain**. State terms: state vector. Context/domain terms: context.
- **Operator or map**. Operator terms: Hamiltonian, observable operator, or generator. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:0912.2823
- arXiv:1706.03846
- arXiv:1708.03640
- arXiv:1605.07654
- arXiv:0908.0752

9.9 Quantum cellular automaton

Central Claim

A quantum cellular automaton can be read as a quantum construction: the chosen basis, pulse sequence, or measurement axis fixes the admissible state space; a Hamiltonian or unitary matrix rotating that state between preparation and measurement defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a quantum cellular automaton is described by a two-dimensional Hilbert space, usually written as a qubit state or a density matrix. The physical question is represented by a Hamiltonian or unitary matrix rotating that state between preparation and measurement; the experimental or mathematical setting is the chosen basis, pulse sequence, or measurement axis. The observable content is obtained from projectors onto the two eigenstates of the measured observable. In the local terminology of this topic, the same construction appears through quantum state or wave function, unitary operator or Hamiltonian, and eigenvalue or energy level. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in preparation, basis, or boundary context, state evolution,

Quantum Mechanism Frame

- **Role**. Quantum cellular automaton contributes a topic-native constructor role to the quantum construction.
- **Placement**. This page is read first as a context-setting move: it fixes the arena in which states, domains, and questions are legal.
- **Carrier or domain**. State terms: quantum state, wave function, or density operator. Context/domain terms: context.
- **Operator or map**. Operator terms: unitary. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.

- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:0912.2823
- arXiv:1109.3239
- arXiv:gr-qc0411110
- arXiv:1706.03846
- arXiv:astro-ph0604157

9.10 Relativistic quantum mechanics

Central Claim

A relativistic quantum mechanics can be read as a quantum construction: the chosen basis, pulse sequence, or measurement axis fixes the admissible state space; a Hamiltonian or unitary matrix rotating that state between preparation and measurement defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a relativistic quantum mechanics is described by a two-dimensional Hilbert space, usually written as a qubit state or a density matrix. The physical question is represented by a Hamiltonian or unitary matrix rotating that state between preparation and measurement; the experimental or mathematical setting is the chosen basis, pulse sequence, or measurement axis. The observable content is obtained from projectors onto the two eigenstates of the measured observable. In the local terminology of this topic, the same construction appears through wavefunction or wave function, Hamiltonian or matrix, and eigenvalue or energy level. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in state evolution, operator-to-spectrum readout, normalization or

Quantum Mechanism Frame

- **Role**. Relativistic quantum mechanics contributes a topic-native constructor role to the quantum construction.
- **Placement**. This page is read first as a context-setting move: it fixes the arena in which states, domains, and questions are legal.
- **Carrier or domain**. State terms: wavefunction. Context/domain terms: potential and context.
- **Operator or map**. Operator terms: Hamiltonian and matrix. Protocol or update terms: projection.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:0912.2823
- arXiv:1604.06537
- arXiv:1604.05385
- arXiv:1801.03283
- arXiv:0908.0752

Chapter 10

State carrier inside Hilbert space

A state is the probability-bearing element of the selected Hilbert space or its density-operator state space. Wave functions, density matrices, superpositions, and coherent states are different representations of this predictive carrier.

10.1 Why This Branch Exists

The state branch should be introduced before particles. It is the predictive carrier; particles, waves, fields, and qubits are later realizations of that carrier.

10.2 Page Map

Page
Density matrix
Quantum superposition
Quantum decoherence
Superposition principle
Coherence (physics)
Wave function
Quantum state
Two-state quantum system
Qubit
Quantum fluctuation
Bose–Einstein statistics
Quantum number
Fourier transform
Wave–particle duality
Quantum statistical mechanics
Quantum machine
Quantum chemistry
Applications of quantum mechanics
Quantum machine learning
Standard Model
Spin (physics)
Quantum simulator

Page
Werner Heisenberg Schrödinger's cat

10.3 Mechanism Pages

Each page below is read as a specialization of the compact quantum constructor. Topic-specific pages use explicit equations or overrides; core-derived pages use the branch-level equation form associated with their mechanism role.

10.4 Density matrix

Central Claim

Density matrix is the mixed-state constructor: it keeps probabilistic preparation, entanglement with unobserved degrees of freedom, and partial information in the same state formalism.

Formal Role

Density matrices generalize pure states without changing the state-to-spectrum readout rule. They are the correct carrier when the preparation is statistical, when a subsystem is traced out, or when decoherence is being described. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its mathematical presentation emphasizes local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Density matrix contributes a state-carrier role to the quantum construction.
- **Placement**. This page is read first as a state-carrier move: it specifies what mathematical object carries prediction.
- **Carrier or domain**. The mathematical state object: vector, wavefunction, density operator, coherent state, field state, or register.
- **Operator or map**. Operators, maps, and observables become meaningful only after this carrier and its domain have been fixed.
- **Admissibility**. Normalization, positivity, inner product, representation, tensor factorization, or superselection conditions define legal states.
- **Readout**. Probability distributions obtained by applying the appropriate observables or measurement maps to the carrier.
- **Check**. Equivalent representations must preserve probabilities and expectation values when the change is only representational.

Topic Equations

Standard constructor skeleton: mixed state, trace rule, and subsystem reduction.

$$
\begin{aligned}
\rho = \sum_a p_a |\psi_a\rangle\langle\psi_a|, \qquad p_a \geq 0, \quad \sum_a p_a = 1 \\
\rho \geq 0, \qquad \operatorname{Tr}\rho = 1 \\
p(i) = \operatorname{Tr}(\rho P_i) \\
\rho_A = \operatorname{Tr}_B(\rho_{AB})
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:0912.2823
- arXiv:1612.00682
- arXiv:1801.03283
- arXiv:1506.05598

10.5 Quantum superposition

Central Claim

A quantum superposition can be read as a quantum construction: the chosen basis, pulse sequence, or measurement axis fixes the admissible state space; a Hamiltonian or unitary matrix rotating that state between preparation and measurement defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a quantum superposition is described by a two-dimensional Hilbert space, usually written as a qubit state or a density matrix. The physical question is represented by a Hamiltonian or unitary matrix rotating that state between preparation and measurement; the experimental or mathematical setting is the chosen basis, pulse sequence, or measurement axis. The observable content is obtained from projectors onto the two eigenstates of the measured observable. In the local terminology of this topic, the same construction appears through quantum state or wave function, observable operator or Hamiltonian, and eigenvalue or energy level. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or

Quantum Mechanism Frame

- **Role**. Quantum superposition contributes a state-carrier role to the quantum construction.
- **Placement**. This page is read first as a state-carrier move: it specifies what mathematical object carries prediction.
- **Carrier or domain**. State terms: quantum state, wave function, or superposition. Context/domain terms: basis.
- **Operator or map**. Operator terms: observable and Hamiltonian. Protocol or update terms: Born rule.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:1604.06537
- arXiv:1612.00682
- arXiv:1801.03283
- arXiv:quant-ph0205159

10.6 Quantum decoherence

Central Claim

a quantum decoherence modifies the interpretation of the probability/readout layer while preserving the formal quantum dynamics.

Formal Role

a quantum decoherence acts on the readout layer of the quantum constructor. The formal ingredients remain the state assignment, the operator or measurement being applied, and the Born-rule map from projectors to probabilities. What changes is the status assigned to those ingredients: for this topic, the state or probability is treated through the agent, measurement context, or interpretive stance attached to the formalism. The page should therefore be read as a statement about the interpretation of state, probability, update, or recorded outcome while the Hamiltonian, spectral resolution, and commutator structure remain the formal reference layer. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its mathematical presentation emphasizes local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum decoherence contributes an open-system transport and coherence role to the quantum construction.
- **Placement**. This page is read first as a state-carrier move: it specifies what mathematical object carries prediction.
- **Carrier or domain**. State terms: quantum state, wave function, wavefunction, density matrix, state vector, or superposition. Context/domain terms: basis.
- **Operator or map**. Operator terms: matrix and unitary. Protocol or update terms: Born rule, collapse, or projection.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:0912.2823
- arXiv:1604.06537
- arXiv:1612.00682
- arXiv:quant-ph0205159

10.7 Superposition principle

Central Claim

The superposition principle can be read as a quantum construction: the chosen basis, pulse sequence, or measurement axis fixes the admissible state space; a Hamiltonian or unitary matrix rotating that state between preparation and measurement defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, the Superposition principle is described by a two-dimensional Hilbert space, usually written as a qubit state or a density matrix. The physical question is represented by a Hamiltonian or unitary matrix rotating that state between preparation and measurement; the experimental or mathematical setting is the chosen basis, pulse sequence, or measurement axis. The observable content is obtained from projectors onto the two eigenstates of the measured observable. In the local terminology of this topic, the same construction appears through wave function or Superposition, Hamiltonian or observable operator, and eigenvalue or energy level. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or

Quantum Mechanism Frame

- **Role**. Superposition principle contributes a state-carrier role to the quantum construction.
- **Placement**. This page is read first as a state-carrier move: it specifies what mathematical object carries prediction.
- **Carrier or domain**. State terms: wave function and Superposition. Context/domain terms: domain.
- **Operator or map**. Operator terms: Hamiltonian, observable operator, or generator. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$\begin{aligned} B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)} \\ \rho_t = U_t \rho_B U_t^\dagger \quad &\text{(unitary evolution from preparation to readout)} \\ O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)} \\ [O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)} \end{aligned}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:0912.2823
- arXiv:1801.03283
- arXiv:quant-ph0205159
- arXiv:1612.00682
- arXiv:1708.03640

10.8 Coherence (physics)

Central Claim

A coherence (physics) can be read as a quantum construction: the measurement basis and experimental arrangement fixes the admissible state space; the self-adjoint observable being asked of that state defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a coherence (physics) is described by a prepared quantum state before the measurement. The physical question is represented by the self-adjoint observable being asked of that state; the experimental or mathematical setting is the measurement basis and experimental arrangement. The observable content is obtained from the observable's spectral projectors and the Born probabilities assigned to them. In the local terminology of this topic, the same construction appears through quantum state or wave function, unitary operator or Hamiltonian, and spectrum or eigenvalue. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, preparation, basis, or boundary context; its mathematical presentation emphasizes

Quantum Mechanism Frame

- **Role**. Coherence (physics) contributes an open-system transport and coherence role to the quantum construction.
- **Placement**. This page is read first as a state-carrier move: it specifies what mathematical object carries prediction.
- **Carrier or domain**. State terms: quantum state, wave function, or density operator. Context/domain terms: potential, detector, or basis.
- **Operator or map**. Operator terms: unitary. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: spectrum. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$
\begin{aligned}
B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)}\\
\rho_t = U_t \rho_B U_t^\dagger \quad &\text{(unitary evolution from preparation to readout)}\\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)}\\
[O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation

- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:0912.2823
- arXiv:1612.00682
- arXiv:quant-ph0205159
- arXiv:2108.07838

10.9 Wave function

Central Claim

Wave function is a basis-dependent representative of a pure-state ray; it is not identical to the abstract state or to physical configuration space.

Formal Role

For a configuration space Q with measure mu, the position wave function is the generalized-basis representative psi(x)=<x|psi> of a ray [psi] in L2(Q,mu). Vectors that differ by a nonzero global phase represent the same pure state. Its modulus squared is a probability density only relative to the stated position measure; spin and particle statistics enlarge or constrain the carrier. The linked equation set is concentrated in state evolution, operator-to-spectrum readout, normalization or admissibility; its mathematical presentation emphasizes local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Wave function contributes a state-carrier role to the quantum construction.
- **Placement**. This page is read first as a state-carrier move: it specifies what mathematical object carries prediction.
- **Carrier or domain**. The mathematical state object: vector, wavefunction, density operator, coherent state, field state, or register.
- **Operator or map**. Operators, maps, and observables become meaningful only after this carrier and its domain have been fixed.
- **Admissibility**. Normalization, positivity, inner product, representation, tensor factorization, or superselection conditions define legal states.
- **Readout**. Probability distributions obtained by applying the appropriate observables or measurement maps to the carrier.
- **Check**. Equivalent representations must preserve probabilities and expectation values when the change is only representational.

Topic Equations

Topic-specific constructor: abstract state ray, position representation, measure-dependent Born probability, and internal spin carrier.

$$\mathcal{H} = L^2(Q, d\mu), \qquad \psi(x) = \langle x|\psi\rangle$$

$$\int_Q |\psi(x)|^2 \, d\mu(x) = 1, \qquad |\psi\rangle \sim e^{i\alpha}|\psi\rangle$$

$$\Pr(X \in \Delta \mid \psi) = \langle\psi|E_X(\Delta)|\psi\rangle = \int_\Delta |\psi(x)|^2 \, d\mu(x)$$

$$\mathcal{H}_{\mathrm{spin}\,s} = L^2(Q, d\mu) \otimes \mathbb{C}^{2s+1}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:0912.2823
- arXiv:1604.06537
- arXiv:1801.03283
- arXiv:1708.03640

10.10 Quantum state

Central Claim

A quantum state can be read as a quantum construction: the potential, domain, initial condition, or boundary condition fixes the admissible state space; the Hamiltonian, whose exponential gives unitary time evolution defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a quantum state is described by a wave function or density operator defined on the Hilbert space allowed by the system's domain. The physical question is represented by the Hamiltonian, whose exponential gives unitary time evolution; the experimental or mathematical setting is the potential, domain, initial condition, or boundary condition. The observable content is obtained from the eigenvalues and eigenfunctions of the relevant observable. In the local terminology of this topic, the same construction appears through Quantum state or wave function, observable operator or matrix, and eigenstate or spectrum. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through uncertainty relation or incompatible observables. The linked equation set is concentrated in state evolution, operator-to-spectrum readout, normalization or

Quantum Mechanism Frame

- **Role**. Quantum state contributes a state-carrier role to the quantum construction.
- **Placement**. This page is read first as a state-carrier move: it specifies what mathematical object carries prediction.
- **Carrier or domain**. State terms: Quantum state, wave function, or superposition. Context/domain terms: detector and preparation.

- **Operator or map**. Operator terms: observable and matrix. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: uncertainty and incompatible. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenstate, spectrum, or eigenvalue. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$
\begin{aligned}
B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)}\\
\rho_t = U_t \rho_B U_t^\dagger \quad &\text{(unitary evolution from preparation to readout)}\\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)}\\
[O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:0912.2823
- arXiv:1604.06537
- arXiv:quant-ph0205159
- arXiv:1708.03640

10.11 Two-state quantum system

Central Claim

A two-state quantum system can be read as a quantum construction: the chosen basis, pulse sequence, or measurement axis fixes the admissible state space; a Hamiltonian or unitary matrix rotating that state between preparation and measurement defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a two-state quantum system is described by a two-dimensional Hilbert space, usually written as a qubit state or a density matrix. The physical question is represented by a Hamiltonian or unitary matrix rotating that state between preparation and measurement; the experimental or mathematical setting is the chosen basis, pulse sequence, or measurement axis. The observable content is obtained from projectors onto the two eigenstates of the measured observable. In the local terminology of this topic, the same construction appears through wavefunction or state vector, observable operator or Hamiltonian, and eigenvalue or energy level. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or

Quantum Mechanism Frame

- **Role**. Two-state quantum system contributes a state-carrier role to the quantum construction.
- **Placement**. This page is read first as a state-carrier move: it specifies what mathematical object carries prediction.
- **Carrier or domain**. State terms: wavefunction, state vector, or superposition. Context/domain terms: basis.
- **Operator or map**. Operator terms: observable, Hamiltonian, or matrix. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.

- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$

$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$

$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$

$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1801.03283
- arXiv:1604.06537
- arXiv:quant-ph0205159
- arXiv:1612.00682
- arXiv:1506.05598

10.12 Qubit

Central Claim

Qubit is the two-dimensional state-carrier constructor used when the admissible state space is \(\mathbb C^2\).

Formal Role

A qubit is the minimal quantum state space with a basis, amplitudes, unitary control, and measurement readout. Bloch-vector language is a representation of the same two-dimensional carrier. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its mathematical presentation emphasizes local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Qubit contributes a state-carrier role to the quantum construction.
- **Placement**. This page is read first as a state-carrier move: it specifies what mathematical object carries prediction.
- **Carrier or domain**. The mathematical state object: vector, wavefunction, density operator, coherent state, field state, or register.
- **Operator or map**. Operators, maps, and observables become meaningful only after this carrier and its domain have been fixed.
- **Admissibility**. Normalization, positivity, inner product, representation, tensor factorization, or superselection conditions define legal states.
- **Readout**. Probability distributions obtained by applying the appropriate observables or measurement maps to the carrier.
- **Check**. Equivalent representations must preserve probabilities and expectation values when the change is only representational.

Topic Equations

Standard constructor skeleton: two-state carrier, Bloch representation, and basis readout.

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \qquad |\alpha|^2 + |\beta|^2 = 1$$

$$\rho = \frac{1}{2}(I + \mathbf{r}\cdot\boldsymbol{\sigma}), \qquad |\mathbf{r}| \le 1$$

$$p(0) = |\langle 0|\psi\rangle|^2, \qquad p(1) = |\langle 1|\psi\rangle|^2$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:1801.03283
- arXiv:1612.00682
- arXiv:quant-ph0205159
- arXiv:1506.05598

10.13 Quantum fluctuation

Central Claim

A quantum fluctuation can be read as a quantum construction: the measurement basis and experimental arrangement fixes the admissible state space; the self-adjoint observable being asked of that state defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a quantum fluctuation is described by a prepared quantum state before the measurement. The physical question is represented by the self-adjoint observable being asked of that state; the experimental or mathematical setting is the measurement basis and experimental arrangement. The observable content is obtained from the observable's spectral projectors and the Born probabilities assigned to them. In the local terminology of this topic, the same construction appears through quantum state or wave function, Hamiltonian or observable operator, and eigenvalue or energy level. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through uncertainty relation or commutator. The linked equation set is concentrated in state evolution, non-commuting compatibility limits, normalization or admissibility; its mathematical presentation emphasizes

Quantum Mechanism Frame

- **Role**. Quantum fluctuation contributes an open-system transport and coherence role to the quantum construction.
- **Placement**. This page is read first as a state-carrier move: it specifies what mathematical object carries prediction.
- **Carrier or domain**. State terms: quantum state, wave function, or density operator. Context/domain terms: preparation condition, measurement setup, or potential or domain.
- **Operator or map**. Operator terms: Hamiltonian, observable operator, or generator. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: uncertainty. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$
\begin{aligned}
B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)} \\
\rho_t = U_t \rho_B U_t^\dagger \quad &\text{(unitary evolution from preparation to readout)} \\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)} \\
[O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.06537
- arXiv:1706.03846
- arXiv:0908.0752
- arXiv:1604.05385
- arXiv:2106.04271

10.14 Bose–Einstein statistics

Central Claim

A bose–einstein statistics can be read as a quantum construction: the chosen basis, pulse sequence, or measurement axis fixes the admissible state space; a Hamiltonian or unitary matrix rotating that state between preparation and measurement defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a bose–Einstein statistics is described by a two-dimensional Hilbert space, usually written as a qubit state or a density matrix. The physical question is represented by a Hamiltonian or unitary matrix rotating that state between preparation and measurement; the experimental or mathematical setting is the chosen basis, pulse sequence, or measurement axis. The observable content is obtained from projectors onto the two eigenstates of the measured observable. In the local terminology of this topic, the same construction appears through quantum state or wave function, Hamiltonian or observable operator, and energy level or eigenvalue. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, preparation,

Quantum Mechanism Frame

- **Role**. Bose–Einstein statistics contributes a topic-native constructor role to the quantum construction.
- **Placement**. This page is read first as a state-carrier move: it specifies what mathematical object carries prediction.

- **Carrier or domain**. State terms: quantum state, wave function, or density operator. Context/domain terms: potential.
- **Operator or map**. Operator terms: Hamiltonian, observable operator, or generator. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: energy level. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$

$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$

$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$

$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:0912.2823
- arXiv:2308.15676
- arXiv:2105.11733
- arXiv:2108.07838
- arXiv:0805.4565

10.15 Quantum number

Central Claim

A quantum number can be read as a quantum construction: the chosen basis, pulse sequence, or measurement axis fixes the admissible state space; a Hamiltonian or unitary matrix rotating that state between preparation and measurement defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a quantum number is described by a two-dimensional Hilbert space, usually written as a qubit state or a density matrix. The physical question is represented by a Hamiltonian or unitary matrix rotating that state between preparation and measurement; the experimental or mathematical setting is the chosen basis, pulse sequence, or measurement axis. The observable content is obtained from projectors onto the two eigenstates of the measured observable. In the local terminology of this topic, the same construction appears through quantum state or wave function, observable operator or Hamiltonian, and spectrum or energy level. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through uncertainty relation or commutator. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or

Quantum Mechanism Frame

- **Role**. Quantum number contributes a topic-native constructor role to the quantum construction.
- **Placement**. This page is read first as a state-carrier move: it specifies what mathematical object carries prediction.
- **Carrier or domain**. State terms: quantum state, wave function, or density operator. Context/domain terms: basis.
- **Operator or map**. Operator terms: observable and Hamiltonian. Protocol or update terms: projection.
- **Admissibility**. Compatibility or closure terms: uncertainty. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: spectrum and energy level. These name the outcome labels, projectors, amplitudes, or records used for testing.

- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$\begin{aligned} B \longmapsto \rho_B \quad & \text{(context specifies an admissible state)} \\ \rho_t = U_t \rho_B U_t^\dagger \quad & \text{(unitary evolution from preparation to readout)} \\ O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad & \text{(spectral probability measure)} \\ [O_1, O_2] \neq 0 \quad & \text{(incompatible observables: no common sharp basis)} \end{aligned}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:1604.06537
- arXiv:0912.2823
- arXiv:1612.00682
- arXiv:1801.03283

10.16 Fourier transform

Central Claim

A fourier transform can be read as a quantum construction: the input encoding, circuit architecture, and final measurement basis fixes the admissible state space; a sequence of unitary gates or quantum channels defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a fourier transform is described by a register state in a finite-dimensional Hilbert space. The physical question is represented by a sequence of unitary gates or quantum channels; the experimental or mathematical setting is the input encoding, circuit architecture, and final measurement basis. The observable content is obtained from measurement probabilities over computational-basis outcomes. In the local terminology of this topic, the same construction appears through quantum state or wave function, unitary operator or Hamiltonian, and eigenvalue or energy level. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through uncertainty relation or commutator. The linked equation set is concentrated in state evolution, normalization or admissibility, non-commuting compatibility limits; its mathematical presentation emphasizes local

Quantum Mechanism Frame

- **Role**. Fourier transform contributes a topic-native constructor role to the quantum construction.
- **Placement**. This page is read first as a state-carrier move: it specifies what mathematical object carries prediction.
- **Carrier or domain**. State terms: quantum state, wave function, or density operator. Context/domain terms: domain.
- **Operator or map**. Operator terms: unitary. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: uncertainty. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1706.03846
- arXiv:1708.03640
- arXiv:0912.2823
- arXiv:1604.06537
- arXiv:0908.0752

10.17 Wave–particle duality

Central Claim

Wave–particle duality can be read as a quantum construction: the potential, domain, initial condition, or boundary condition fixes the admissible state space; the Hamiltonian, whose exponential gives unitary time evolution defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, wave–particle duality is described by a wave function or density operator defined on the Hilbert space allowed by the system's domain. The physical question is represented by the Hamiltonian, whose exponential gives unitary time evolution; the experimental or mathematical setting is the potential, domain, initial condition, or boundary condition. The observable content is obtained from the eigenvalues and eigenfunctions of the relevant observable. In the local terminology of this topic, the same construction appears through probability amplitude or wave function, Hamiltonian or observable operator, and spectrum or eigenvalue. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in state evolution, operator-to-spectrum readout, normalization or

Quantum Mechanism Frame

- **Role**. Wave–particle duality contributes a many-mode field or particle-realization role to the quantum construction.
- **Placement**. This page is read first as a state-carrier move: it specifies what mathematical object carries prediction.
- **Carrier or domain**. State terms: probability amplitude. Context/domain terms: potential and detector.
- **Operator or map**. Operator terms: Hamiltonian, observable operator, or generator. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: spectrum. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:0912.2823
- arXiv:1604.06537
- arXiv:1708.03640
- arXiv:quant-ph0205159

10.18 Quantum statistical mechanics

Central Claim

A quantum statistical mechanics can be read as a quantum construction: the chosen basis, pulse sequence, or measurement axis fixes the admissible state space; a Hamiltonian or unitary matrix rotating that state between preparation and measurement defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a quantum statistical mechanics is described by a two-dimensional Hilbert space, usually written as a qubit state or a density matrix. The physical question is represented by a Hamiltonian or unitary matrix rotating that state between preparation and measurement; the experimental or mathematical setting is the chosen basis, pulse sequence, or measurement axis. The observable content is obtained from projectors onto the two eigenstates of the measured observable. In the local terminology of this topic, the same construction appears through quantum state or density matrix, observable operator or matrix, and spectrum or eigenvalue. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or

Quantum Mechanism Frame

- **Role**. Quantum statistical mechanics contributes a topic-native constructor role to the quantum construction.
- **Placement**. This page is read first as a state-carrier move: it specifies what mathematical object carries prediction.
- **Carrier or domain**. State terms: quantum state and density matrix. Context/domain terms: basis.
- **Operator or map**. Operator terms: observable, matrix, or Hamiltonian. Protocol or update terms: projection.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: spectrum and eigenvalue. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:0912.2823
- arXiv:1506.05598
- arXiv:1801.03283
- arXiv:1612.00682

10.19 Quantum machine

Central Claim

A quantum machine can be read as a quantum construction: the chosen basis, pulse sequence, or measurement axis fixes the admissible state space; a Hamiltonian or unitary matrix rotating that state between preparation and measurement defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a quantum machine is described by a two-dimensional Hilbert space, usually written as a qubit state or a density matrix. The physical question is represented by a Hamiltonian or unitary matrix rotating that state between preparation and measurement; the experimental or mathematical setting is the chosen basis, pulse sequence, or measurement axis. The observable content is obtained from projectors onto the two eigenstates of the measured observable. In the local terminology of this topic, the same construction appears through quantum state or superposition, observable operator or Hamiltonian, and eigenvalue or energy level. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, controlled update

Quantum Mechanism Frame

- **Role**. Quantum machine contributes a topic-native constructor role to the quantum construction.
- **Placement**. This page is read first as a state-carrier move: it specifies what mathematical object carries prediction.
- **Carrier or domain**. State terms: quantum state and superposition. Context/domain terms: preparation condition, measurement setup, or potential or domain.
- **Operator or map**. Operator terms: observable. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:0805.4565
- arXiv:2108.07838
- arXiv:1612.00682
- arXiv:quant-ph0205159

10.20 Quantum chemistry

Central Claim

Quantum chemistry can be read as a quantum construction: the chosen basis, pulse sequence, or measurement axis fixes the admissible state space; a Hamiltonian or unitary matrix rotating that state between preparation and measurement defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, quantum chemistry is described by a two-dimensional Hilbert space, usually written as a qubit state or a density matrix. The physical question is represented by a Hamiltonian or unitary matrix rotating that state between preparation and measurement; the experimental or mathematical setting is the chosen basis, pulse sequence, or measurement axis. The observable content is obtained from projectors onto the two eigenstates of the measured observable. In the local terminology of this topic, the same construction appears through wave function or quantum state, observable operator or Hamiltonian, and eigenvalue or energy level. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or

Quantum Mechanism Frame

- **Role**. Quantum chemistry contributes a topic-native constructor role to the quantum construction.
- **Placement**. This page is read first as a state-carrier move: it specifies what mathematical object carries prediction.
- **Carrier or domain**. State terms: wave function and quantum state. Context/domain terms: potential and basis.
- **Operator or map**. Operator terms: observable and Hamiltonian. Protocol or update terms: path integral.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$

$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$

$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$

$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:0912.2823
- arXiv:1612.00682
- arXiv:quant-ph0205159
- arXiv:1801.03283
- arXiv:1506.05598

10.21 Applications of quantum mechanics

Central Claim

An applications of quantum mechanics can be read as a quantum construction: the chosen basis, pulse sequence, or measurement axis fixes the admissible state space; a Hamiltonian or unitary matrix rotating that state between preparation and measurement defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, an applications of quantum mechanics is described by a two-dimensional Hilbert space, usually written as a qubit state or a density matrix. The physical question is represented by a Hamiltonian or unitary matrix rotating that state between preparation and measurement; the experimental or mathematical setting is the chosen basis, pulse sequence, or measurement axis. The observable content is obtained from projectors onto the two eigenstates of the measured observable. In the local terminology of this topic, the same construction appears through superposition or wave function, Hamiltonian or observable operator, and spectrum or eigenstate. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in operator-to-spectrum readout, state evolution,

Quantum Mechanism Frame

- **Role**. Applications of quantum mechanics contributes a topic-native constructor role to the quantum construction.
- **Placement**. This page is read first as a state-carrier move: it specifies what mathematical object carries prediction.
- **Carrier or domain**. State terms: superposition. Context/domain terms: potential.
- **Operator or map**. Operator terms: Hamiltonian, observable operator, or generator. Protocol or update terms: collapse.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: spectrum and eigenstate. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^{\dagger} \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:0912.2823
- arXiv:quant-ph0205159
- arXiv:0805.4565
- arXiv:1612.00682

10.22 Quantum machine learning

Central Claim

A quantum machine learning can be read as a quantum construction: the chosen basis, pulse sequence, or measurement axis fixes the admissible state space; a Hamiltonian or unitary matrix rotating that state between preparation and measurement defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a quantum machine learning is described by a two-dimensional Hilbert space, usually written as a qubit state or a density matrix. The physical question is represented by a Hamiltonian or unitary matrix rotating that state between preparation and measurement; the experimental or mathematical setting is the chosen basis, pulse sequence, or measurement axis. The observable content is obtained from projectors onto the two eigenstates of the measured observable. In the local terminology of this topic, the same construction appears through quantum state or superposition, matrix or Hamiltonian, and eigenvalue or energy level. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or

Quantum Mechanism Frame

- **Role**. Quantum machine learning contributes a topic-native constructor role to the quantum construction.
- **Placement**. This page is read first as a state-carrier move: it specifies what mathematical object carries prediction.
- **Carrier or domain**. State terms: quantum state and superposition. Context/domain terms: preparation and basis.
- **Operator or map**. Operator terms: matrix, Hamiltonian, or unitary. Protocol or update terms: unitary evolution.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^{\dagger} \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:0912.2823
- arXiv:1604.05385
- arXiv:1612.00682
- arXiv:quant-ph0205159
- arXiv:1801.03283

10.23 Standard Model

Central Claim

The standard model can be read as a quantum construction: the chosen basis, pulse sequence, or measurement axis fixes the admissible state space; a Hamiltonian or unitary matrix rotating that state between preparation and measurement defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, the Standard Model is described by a two-dimensional Hilbert space, usually written as a qubit state or a density matrix. The physical question is represented by a Hamiltonian or unitary matrix rotating that state between preparation and measurement; the experimental or mathematical setting is the chosen basis, pulse sequence, or measurement axis. The observable content is obtained from projectors onto the two eigenstates of the measured observable. In the local terminology of this topic, the same construction appears through quantum state or wave function, Hamiltonian or observable operator, and eigenvalue or energy level. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or

Quantum Mechanism Frame

- **Role**. Standard Model contributes a many-mode field or particle-realization role to the quantum construction.
- **Placement**. This page is read first as a state-carrier move: it specifies what mathematical object carries prediction.
- **Carrier or domain**. State terms: quantum state. Context/domain terms: basis.
- **Operator or map**. Operator terms: Hamiltonian, observable operator, or generator. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:0912.2823
- arXiv:1604.06537
- arXiv:1801.03283
- arXiv:quant-ph0205159
- arXiv:1612.00682

10.24 Spin (physics)

Central Claim

A spin (physics) can be read as a quantum construction: the chosen basis, pulse sequence, or measurement axis fixes the admissible state space; a Hamiltonian or unitary matrix rotating that state between preparation and measurement defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a spin (physics) is described by a two-dimensional Hilbert space, usually written as a qubit state or a density matrix. The physical question is represented by a Hamiltonian or unitary matrix rotating that state between preparation and measurement; the experimental or mathematical setting is the chosen basis, pulse sequence, or measurement axis. The observable content is obtained from projectors onto the two eigenstates of the measured observable. In the local terminology of this topic, the same construction appears through wave function or quantum state, Hamiltonian or observable operator, and eigenvalue or energy level. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or

Quantum Mechanism Frame

- **Role**. Spin (physics) contributes a topic-native constructor role to the quantum construction.
- **Placement**. This page is read first as a state-carrier move: it specifies what mathematical object carries prediction.
- **Carrier or domain**. State terms: wave function and quantum state. Context/domain terms: context.
- **Operator or map**. Operator terms: Hamiltonian. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:0912.2823
- arXiv:1604.05385
- arXiv:quant-ph0205159
- arXiv:1801.03283
- arXiv:1708.03640

10.25 Quantum simulator

Central Claim

Quantum simulator is a target–carrier–validation construction: a controllable physical system encodes another model, and selected observables test whether the encoded dynamics is faithful.

Formal Role

The simulator Hamiltonian is not by itself the target theory. The claim also needs an encoding between target and device states, a correspondence between their generators or channels, and validation observables with an error budget over the stated time and parameter range. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its mathematical presentation emphasizes local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum simulator contributes an unresolved constructor role to the quantum construction.
- **Placement**. This page is read first as a state-carrier move: it specifies what mathematical object carries prediction.
- **Carrier or domain**. A controllable device state space together with an explicit encoding of the target state space.
- **Operator or map**. Device Hamiltonians, channels, or gate sequences intended to reproduce target dynamics under the encoding.
- **Admissibility**. Control errors, leakage, finite size, noise, and approximation order define the regime in which the correspondence is claimed.
- **Readout**. Encoded target observables compared with independently predicted or calibrated device measurements.
- **Check**. Validation requires observables and error bounds beyond agreement with the programmed control Hamiltonian.

Topic Equations

Topic-specific constructor: encoding, dynamical correspondence, and observable validation are separate obligations.

$$
\begin{aligned}
V : \mathcal{H}_{\text{target}} &\hookrightarrow \mathcal{H}_{\text{device}} \\
\|U_{\text{device}}(t)V - VU_{\text{target}}(t)\| &\leq \varepsilon(t) \\
\left|\langle O\rangle_{\text{target}} - \langle VOV^{\dagger}\rangle_{\text{device}}\right| &\leq \delta_O
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels

- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:0912.2823
- arXiv:1604.05385
- arXiv:quant-ph0205159
- arXiv:1708.03640
- arXiv:1801.03283

10.26 Werner Heisenberg

Central Claim

A werner heisenberg can be read as a quantum construction: the chosen basis, pulse sequence, or measurement axis fixes the admissible state space; a Hamiltonian or unitary matrix rotating that state between preparation and measurement defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a werner Heisenberg is described by a two-dimensional Hilbert space, usually written as a qubit state or a density matrix. The physical question is represented by a Hamiltonian or unitary matrix rotating that state between preparation and measurement; the experimental or mathematical setting is the chosen basis, pulse sequence, or measurement axis. The observable content is obtained from projectors onto the two eigenstates of the measured observable. In the local terminology of this topic, the same construction appears through wave function or state vector, matrix or Hamiltonian, and eigenvalue or energy level. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through uncertainty relation or non-commuting observables. The linked equation set is concentrated in state evolution, operator-to-spectrum readout, non-commuting

Quantum Mechanism Frame

- **Role**. Werner Heisenberg contributes a topic-native constructor role to the quantum construction.
- **Placement**. This page is read first as a state-carrier move: it specifies what mathematical object carries prediction.
- **Carrier or domain**. State terms: wave function. Context/domain terms: preparation condition, measurement setup, or potential or domain.
- **Operator or map**. Operator terms: matrix. Protocol or update terms: collapse.
- **Admissibility**. Compatibility or closure terms: uncertainty and non-commuting. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.06537
- arXiv:1604.05385
- arXiv:0912.2823
- arXiv:1708.03640
- arXiv:1801.03283

10.27 Schrödinger's cat

Central Claim

a schrödinger's cat modifies the interpretation of the probability/readout layer while preserving the formal quantum dynamics.

Formal Role

a schrödinger's cat acts on the readout layer of the quantum constructor. The formal ingredients remain the state assignment, the operator or measurement being applied, and the Born-rule map from projectors to probabilities. What changes is the status assigned to those ingredients: for this topic, the state or probability is treated through the agent, measurement context, or interpretive stance attached to the formalism. The page should therefore be read as a statement about the interpretation of state, probability, update, or recorded outcome while the Hamiltonian, spectral resolution, and commutator structure remain the formal reference layer. The linked equation set is concentrated in state evolution, operator-to-spectrum readout, normalization or admissibility; its mathematical presentation emphasizes local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Schrödinger's cat contributes a lawful state-transport role to the quantum construction.
- **Placement**. This page is read first as a state-carrier move: it specifies what mathematical object carries prediction.
- **Carrier or domain**. State terms: wave function, quantum state, wavefunction, or superposition. Context/domain terms: domain.
- **Operator or map**. Operator terms: observable. Protocol or update terms: collapse.

- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$\begin{aligned} B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)} \\ \rho_t = U_t \rho_B U_t^\dagger \quad &\text{(unitary evolution from preparation to readout)} \\ O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)} \\ [O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)} \end{aligned}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:0912.2823
- arXiv:1604.06537
- arXiv:quant-ph0205159
- arXiv:1708.03640

Chapter 11

Generator: lawful change before readout

Hamiltonians, unitary maps, equations of motion, and path weights describe the lawful transport of the state before a question is resolved.

11.1 Why This Branch Exists

Time evolution is a transport problem over states. The Hamiltonian and path integral are two views of the same generator role rather than unrelated formalisms.

11.2 Page Map

Page
Unitary operator
Perturbation theory
Quantum dynamics
Path integral formulation
Hamiltonian mechanics
Path integral
Hamiltonian (quantum mechanics)
Perturbation theory (quantum mechanics)
Schrödinger picture
Creation and annihilation operators
Schrödinger equation
Quantum stochastic calculus
Quantum chaos
Symmetry in quantum mechanics
Quantum amplifier
Quantum mechanics
Heisenberg picture
Heisenberg group
Canonical commutation relation
Quantum biology
Quantum engineering

Page
Quantum logic

11.3 Mechanism Pages

Each page below is read as a specialization of the compact quantum constructor. Topic-specific pages use explicit equations or overrides; core-derived pages use the branch-level equation form associated with their mechanism role.

11.4 Unitary operator

Central Claim

Unitary operator is the reversible-map constructor: it changes the state while preserving inner products and probabilities.

Formal Role

Unitary maps are the admissible reversible transformations of closed-system quantum theory. They preserve normalization and carry projective geometry of the state space through time or controlled operations. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its mathematical presentation emphasizes local notation, information profile, spectral profile.

Quantum Mechanism Frame

- **Role**. Unitary operator contributes a lawful state-transport role to the quantum construction.
- **Placement**. This page is read first as a lawful-transport move: it identifies what changes the state before readout.
- **Carrier or domain**. A state vector, density operator, wavefunction, field state, or register on a specified domain.
- **Operator or map**. Hamiltonian, unitary map, channel generator, action, constraint, or differential operator that transports the state.
- **Admissibility**. Self-adjointness, complete positivity, trace preservation, gauge constraints, and boundary/domain conditions decide whether the evolution is legal.
- **Readout**. Time-dependent probabilities, spectra, transition amplitudes, conserved quantities, or response functions.
- **Check**. The generator must predict the observed evolution while preserving the relevant normalization, positivity, symmetry, or conservation constraint.

Topic Equations

Standard constructor skeleton: reversible state transformation and inner-product preservation.

$$
\begin{aligned}
U^\dagger U = UU^\dagger &= I \\
|\psi'\rangle &= U|\psi\rangle \\
\langle\psi'|\phi'\rangle &= \langle\psi|\phi\rangle
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:0912.2823
- arXiv:1612.00682
- arXiv:1801.03283
- arXiv:quant-ph0205159
- arXiv:1506.05598

11.5 Perturbation theory

Central Claim

Perturbation theory can be read as a quantum construction: the potential, domain, initial condition, or boundary condition fixes the admissible state space; the Hamiltonian, whose exponential gives unitary time evolution defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, perturbation theory is described by a wave function or density operator defined on the Hilbert space allowed by the system's domain. The physical question is represented by the Hamiltonian, whose exponential gives unitary time evolution; the experimental or mathematical setting is the potential, domain, initial condition, or boundary condition. The observable content is obtained from the eigenvalues and eigenfunctions of the relevant observable. In the local terminology of this topic, the same construction appears through quantum state or wave function, Hamiltonian or observable operator, and eigenvalue or energy level. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or

Quantum Mechanism Frame

- **Role**. Perturbation theory contributes a topic-native constructor role to the quantum construction.
- **Placement**. This page is read first as a lawful-transport move: it identifies what changes the state before readout.
- **Carrier or domain**. State terms: quantum state, wave function, or density operator. Context/domain terms: preparation condition, measurement setup, or potential or domain.
- **Operator or map**. Operator terms: Hamiltonian, observable operator, or generator. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$\begin{aligned} B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)} \\ \rho_t = U_t \rho_B U_t^\dagger \quad &\text{(unitary evolution from preparation to readout)} \\ O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)} \\ [O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)} \end{aligned}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:0912.2823
- arXiv:1612.00682
- arXiv:0805.4565
- arXiv:2108.07838
- arXiv:quant-ph0205159

11.6 Quantum dynamics

Central Claim

A quantum dynamics can be read as a quantum construction: the potential, domain, initial condition, or boundary condition fixes the admissible state space; the Hamiltonian, whose exponential gives unitary time evolution defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a quantum dynamics is described by a wave function or density operator defined on the Hilbert space allowed by the system's domain. The physical question is represented by the Hamiltonian, whose exponential gives unitary time evolution; the experimental or mathematical setting is the potential, domain, initial condition, or boundary condition. The observable content is obtained from the eigenvalues and eigenfunctions of the relevant observable. In the local terminology of this topic, the same construction appears through state vector or density matrix, Hamiltonian or observable operator, and eigenvalue or energy level. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or

Quantum Mechanism Frame

- **Role**. Quantum dynamics contributes a lawful state-transport role to the quantum construction.
- **Placement**. This page is read first as a lawful-transport move: it identifies what changes the state before readout.
- **Carrier or domain**. State terms: state vector, density matrix, quantum state, or superposition. Context/domain terms: preparation condition, measurement setup, or potential or domain.
- **Operator or map**. Operator terms: Hamiltonian, observable, matrix, unitary, or commutator. Protocol or update terms: unitary evolution.
- **Admissibility**. Compatibility or closure terms: commutator and uncertainty. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation

- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.06537
- arXiv:1801.03283
- arXiv:1612.00682
- arXiv:0912.2823
- arXiv:1506.05598

11.7 Path integral formulation

Central Claim

Path integral formulation is the formulation in which the generator is represented by action-weighted histories.

Formal Role

This page belongs with lawful change because it changes how the transport step is calculated. The observable predictions remain probabilities after amplitudes are composed and squared or traced. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its mathematical presentation emphasizes local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Path integral formulation contributes a lawful state-transport role to the quantum construction.
- **Placement**. This page is read first as a lawful-transport move: it identifies what changes the state before readout.
- **Carrier or domain**. A state vector, density operator, wavefunction, field state, or register on a specified domain.
- **Operator or map**. Hamiltonian, unitary map, channel generator, action, constraint, or differential operator that transports the state.
- **Admissibility**. Self-adjointness, complete positivity, trace preservation, gauge constraints, and boundary/domain conditions decide whether the evolution is legal.
- **Readout**. Time-dependent probabilities, spectra, transition amplitudes, conserved quantities, or response functions.
- **Check**. The generator must predict the observed evolution while preserving the relevant normalization, positivity, symmetry, or conservation constraint.

Topic Equations

Standard constructor skeleton: transition amplitudes and generating functional.

$$\langle q_f, t_f | q_i, t_i \rangle = \int \mathcal{D}q \, e^{iS[q]/\hbar}$$
$$Z[J] = \int \mathcal{D}\phi \, \exp\left(\frac{i}{\hbar} (S[\phi] + \int J\phi) \right)$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:0912.2823
- arXiv:1604.06537
- arXiv:1801.03283
- arXiv:1612.00682
- arXiv:quant-ph0205159

11.8 Hamiltonian mechanics

Central Claim

A hamiltonian mechanics can be read as a quantum construction: the potential, domain, initial condition, or boundary condition fixes the admissible state space; the Hamiltonian, whose exponential gives unitary time evolution defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a hamiltonian mechanics is described by a wave function or density operator defined on the Hilbert space allowed by the system's domain. The physical question is represented by the Hamiltonian, whose exponential gives unitary time evolution; the experimental or mathematical setting is the potential, domain, initial condition, or boundary condition. The observable content is obtained from the eigenvalues and eigenfunctions of the relevant observable. In the local terminology of this topic, the same construction appears through quantum state or wave function, Hamiltonian or observable operator, and eigenvalue or energy level. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or

Quantum Mechanism Frame

- **Role**. Hamiltonian mechanics contributes a lawful state-transport role to the quantum construction.
- **Placement**. This page is read first as a lawful-transport move: it identifies what changes the state before readout.

- **Carrier or domain**. State terms: quantum state, wave function, or density operator. Context/domain terms: preparation condition, measurement setup, or potential or domain.
- **Operator or map**. Operator terms: Hamiltonian. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$
\begin{aligned}
B \longmapsto \rho_B \quad & \text{(context specifies an admissible state)} \\
\rho_t = U_t \rho_B U_t^{\dagger} \quad & \text{(unitary evolution from preparation to readout)} \\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad & \text{(spectral probability measure)} \\
[O_1, O_2] \neq 0 \quad & \text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.06537
- arXiv:2108.07838
- arXiv:1506.05598
- arXiv:1612.00682
- arXiv:2111.12617

11.9 Path integral

Central Claim

Path integral is an alternate generator constructor: transition amplitudes are obtained by summing phase weights over histories.

Formal Role

The path integral does not replace the operator constructor. It repackages the generator step as a weighted sum over histories between boundary conditions. It is especially useful when action, symmetry, and field degrees of freedom are more natural than state-vector evolution. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its mathematical presentation emphasizes local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Path integral contributes a lawful state-transport role to the quantum construction.
- **Placement**. This page is read first as a lawful-transport move: it identifies what changes the state before readout.
- **Carrier or domain**. A state vector, density operator, wavefunction, field state, or register on a specified domain.
- **Operator or map**. Hamiltonian, unitary map, channel generator, action, constraint, or differential operator that transports the state.
- **Admissibility**. Self-adjointness, complete positivity, trace preservation, gauge constraints, and boundary/domain conditions decide whether the evolution is legal.
- **Readout**. Time-dependent probabilities, spectra, transition amplitudes, conserved quantities, or response functions.
- **Check**. The generator must predict the observed evolution while preserving the relevant normalization, positivity, symmetry, or conservation constraint.

Topic Equations

Standard constructor skeleton: boundary-to-boundary transition amplitude through action weights.

$$K(x_f, t_f; x_i, t_i) = \int_{x_i}^{x_f} \mathcal{D}x(t)\, \exp\left(\frac{i}{\hbar} S[x]\right)$$
$$\psi(x_f, t_f) = \int K(x_f, t_f; x_i, t_i)\psi(x_i, t_i)\, dx_i$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:2107.01923
- arXiv:0801.3568
- arXiv:2105.11733
- arXiv:1706.07300
- arXiv:cond-mat0108470

11.10 Hamiltonian (quantum mechanics)

Central Claim

Hamiltonian (quantum mechanics) is the generator observable: it both transports states and supplies the energy spectrum.

Formal Role

The Hamiltonian has a dual role. Dynamically, it generates unitary time evolution. Spectrally, its eigenvalues are admissible energy readouts. This dual role is one reason the operator/spectrum branch is central. The linked equation set is concentrated in state evolution, operator-to-spectrum readout, preparation, basis, or boundary context; its mathematical presentation emphasizes local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Hamiltonian (quantum mechanics) contributes a lawful state-transport role to the quantum construction.
- **Placement**. This page is read first as a lawful-transport move: it identifies what changes the state before readout.
- **Carrier or domain**. A state vector, density operator, wavefunction, field state, or register on a specified domain.
- **Operator or map**. Hamiltonian, unitary map, channel generator, action, constraint, or differential operator that transports the state.
- **Admissibility**. Self-adjointness, complete positivity, trace preservation, gauge constraints, and boundary/domain conditions decide whether the evolution is legal.
- **Readout**. Time-dependent probabilities, spectra, transition amplitudes, conserved quantities, or response functions.
- **Check**. The generator must predict the observed evolution while preserving the relevant normalization, positivity, symmetry, or conservation constraint.

Topic Equations

Standard constructor skeleton: energy spectrum and unitary generation.

$$\begin{aligned} H|E_n\rangle &= E_n|E_n\rangle \\ U(t) &= e^{-iHt/\hbar} \\ \rho(t) &= U(t)\rho(0)U(t)^\dagger \end{aligned}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:0912.2823
- arXiv:1604.05385
- arXiv:0908.0752
- arXiv:2108.07838
- arXiv:1612.00682

11.11 Perturbation theory (quantum mechanics)

Central Claim

A perturbation theory (quantum mechanics) can be read as a quantum construction: the potential, domain, initial condition, or boundary condition fixes the admissible state space; the Hamiltonian, whose exponential gives unitary time evolution defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a perturbation theory (quantum mechanics) is described by a wave function or density operator defined on the Hilbert space allowed by the system's domain. The physical question is represented by the Hamiltonian, whose exponential gives unitary time evolution; the experimental or mathematical setting is the potential, domain, initial condition, or boundary condition. The observable content is obtained from the eigenvalues and eigenfunctions of the relevant observable. In the local terminology of this topic, the same construction appears through quantum state or wave function, Hamiltonian or observable operator, and spectrum or eigenvalue. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in operator-to-spectrum readout, preparation, basis, or

Quantum Mechanism Frame

- **Role**. Perturbation theory (quantum mechanics) contributes a topic-native constructor role to the quantum construction.
- **Placement**. This page is read first as a lawful-transport move: it identifies what changes the state before readout.
- **Carrier or domain**. State terms: quantum state, wave function, or density operator. Context/domain terms: potential.
- **Operator or map**. Operator terms: Hamiltonian. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: spectrum. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$\begin{aligned} B &\longmapsto \rho_B \quad \text{(context specifies an admissible state)} \\ \rho_t = U_t \rho_B U_t^\dagger &\quad \text{(unitary evolution from preparation to readout)} \\ O = \sum_i \lambda_i P_i, &\quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)} \\ [O_1, O_2] \neq 0 &\quad \text{(incompatible observables: no common sharp basis)} \end{aligned}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:0912.2823
- arXiv:1604.05385
- arXiv:1612.00682
- arXiv:1506.05598
- arXiv:quant-ph0205159

11.12 Schrödinger picture

Central Claim

The schrödinger picture can be read as a quantum construction: the potential, domain, initial condition, or boundary condition fixes the admissible state space; the Hamiltonian, whose exponential gives unitary time evolution defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, the Schrödinger picture is described by a wave function or density operator defined on the Hilbert space allowed by the system's domain. The physical question is represented by the Hamiltonian, whose exponential gives unitary time evolution; the experimental or mathematical setting is the potential, domain, initial condition, or boundary condition. The observable content is obtained from the eigenvalues and eigenfunctions of the relevant observable. In the local terminology of this topic, the same construction appears through state vector or wavefunction, Hamiltonian or unitary operator, and eigenvalue or energy level. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, preparation, basis, or

Quantum Mechanism Frame

- **Role**. Schrödinger picture contributes a lawful state-transport role to the quantum construction.
- **Placement**. This page is read first as a lawful-transport move: it identifies what changes the state before readout.

- **Carrier or domain**. State terms: state vector and wavefunction. Context/domain terms: potential.
- **Operator or map**. Operator terms: Hamiltonian and unitary. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:0912.2823
- arXiv:1604.06537
- arXiv:1801.03283
- arXiv:1612.00682
- arXiv:1506.05598

11.13 Creation and annihilation operators

Central Claim

Creation and annihilation operators are sector-changing operators: they add or remove one quantum from a mode and make many-body or field descriptions executable.

Formal Role

The page is about the algebraic move that changes occupation number. Creation raises the population of a mode, annihilation lowers it, and the commutation or anticommutation rule determines the statistics. The number operator gives the spectral readout. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, non-commuting compatibility limits; its mathematical presentation emphasizes local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Creation and annihilation operators contributes an unresolved constructor role to the quantum construction.
- **Placement**. This page is read first as a lawful-transport move: it identifies what changes the state before readout.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

Topic-specific constructor: the equations express raising, lowering, and occupation-number readout.

$$\begin{aligned} a_i^\dagger|\ldots,n_i,\ldots\rangle &= \sqrt{n_i+1}|\ldots,n_i+1,\ldots\rangle \\ a_i|\ldots,n_i,\ldots\rangle &= \sqrt{n_i}|\ldots,n_i-1,\ldots\rangle \\ N_i &= a_i^\dagger a_i \end{aligned}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.06537
- arXiv:0912.2823
- arXiv:1801.03283
- arXiv:2111.12617
- arXiv:1506.05598

11.14 Schrödinger equation

Central Claim

Schrödinger equation is the state-transport constructor: the Hamiltonian generates lawful change of the state before readout.

Formal Role

The Schrödinger equation is not a measurement rule. It is the generator step of the quantum constructor. It evolves the predictive carrier while preserving normalization when the Hamiltonian is self-adjoint. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its mathematical presentation emphasizes local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Schrödinger equation contributes a lawful state-transport role to the quantum construction.
- **Placement**. This page is read first as a lawful-transport move: it identifies what changes the state before readout.
- **Carrier or domain**. A state vector, density operator, wavefunction, field state, or register on a specified domain.
- **Operator or map**. Hamiltonian, unitary map, channel generator, action, constraint, or differential operator that transports the state.
- **Admissibility**. Self-adjointness, complete positivity, trace preservation, gauge constraints, and boundary/domain conditions decide whether the evolution is legal.
- **Readout**. Time-dependent probabilities, spectra, transition amplitudes, conserved quantities, or response functions.
- **Check**. The generator must predict the observed evolution while preserving the relevant normalization, positivity, symmetry, or conservation constraint.

Topic Equations

Standard constructor skeleton: Hamiltonian transport and norm preservation.

$$i\hbar\,\partial_t|\psi(t)\rangle = H|\psi(t)\rangle$$

$$|\psi(t)\rangle = U(t)|\psi(0)\rangle, \qquad U(t) = e^{-iHt/\hbar}$$

$$\frac{d}{dt}\langle\psi(t)|\psi(t)\rangle = 0 \quad (H = H^\dagger)$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:0912.2823
- arXiv:1604.06537
- arXiv:1604.05385
- arXiv:1801.03283
- arXiv:1612.00682

11.15 Quantum stochastic calculus

Central Claim

A quantum stochastic calculus can be read as a quantum construction: the potential, domain, initial condition, or boundary condition fixes the admissible state space; the Hamiltonian, whose exponential gives unitary time evolution defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a quantum stochastic calculus is described by a wave function or density operator defined on the Hilbert space allowed by the system's domain. The physical question is represented by the Hamiltonian, whose exponential gives unitary time evolution; the experimental or mathematical setting is the potential, domain, initial condition, or boundary condition. The observable content is obtained from the eigenvalues and eigenfunctions of the relevant observable. In the local terminology of this topic, the same construction appears through quantum state or wave function, Hamiltonian or commutator, and mode or eigenvalue. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or non-commuting observables. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility;

Quantum Mechanism Frame

- **Role**. Quantum stochastic calculus contributes a topic-native constructor role to the quantum construction.
- **Placement**. This page is read first as a lawful-transport move: it identifies what changes the state before readout.
- **Carrier or domain**. State terms: quantum state, wave function, or density operator. Context/domain terms: preparation condition, measurement setup, or potential or domain.
- **Operator or map**. Operator terms: Hamiltonian and commutator. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: commutator. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: mode. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$
\begin{aligned}
B \longmapsto \rho_B &\quad \text{(context specifies an admissible state)}\\
\rho_t = U_t \rho_B U_t^\dagger &\quad \text{(unitary evolution from preparation to readout)}\\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) &\quad \text{(spectral probability measure)}\\
[O_1, O_2] \neq 0 &\quad \text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.06537

- arXiv:1604.05385

- arXiv:1612.00682

- arXiv:2108.07838

- arXiv:0912.2823

11.16 Quantum chaos

Central Claim

A quantum chaos can be read as a quantum construction: the potential, domain, initial condition, or boundary condition fixes the admissible state space; the Hamiltonian, whose exponential gives unitary time evolution defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a quantum chaos is described by a wave function or density operator defined on the Hilbert space allowed by the system's domain. The physical question is represented by the Hamiltonian, whose exponential gives unitary time evolution; the experimental or mathematical setting is the potential, domain, initial condition, or boundary condition. The observable content is obtained from the eigenvalues and eigenfunctions of the relevant observable. In the local terminology of this topic, the same construction appears through quantum state or wave function, Hamiltonian or observable operator, and eigenvalue or energy level. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility;

Quantum Mechanism Frame

- **Role**. Quantum chaos contributes a topic-native constructor role to the quantum construction.

- **Placement**. This page is read first as a lawful-transport move: it identifies what changes the state before readout.

- **Carrier or domain**. State terms: quantum state, wave function, or density operator. Context/domain terms: basis.
- **Operator or map**. Operator terms: Hamiltonian, observable, or matrix. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue and energy level. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$
\begin{aligned}
B \longmapsto \rho_B &\quad \text{(context specifies an admissible state)} \\
\rho_t = U_t \rho_B U_t^\dagger &\quad \text{(unitary evolution from preparation to readout)} \\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) &\quad \text{(spectral probability measure)} \\
[O_1, O_2] \neq 0 &\quad \text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:0912.2823
- arXiv:1801.03283
- arXiv:1612.00682
- arXiv:quant-ph0205159
- arXiv:1506.05598

11.17 Symmetry in quantum mechanics

Central Claim

A symmetry in quantum mechanics can be read as a quantum construction: the potential, domain, initial condition, or boundary condition fixes the admissible state space; the Hamiltonian, whose exponential gives unitary time evolution defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a symmetry in quantum mechanics is described by a wave function or density operator defined on the Hilbert space allowed by the system's domain. The physical question is represented by the Hamiltonian, whose exponential gives unitary time evolution; the experimental or mathematical setting is the potential, domain, initial condition, or boundary condition. The observable content is obtained from the eigenvalues and eigenfunctions of the relevant observable. In the local terminology of this topic, the same construction appears through wavefunction or quantum state, matrix or generator, and eigenvalue or energy level. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through non-commuting observables or commutator. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or

Quantum Mechanism Frame

- **Role**. Symmetry in quantum mechanics contributes a topic-native constructor role to the quantum construction.
- **Placement**. This page is read first as a lawful-transport move: it identifies what changes the state before readout.
- **Carrier or domain**. State terms: wavefunction and quantum state. Context/domain terms: context.
- **Operator or map**. Operator terms: matrix, generator, unitary, or commutator. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: non-commuting and commutator. These determine which questions, states, or updates are legal.

- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$

$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$

$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$

$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.06537
- arXiv:1801.03283
- arXiv:0912.2823
- arXiv:1506.05598
- arXiv:1612.00682

11.18 Quantum amplifier

Central Claim

A quantum amplifier can be read as a quantum construction: the potential, domain, initial condition, or boundary condition fixes the admissible state space; the Hamiltonian, whose exponential gives unitary time evolution defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a quantum amplifier is described by a wave function or density operator defined on the Hilbert space allowed by the system's domain. The physical question is represented by the Hamiltonian, whose exponential gives unitary time evolution; the experimental or mathematical setting is the potential, domain, initial condition, or boundary condition. The observable content is obtained from the eigenvalues and eigenfunctions of the relevant observable. In the local terminology of this topic, the same construction appears through quantum state or state vector, unitary operator or Hamiltonian, and mode or eigenvalue. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through uncertainty relation or commutator. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its

Quantum Mechanism Frame

- **Role**. Quantum amplifier contributes an instrument-mediated readout role to the quantum construction.
- **Placement**. This page is read first as a lawful-transport move: it identifies what changes the state before readout.
- **Carrier or domain**. State terms: quantum state, state vector, or superposition. Context/domain terms: preparation condition, measurement setup, or potential or domain.
- **Operator or map**. Operator terms: unitary. Protocol or update terms: projection.
- **Admissibility**. Compatibility or closure terms: uncertainty. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: mode. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.06537
- arXiv:1604.05385
- arXiv:1612.00682
- arXiv:2108.07838
- arXiv:1506.05598

11.19 Quantum mechanics

Central Claim

Quantum mechanics is the baseline constructor: states live in Hilbert space, physical questions are represented by operators, and probabilities are assigned to spectral projectors.

Formal Role

The page supplies the general quantum assembly. A preparation gives a state vector or density operator. A self-adjoint observable or measurement operator family gives the possible readout channels. The Born or trace rule assigns probabilities, while Hamiltonian evolution transports the state between preparation and readout. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its mathematical presentation emphasizes local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum mechanics contributes an unresolved constructor role to the quantum construction.
- **Placement**. This page is read first as a lawful-transport move: it identifies what changes the state before readout.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

Topic-specific constructor: the equations express state admissibility, spectral readout, unitary evolution, and incompatibility.

$$\begin{aligned} \rho \geq 0, \qquad \operatorname{Tr}\rho = 1 \\ A = \sum_a a P_a, \qquad p(a) = \operatorname{Tr}(\rho P_a) \\ \rho(t) = U(t)\rho(0)U(t)^\dagger, \qquad U(t) = e^{-iHt/\hbar} \\ [A,B] \neq 0 \quad \Rightarrow \quad \text{no generic common sharp eigenbasis} \end{aligned}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation

- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:0912.2823
- arXiv:1604.06537
- arXiv:1612.00682
- arXiv:1801.03283

11.20 Heisenberg picture

Central Claim

The heisenberg picture can be read as a quantum construction: the potential, domain, initial condition, or boundary condition fixes the admissible state space; the Hamiltonian, whose exponential gives unitary time evolution defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, the Heisenberg picture is described by a wave function or density operator defined on the Hilbert space allowed by the system's domain. The physical question is represented by the Hamiltonian, whose exponential gives unitary time evolution; the experimental or mathematical setting is the potential, domain, initial condition, or boundary condition. The observable content is obtained from the eigenvalues and eigenfunctions of the relevant observable. In the local terminology of this topic, the same construction appears through quantum state or wave function, Hamiltonian or observable operator, and eigenvalue or energy level. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or non-commuting observables. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or

Quantum Mechanism Frame

- **Role**. Heisenberg picture contributes a topic-native constructor role to the quantum construction.
- **Placement**. This page is read first as a lawful-transport move: it identifies what changes the state before readout.
- **Carrier or domain**. State terms: quantum state. Context/domain terms: basis.
- **Operator or map**. Operator terms: Hamiltonian, observable, or commutator. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: commutator. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.06537
- arXiv:1612.00682
- arXiv:2108.07838
- arXiv:0912.2823
- arXiv:1801.03283

11.21 Heisenberg group

Central Claim

A heisenberg group can be read as a quantum construction: the potential, domain, initial condition, or boundary condition fixes the admissible state space; the Hamiltonian, whose exponential gives unitary time evolution defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a heisenberg group is described by a wave function or density operator defined on the Hilbert space allowed by the system's domain. The physical question is represented by the Hamiltonian, whose exponential gives unitary time evolution; the experimental or mathematical setting is the potential, domain, initial condition, or boundary condition. The observable content is obtained from the eigenvalues and eigenfunctions of the relevant observable. In the local terminology of this topic, the same construction appears through quantum state or wave function, matrix or unitary operator, and eigenvalue or energy level. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its

Quantum Mechanism Frame

- **Role**. Heisenberg group contributes a topic-native constructor role to the quantum construction.
- **Placement**. This page is read first as a lawful-transport move: it identifies what changes the state before readout.
- **Carrier or domain**. State terms: quantum state, wave function, or density operator. Context/domain terms: context.
- **Operator or map**. Operator terms: matrix and unitary. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:0912.2823
- arXiv:1604.06537
- arXiv:1612.00682
- arXiv:2108.07838
- arXiv:0805.4565

11.22 Canonical commutation relation

Central Claim

The canonical commutation relation can be read as a quantum construction: the potential, domain, initial condition, or boundary condition fixes the admissible state space; the Hamiltonian, whose exponential gives unitary time evolution defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, the Canonical commutation relation is described by a wave function or density operator defined on the Hilbert space allowed by the system's domain. The physical question is represented by the Hamiltonian, whose exponential gives unitary time evolution; the experimental or mathematical setting is the potential, domain, initial condition, or boundary condition. The observable content is obtained from the eigenvalues and eigenfunctions of the relevant observable. In the local terminology of this topic, the same construction appears through quantum state or wave function, Hamiltonian or commutator, and eigenvalue or energy level. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, non-commuting

Quantum Mechanism Frame

- **Role**. Canonical commutation relation contributes a topic-native constructor role to the quantum construction.
- **Placement**. This page is read first as a lawful-transport move: it identifies what changes the state before readout.

- **Carrier or domain**. State terms: quantum state. Context/domain terms: preparation condition, measurement setup, or potential or domain.
- **Operator or map**. Operator terms: Hamiltonian and commutator. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: commutator and uncertainty. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$
\begin{aligned}
B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)}\\
\rho_t = U_t \rho_B U_t^\dagger \quad &\text{(unitary evolution from preparation to readout)}\\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)}\\
[O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.06537
- arXiv:2111.12617
- arXiv:0912.2823
- arXiv:0908.0752
- arXiv:1506.05598

11.23 Quantum biology

Central Claim

Quantum biology can be read as a quantum construction: the potential, domain, initial condition, or boundary condition fixes the admissible state space; the Hamiltonian, whose exponential gives unitary time evolution defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, quantum biology is described by a wave function or density operator defined on the Hilbert space allowed by the system's domain. The physical question is represented by the Hamiltonian, whose exponential gives unitary time evolution; the experimental or mathematical setting is the potential, domain, initial condition, or boundary condition. The observable content is obtained from the eigenvalues and eigenfunctions of the relevant observable. In the local terminology of this topic, the same construction appears through quantum state or wave function, Hamiltonian or observable operator, and eigenvalue or energy level. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through complementarity or uncertainty relation. The linked equation set is concentrated in state evolution, normalization or admissibility, operator-to-spectrum

Quantum Mechanism Frame

- **Role**. Quantum biology contributes an open-system transport and coherence role to the quantum construction.
- **Placement**. This page is read first as a lawful-transport move: it identifies what changes the state before readout.
- **Carrier or domain**. State terms: quantum state, wave function, or density operator. Context/domain terms: potential.
- **Operator or map**. Operator terms: Hamiltonian, observable operator, or generator. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: complementarity and Uncertainty. These determine which questions, states, or updates are legal.

- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1706.03846
- arXiv:0912.2823
- arXiv:1604.05385
- arXiv:1604.06537
- arXiv:1708.03640

11.24 Quantum engineering

Central Claim

A quantum engineering can be read as a quantum construction: the chosen basis, pulse sequence, or measurement axis fixes the admissible state space; a Hamiltonian or unitary matrix rotating that state between preparation and measurement defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a quantum engineering is described by a two-dimensional Hilbert space, usually written as a qubit state or a density matrix. The physical question is represented by a Hamiltonian or unitary matrix rotating that state between preparation and measurement; the experimental or mathematical setting is the chosen basis, pulse sequence, or measurement axis. The observable content is obtained from projectors onto the two eigenstates of the measured observable. In the local terminology of this topic, the same construction appears through quantum state or superposition, generator or Hamiltonian, and eigenvalue or energy level. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in state evolution, operator-to-spectrum readout, normalization or admissibility;

Quantum Mechanism Frame

- **Role**. Quantum engineering contributes a topic-native constructor role to the quantum construction.
- **Placement**. This page is read first as a lawful-transport move: it identifies what changes the state before readout.
- **Carrier or domain**. State terms: quantum state and superposition. Context/domain terms: preparation condition, measurement setup, or potential or domain.
- **Operator or map**. Operator terms: generator. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1706.03846
- arXiv:1708.03640
- arXiv:0805.4565
- arXiv:0908.0752
- arXiv:quant-ph0205159

11.25 Quantum logic

Central Claim

a quantum logic modifies the interpretation of the probability/readout layer while preserving the formal quantum dynamics.

Formal Role

a quantum logic acts on the readout layer of the quantum constructor. The formal ingredients remain the state assignment, the operator or measurement being applied, and the Born-rule map from projectors to probabilities. What changes is the status assigned to those ingredients: for this topic, the state or probability is treated through the agent, measurement context, or interpretive stance attached to the formalism. The page should therefore be read as a statement about the interpretation of state, probability, update, or recorded outcome while the Hamiltonian, spectral resolution, and commutator structure remain the formal reference layer. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its mathematical presentation emphasizes local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum logic contributes a topic-native constructor role to the quantum construction.
- **Placement**. This page is read first as a lawful-transport move: it identifies what changes the state before readout.
- **Carrier or domain**. State terms: wavefunction and superposition. Context/domain terms: potential and basis.
- **Operator or map**. Operator terms: observable and Hamiltonian. Protocol or update terms: collapse and projection.
- **Admissibility**. Compatibility or closure terms: uncertainty. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.06537
- arXiv:1604.05385
- arXiv:0912.2823
- arXiv:1612.00682
- arXiv:quant-ph0205159

Chapter 12

Spectral question: what can be asked

An observable is a permitted question whose operator form determines the possible numerical answers.

12.1 Why This Branch Exists

The central unit is a legal question posed to a state. Spectra make the possible answers visible.

12.2 Page Map

Page
Angular momentum operator
Observable
Self-adjoint operator
Spectral theory
Pauli matrices
Operator theory
Operator (physics)
Eigenvalues and eigenvectors
Von Neumann algebra
Electron microscope

12.3 Mechanism Pages

Each page below is read as a specialization of the compact quantum constructor. Topic-specific pages use explicit equations or overrides; core-derived pages use the branch-level equation form associated with their mechanism role.

12.4 Angular momentum operator

Central Claim

The angular momentum operator can be read as a quantum construction: the measurement basis and experimental arrangement fixes the admissible state space; the self-adjoint observable being asked of that state defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, the Angular momentum operator is described by a prepared quantum state before the measurement. The physical question is represented by the self-adjoint observable being asked of that state; the experimental or mathematical setting is the measurement basis and experimental arrangement. The observable content is obtained from the observable's spectral projectors and the Born probabilities assigned to them. In the local terminology of this topic, the same construction appears through quantum state or wave function, observable operator or commutator, and eigenstate or eigenvalue. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or non-commuting observables. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its mathematical presentation

Quantum Mechanism Frame

- **Role**. Angular momentum operator contributes a topic-native constructor role to the quantum construction.
- **Placement**. This page is read first as a question-selection move: it identifies the spectrum or answer set being read.
- **Carrier or domain**. State terms: quantum state, wave function, or density operator. Context/domain terms: basis.
- **Operator or map**. Operator terms: observable and commutator. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: commutator. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenstate. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.06537
- arXiv:0912.2823
- arXiv:1604.05385
- arXiv:1612.00682
- arXiv:1801.03283

12.5 Observable

Central Claim

Observable is the legal-question constructor: it turns a physical question into an operator with spectral outcome channels.

Formal Role

An observable is the mathematical form of a question that can be asked of a state. Its spectral decomposition defines the possible answers. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its mathematical presentation emphasizes local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Observable contributes an unresolved constructor role to the quantum construction.
- **Placement**. This page is read first as a question-selection move: it identifies the spectrum or answer set being read.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

Standard constructor skeleton: self-adjoint question, spectral projectors, and Born probabilities.

$$A = A^\dagger$$
$$A = \sum_i a_i P_i$$
$$p(a_i) = \mathrm{Tr}(\rho P_i)$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:1604.06537
- arXiv:1612.00682
- arXiv:1801.03283
- arXiv:quant-ph0205159

12.6 Self-adjoint operator

Central Claim

Self-adjoint operator is the admissible-observable condition: it gives real spectra and well-defined spectral measures.

Formal Role

Self-adjointness is not a technical decoration. It is the condition that makes an operator a legitimate spectral question in ordinary quantum mechanics. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its mathematical presentation emphasizes local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Self-adjoint operator contributes an unresolved constructor role to the quantum construction.
- **Placement**. This page is read first as a question-selection move: it identifies the spectrum or answer set being read.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

Standard constructor skeleton: spectral theorem form of a legitimate observable.

$$A = A^\dagger$$
$$A = \int_{\sigma(A)} \lambda \, dE_A(\lambda)$$
$$\Pr(\Delta) = \mathrm{Tr}(\rho E_A(\Delta))$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:0912.2823
- arXiv:1612.00682
- arXiv:1801.03283
- arXiv:quant-ph0205159
- arXiv:1506.05598

12.7 Spectral theory

Central Claim

Spectral theory can be read as a quantum construction: the potential, domain, initial condition, or boundary condition fixes the admissible state space; the Hamiltonian, whose exponential gives unitary time evolution defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, spectral theory is described by a wave function or density operator defined on the Hilbert space allowed by the system's domain. The physical question is represented by the Hamiltonian, whose exponential gives unitary time evolution; the experimental or mathematical setting is the potential, domain, initial condition, or boundary condition. The observable content is obtained from the eigenvalues and eigenfunctions of the relevant observable. In the local terminology of this topic, the same construction appears through quantum state or wave function, matrix or Hamiltonian, and eigenvalue or spectrum. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its mathematical

Quantum Mechanism Frame

- **Role**. Spectral theory contributes a topic-native constructor role to the quantum construction.
- **Placement**. This page is read first as a question-selection move: it identifies the spectrum or answer set being read.
- **Carrier or domain**. State terms: quantum state, wave function, or density operator. Context/domain terms: basis.
- **Operator or map**. Operator terms: matrix. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue and spectrum. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:2108.07838
- arXiv:1612.00682
- arXiv:0805.4565
- arXiv:quant-ph0205159
- arXiv:1506.05598

12.8 Pauli matrices

Central Claim

The pauli matrices can be read as a quantum construction: the chosen basis, pulse sequence, or measurement axis fixes the admissible state space; a Hamiltonian or unitary matrix rotating that state between preparation and measurement defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, the Pauli matrices is described by a two-dimensional Hilbert space, usually written as a qubit state or a density matrix. The physical question is represented by a Hamiltonian or unitary matrix rotating that state between preparation and measurement; the experimental or mathematical setting is the chosen basis, pulse sequence, or measurement axis. The observable content is obtained from projectors onto the two eigenstates of the measured observable. In the local terminology of this topic, the same construction appears through quantum state or wave function, matrix or observable operator, and eigenvalue or energy level. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or

Quantum Mechanism Frame

- **Role**. Pauli matrices contributes a topic-native constructor role to the quantum construction.
- **Placement**. This page is read first as a question-selection move: it identifies the spectrum or answer set being read.
- **Carrier or domain**. State terms: quantum state, wave function, or density operator. Context/domain terms: basis and context.
- **Operator or map**. Operator terms: matrix, observable, or unitary. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$
\begin{aligned}
B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)} \\
\rho_t = U_t \rho_B U_t^\dagger \quad &\text{(unitary evolution from preparation to readout)} \\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)} \\
[O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:0912.2823
- arXiv:2108.07838
- arXiv:1612.00682
- arXiv:0805.4565
- arXiv:1506.05598

12.9 Operator theory

Central Claim

Operator theory can be read as a quantum construction: the potential, domain, initial condition, or boundary condition fixes the admissible state space; the Hamiltonian, whose exponential gives unitary time evolution defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, operator theory is described by a wave function or density operator defined on the Hilbert space allowed by the system's domain. The physical question is represented by the Hamiltonian, whose exponential gives unitary time evolution; the experimental or mathematical setting is the potential, domain, initial condition, or boundary condition. The observable content is obtained from the eigenvalues and eigenfunctions of the relevant observable. In the local terminology of this topic, the same construction appears through quantum state or wave function, matrix or unitary operator, and Spectrum or eigenvalue. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its

Quantum Mechanism Frame

- **Role**. Operator theory contributes a topic-native constructor role to the quantum construction.
- **Placement**. This page is read first as a question-selection move: it identifies the spectrum or answer set being read.
- **Carrier or domain**. State terms: quantum state, wave function, or density operator. Context/domain terms: basis.
- **Operator or map**. Operator terms: matrix and unitary. Protocol or update terms: projection.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: Spectrum and eigenvalue. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:0912.2823
- arXiv:1604.05385
- arXiv:1612.00682
- arXiv:1506.05598
- arXiv:quant-ph0205159

12.10 Operator (physics)

Central Claim

An operator (physics) can be read as a quantum construction: the potential, domain, initial condition, or boundary condition fixes the admissible state space; the Hamiltonian, whose exponential gives unitary time evolution defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, an operator (physics) is described by a wave function or density operator defined on the Hilbert space allowed by the system's domain. The physical question is represented by the Hamiltonian, whose exponential gives unitary time evolution; the experimental or mathematical setting is the potential, domain, initial condition, or boundary condition. The observable content is obtained from the eigenvalues and eigenfunctions of the relevant observable. In the local terminology of this topic, the same construction appears through wavefunction or wave function, Hamiltonian or matrix, and eigenvalue or eigenstate. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its

Quantum Mechanism Frame

- **Role**. Operator (physics) contributes a topic-native constructor role to the quantum construction.
- **Placement**. This page is read first as a question-selection move: it identifies the spectrum or answer set being read.
- **Carrier or domain**. State terms: wavefunction. Context/domain terms: context.
- **Operator or map**. Operator terms: Hamiltonian, matrix, generator, observable, or unitary. Protocol or update terms: projection.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$

$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$

$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$

$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:1801.03283
- arXiv:0912.2823
- arXiv:1612.00682
- arXiv:1506.05598

12.11 Eigenvalues and eigenvectors

Central Claim

An eigenvalues and eigenvectors can be read as a quantum construction: the measurement basis and experimental arrangement fixes the admissible state space; the self-adjoint observable being asked of that state defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, an eigenvalues and eigenvectors is described by a prepared quantum state before the measurement. The physical question is represented by the self-adjoint observable being asked of that state; the experimental or mathematical setting is the measurement basis and experimental arrangement. The observable content is obtained from the observable's spectral projectors and the Born probabilities assigned to them. In the local terminology of this topic, the same construction appears through quantum state or wave function, matrix or Hamiltonian, and eigenvalue or eigenstate. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, preparation, basis, or boundary context; its mathematical presentation emphasizes

Quantum Mechanism Frame

- **Role**. Eigenvalues and eigenvectors contributes a topic-native constructor role to the quantum construction.
- **Placement**. This page is read first as a question-selection move: it identifies the spectrum or answer set being read.
- **Carrier or domain**. State terms: quantum state, wave function, or density operator. Context/domain terms: basis, domain, or context.
- **Operator or map**. Operator terms: matrix. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:0912.2823
- arXiv:1604.05385
- arXiv:1612.00682
- arXiv:2308.15676
- arXiv:2108.07838

12.12 Von Neumann algebra

Central Claim

The von neumann algebra can be read as a quantum construction: the measurement basis and experimental arrangement fixes the admissible state space; the self-adjoint observable being asked of that state defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, the Von Neumann algebra is described by a prepared quantum state before the measurement. The physical question is represented by the self-adjoint observable being asked of that state; the experimental or mathematical setting is the measurement basis and experimental arrangement. The observable content is obtained from the observable's spectral projectors and the Born probabilities assigned to them. In the local terminology of this topic, the same construction appears through quantum state or wave function, unitary operator or Hamiltonian, and eigenvalue or energy level. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its mathematical presentation emphasizes local

Quantum Mechanism Frame

- **Role**. Von Neumann algebra contributes a topic-native constructor role to the quantum construction.
- **Placement**. This page is read first as a question-selection move: it identifies the spectrum or answer set being read.
- **Carrier or domain**. State terms: quantum state, wave function, or density operator. Context/domain terms: preparation condition, measurement setup, or potential or domain.
- **Operator or map**. Operator terms: unitary. Protocol or update terms: projection.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:0912.2823
- arXiv:1604.05385
- arXiv:1604.06537
- arXiv:1612.00682
- arXiv:quant-ph0205159

12.13 Electron microscope

Central Claim

Electron microscope can be represented as a formal construction: a context specifies a state, an operator transforms it, and admissible outcomes are read through a spectrum.

Formal Role

Electron microscope is organized by the relation between state terms (quantum state, wave function, density operator), operator terms (Hamiltonian, observable operator, generator), context or boundary terms (detector), and spectral readout terms (mode). The local vocabulary can change across formulations, but the same state-to-transformation-to-readout structure can remain identifiable. Across equation-level evidence the dominant pattern is operator and spectrum, state evolution / transport, closure / conservation, and boundary or preparation, carried by field-specific vocabulary, probability / information, and symbolic structure. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its mathematical presentation emphasizes local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Electron microscope contributes an instrument-mediated readout role to the quantum construction.
- **Placement**. This page is read first as a question-selection move: it identifies the spectrum or answer set being read.
- **Carrier or domain**. State terms: quantum state, wave function, or density operator. Context/domain terms: detector.
- **Operator or map**. Operator terms: Hamiltonian, observable operator, or generator. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: mode. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. Candidate transfer targets are systems where the same state-to-operator-to-spectrum conversion appears, but one edge of the construction remains experimentally unresolved.

Core-Derived Role Equations

This equation block is the branch-level quantum mechanism attached to the page by the compact constructor. It gives the role equation to check against the page's source evidence; source notation may differ.

$$\rho_{\text{probe}} \mapsto \mathcal{E}_{\text{sample}}(\rho_{\text{probe}}), \quad p(y) = \mathrm{Tr}(M_y \mathcal{E}_{\text{sample}}(\rho_{\text{probe}})), \quad \hat{s} = R(\{y_i\})$$

What Remains Stable

- The mechanism is an apparatus-coupled readout: a prepared probe state interacts with a sample or field, the interaction changes phase, momentum, intensity, or counting statistics, and the instrument reconstructs an image, spectrum, trajectory, or estimate.
- Electron microscope defines the legal question being asked of the state.
- The measurable answers are encoded by the operator spectrum, projectors, or spectral measure.
- The operator role is preserved across equivalent bases even when matrix entries change.

What Changes With Realization

- The local title, representation, and physical realization may change while the constructor role is preserved.
- The same observable may be represented by matrices, differential operators, projectors, or algebraic elements.
- Degeneracy, basis choice, and domain conditions can change how the spectrum is displayed.
- Detector implementation changes the physical realization, not the operator role itself.

Checks

- Use this role by specifying the page's state carrier, operator or map, readout, and compatibility condition in its own quantum language.
- Self-adjointness, or the appropriate POVM positivity condition, is what makes the question a legal readout.
- A complete spectral resolution supplies all outcome channels for the question being asked.

Source Pointers

- arXiv:0912.2823
- arXiv:1612.00682
- arXiv:0805.4565
- arXiv:2108.07838
- arXiv:quant-ph0205159

Chapter 13

Readout rule: how answers become probabilities

Measurement connects a state and an observable to recorded frequencies. Projection, POVMs, Born weights, and collapse language are alternative ways of presenting this state-to-spectrum readout step.

13.1 Why This Branch Exists

Measurement is best placed after observables and spectra: it is the rule that turns spectral resolution into recorded probability, not the mystical starting point of the theory.

13.2 Page Map

Page
POVM
Wave function collapse
Born rule
Measurement in quantum mechanics
Quantum jump
Measurement problem
Quantum eraser experiment
Projection-valued measure
Delayed-choice quantum eraser
Quantum metrology

13.3 Mechanism Pages

Each page below is read as a specialization of the compact quantum constructor. Topic-specific pages use explicit equations or overrides; core-derived pages use the branch-level equation form associated with their mechanism role.

13.4 POVM

Central Claim

POVM is the generalized-readout constructor: outcome effects need not be orthogonal projectors.

Formal Role

POVMs separate the probability readout from the idealized projection assumption. They are the natural mechanism for noisy, coarse-grained, indirect, or open-system measurements. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its mathematical presentation emphasizes local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. POVM contributes a probability/readout role to the quantum construction.
- **Placement**. This page is read first as a readout move: it connects the state and question to recorded outcomes.
- **Carrier or domain**. A state vector or density operator together with the measurement context in which outcome channels are defined.
- **Operator or map**. A projection-valued measure, POVM, update map, or instrument map connecting state to record.
- **Admissibility**. Outcome probabilities must be positive, normalized, and tied to a specified readout map rather than to informal observer language.
- **Readout**. Born probabilities, detector records, post-measurement states, ensemble frequencies, or decision probabilities.
- **Check**. The interpretation is constrained by whether it changes the probability rule, the update rule, the detector model, or only the language used for them.

Topic Equations

Standard constructor skeleton: positive effects and generalized Born rule.

$$E_i \geq 0, \qquad \sum_i E_i = I$$

$$p(i) = \mathrm{Tr}(\rho E_i)$$

$$E_i = \sum_\alpha K_{i\alpha}^\dagger K_{i\alpha}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:1801.03283
- arXiv:1612.00682
- arXiv:1506.05598
- arXiv:quant-ph0205159

13.5 Wave function collapse

Central Claim

a wave function collapse modifies the interpretation of the probability/readout layer while preserving the formal quantum dynamics.

Formal Role

a wave function collapse acts on the readout layer of the quantum constructor. The formal ingredients remain the state assignment, the operator or measurement being applied, and the Born-rule map from projectors to probabilities. What changes is the status assigned to those ingredients: for this topic, the state or probability is treated through the agent, measurement context, or interpretive stance attached to the formalism. The page should therefore be read as a statement about the interpretation of state, probability, update, or recorded outcome while the Hamiltonian, spectral resolution, and commutator structure remain the formal reference layer. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its mathematical presentation emphasizes local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Wave function collapse contributes a probability/readout role to the quantum construction.
- **Placement**. This page is read first as a readout move: it connects the state and question to recorded outcomes.
- **Carrier or domain**. State terms: Wave function, state vector, quantum state, or superposition. Context/domain terms: detector, measurement apparatus, or basis.
- **Operator or map**. Operator terms: observable. Protocol or update terms: collapse and Born rule.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenstate and eigenvalue. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:0912.2823
- arXiv:1604.06537
- arXiv:1612.00682
- arXiv:1801.03283

13.6 Born rule

Central Claim

Born rule is the probability-readout constructor: it maps a state and a spectral channel to an observed probability.

Formal Role

The Born rule is the point where the constructor becomes predictive. It does not name an object; it connects state preparation and a legal question to frequencies over outcome channels. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its mathematical presentation emphasizes local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Born rule contributes a probability/readout role to the quantum construction.
- **Placement**. This page is read first as a readout move: it connects the state and question to recorded outcomes.
- **Carrier or domain**. A state vector or density operator together with the measurement context in which outcome channels are defined.
- **Operator or map**. A projection-valued measure, POVM, update map, or instrument map connecting state to record.
- **Admissibility**. Outcome probabilities must be positive, normalized, and tied to a specified readout map rather than to informal observer language.
- **Readout**. Born probabilities, detector records, post-measurement states, ensemble frequencies, or decision probabilities.
- **Check**. The interpretation is constrained by whether it changes the probability rule, the update rule, the detector model, or only the language used for them.

Topic Equations

Standard constructor skeleton: probability assignment for projective and position readouts.

$$p(i|\rho, \{P_i\}) = \mathrm{Tr}(\rho P_i)$$
$$\Pr(X \in \Delta|\psi) = \int_{\Delta} |\psi(x)|^2 \, d\mu(x)$$
$$\sum_i p(i) = 1$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:1604.06537
- arXiv:1801.03283
- arXiv:1612.00682
- arXiv:0912.2823

13.7 Measurement in quantum mechanics

Central Claim

Measurement in quantum mechanics is the complete readout junction: it combines a state, a measurement model, probabilities, and sometimes an update rule.

Formal Role

Measurement is not the root of quantum theory in this book. It is the junction where a prepared state and an observable or POVM are converted into probabilities and recorded outcomes. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its mathematical presentation emphasizes local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Measurement in quantum mechanics contributes a probability/readout role to the quantum construction.
- **Placement**. This page is read first as a readout move: it connects the state and question to recorded outcomes.
- **Carrier or domain**. A state vector or density operator together with the measurement context in which outcome channels are defined.
- **Operator or map**. A projection-valued measure, POVM, update map, or instrument map connecting state to record.
- **Admissibility**. Outcome probabilities must be positive, normalized, and tied to a specified readout map rather than to informal observer language.
- **Readout**. Born probabilities, detector records, post-measurement states, ensemble frequencies, or decision probabilities.
- **Check**. The interpretation is constrained by whether it changes the probability rule, the update rule, the detector model, or only the language used for them.

Topic Equations

Standard constructor skeleton: generalized measurement probability and conditional update.

$$p(i) = \mathrm{Tr}(\rho E_i)$$
$$\rho \mapsto \rho_i = \frac{K_i \rho K_i^\dagger}{\mathrm{Tr}(K_i \rho K_i^\dagger)}$$
$$E_i = K_i^\dagger K_i$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels

- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:1604.06537
- arXiv:0912.2823
- arXiv:1612.00682
- arXiv:quant-ph0205159

13.8 Quantum jump

Central Claim

A quantum jump can be read as a quantum construction: the measurement basis and experimental arrangement fixes the admissible state space; the self-adjoint observable being asked of that state defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a quantum jump is described by a prepared quantum state before the measurement. The physical question is represented by the self-adjoint observable being asked of that state; the experimental or mathematical setting is the measurement basis and experimental arrangement. The observable content is obtained from the observable's spectral projectors and the Born probabilities assigned to them. In the local terminology of this topic, the same construction appears through quantum state or wave function, unitary operator or Hamiltonian, and energy level or eigenvalue. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its mathematical presentation emphasizes local notation,

Quantum Mechanism Frame

- **Role**. Quantum jump contributes a topic-native constructor role to the quantum construction.
- **Placement**. This page is read first as a readout move: it connects the state and question to recorded outcomes.
- **Carrier or domain**. State terms: quantum state. Context/domain terms: preparation condition, measurement setup, or potential or domain.
- **Operator or map**. Operator terms: unitary. Protocol or update terms: unitary evolution.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: energy level. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$
\begin{aligned}
B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)}\\
\rho_t = U_t \rho_B U_t^\dagger \quad &\text{(unitary evolution from preparation to readout)}\\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)}\\
[O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:2108.07838
- arXiv:1612.00682
- arXiv:0805.4565
- arXiv:quant-ph0205159

13.9 Measurement problem

Central Claim

Measurement problem is the junction between unitary system–apparatus coupling, probability readout, and conditional state update; these are distinct maps and need not be identified.

Formal Role

A measurement model first couples the system to an apparatus or environment. A POVM or instrument then assigns outcome probabilities, and a conditional map specifies the post-record state. The foundational problem concerns the relation between these operations and a definite record, not the absence of a probability formula. The linked equation set is concentrated in state evolution, preparation, basis, or boundary context, normalization or admissibility; its mathematical presentation emphasizes local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Measurement problem contributes a probability/readout role to the quantum construction.
- **Placement**. This page is read first as a readout move: it connects the state and question to recorded outcomes.

- **Carrier or domain**. A joint system–apparatus state, possibly enlarged by environmental degrees of freedom.
- **Operator or map**. A premeasurement interaction followed by a measurement instrument whose components label possible records.
- **Admissibility**. The instrument maps are completely positive and their sum is trace preserving; the outcome effects sum to the identity.
- **Readout**. Outcome probabilities and conditional post-record states must be stated separately.
- **Check**. A proposed resolution must identify where a definite record enters and how its prediction differs from the unconditioned state evolution.

Topic Equations

Topic-specific constructor: premeasurement coupling, outcome probability, conditional update, and unconditioned evolution are separated.

$$\rho'_{SA} = U_{SA}(\rho_S \otimes \rho_A)U_{SA}^{\dagger}$$
$$p(i) = \mathrm{Tr}[\mathcal{I}_i(\rho_S)] = \mathrm{Tr}(\rho_S E_i)$$
$$\rho_{S|i} = \frac{\mathcal{I}_i(\rho_S)}{p(i)}, \qquad \rho'_S = \sum_i \mathcal{I}_i(\rho_S)$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:0912.2823
- arXiv:1706.03846
- arXiv:2501.07524
- arXiv:1604.06537

13.10 Quantum eraser experiment

Central Claim

A quantum eraser experiment can be read as a quantum construction: the measurement basis and experimental arrangement fixes the admissible state space; the self-adjoint observable being asked of that state defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a quantum eraser experiment is described by a prepared quantum state before the measurement. The physical question is represented by the self-adjoint observable being asked of that state; the experimental or mathematical setting is the measurement basis and experimental arrangement. The observable content is obtained from the observable's spectral projectors and the Born probabilities assigned to them. In the local terminology of this topic, the same construction appears through quantum state or wave function, Hamiltonian or observable operator, and mode or eigenvalue. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through complementarity or commutator. The linked equation set is concentrated in operator-to-spectrum readout, preparation, basis, or boundary context, state evolution; its mathematical presentation emphasizes

Quantum Mechanism Frame

- **Role**. Quantum eraser experiment contributes a compatibility or joint-readout role to the quantum construction.
- **Placement**. This page is read first as a readout move: it connects the state and question to recorded outcomes.
- **Carrier or domain**. State terms: quantum state, wave function, or density operator. Context/domain terms: detector and experimental setup.
- **Operator or map**. Operator terms: Hamiltonian, observable operator, or generator. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: complementarity. These determine which questions, states, or updates are legal.

- **Readout**. Readout terms: mode. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$

$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$

$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$

$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:0912.2823
- arXiv:gr-qc0303063
- arXiv:1604.06537
- arXiv:2110.09771

13.11 Projection-valued measure

Central Claim

Projection-valued measure is the sharp-readout constructor: mutually exclusive outcome projectors partition the identity.

Formal Role

A projection-valued measure encodes an ideal sharp measurement. It defines outcome channels that are orthogonal and exhaustive. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its mathematical presentation emphasizes local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Projection-valued measure contributes a probability/readout role to the quantum construction.
- **Placement**. This page is read first as a readout move: it connects the state and question to recorded outcomes.
- **Carrier or domain**. A state vector or density operator together with the measurement context in which outcome channels are defined.
- **Operator or map**. A projection-valued measure, POVM, update map, or instrument map connecting state to record.
- **Admissibility**. Outcome probabilities must be positive, normalized, and tied to a specified readout map rather than to informal observer language.
- **Readout**. Born probabilities, detector records, post-measurement states, ensemble frequencies, or decision probabilities.
- **Check**. The interpretation is constrained by whether it changes the probability rule, the update rule, the detector model, or only the language used for them.

Topic Equations

Standard constructor skeleton: sharp outcome channels, probability, and projective update.

$$P_i P_j = \delta_{ij} P_i, \qquad \sum_i P_i = I$$

$$p(i) = \mathrm{Tr}(\rho P_i)$$

$$\rho \mapsto \frac{P_i \rho P_i}{\mathrm{Tr}(\rho P_i)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:1612.00682
- arXiv:1801.03283
- arXiv:1506.05598
- arXiv:quant-ph0205159

13.12 Delayed-choice quantum eraser

Central Claim

a delayed-choice quantum eraser modifies the interpretation of the probability/readout layer while preserving the formal quantum dynamics.

Formal Role

a delayed-choice quantum eraser acts on the readout layer of the quantum constructor. The formal ingredients remain the state assignment, the operator or measurement being applied, and the Born-rule map from projectors to probabilities. What changes is the status assigned to those ingredients: for this topic, the state or probability is treated through the agent, measurement context, or interpretive stance attached to the formalism. The page should therefore be read as a statement about the interpretation of state, probability, update, or recorded outcome while the Hamiltonian, spectral resolution, and commutator structure remain the formal reference layer. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its mathematical presentation emphasizes local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Delayed-choice quantum eraser contributes a compatibility or joint-readout role to the quantum construction.
- **Placement**. This page is read first as a readout move: it connects the state and question to recorded outcomes.
- **Carrier or domain**. State terms: superposition. Context/domain terms: detector, experimental setup, or basis.
- **Operator or map**. Operator terms: Hamiltonian, observable operator, or generator. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: complementarity and uncertainty. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: mode. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$
\begin{aligned}
B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)} \\
\rho_t = U_t \rho_B U_t^\dagger \quad &\text{(unitary evolution from preparation to readout)} \\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)} \\
[O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:0912.2823
- arXiv:1604.06537
- arXiv:quant-ph0205159
- arXiv:1708.03640

13.13 Quantum metrology

Central Claim

A quantum metrology can be read as a quantum construction: the potential, domain, initial condition, or boundary condition fixes the admissible state space; the Hamiltonian, whose exponential gives unitary time evolution defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a quantum metrology is described by a wave function or density operator defined on the Hilbert space allowed by the system's domain. The physical question is represented by the Hamiltonian, whose exponential gives unitary time evolution; the experimental or mathematical setting is the potential, domain, initial condition, or boundary condition. The observable content is obtained from the eigenvalues and eigenfunctions of the relevant observable. In the local terminology of this topic, the same construction appears through quantum state or wave function, Hamiltonian or unitary operator, and mode or eigenvalue. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its

Quantum Mechanism Frame

- **Role**. Quantum metrology contributes an instrument-mediated readout role to the quantum construction.
- **Placement**. This page is read first as a readout move: it connects the state and question to recorded outcomes.

- **Carrier or domain**. State terms: quantum state, wave function, or density operator. Context/domain terms: basis.
- **Operator or map**. Operator terms: Hamiltonian and unitary. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: mode. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$\begin{aligned} B &\longmapsto \rho_B \quad \text{(context specifies an admissible state)} \\ \rho_t = U_t \rho_B U_t^\dagger & \quad \text{(unitary evolution from preparation to readout)} \\ O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) & \quad \text{(spectral probability measure)} \\ [O_1, O_2] \neq 0 & \quad \text{(incompatible observables: no common sharp basis)} \end{aligned}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.06537
- arXiv:1604.05385
- arXiv:0912.2823
- arXiv:1612.00682
- arXiv:1506.05598

Chapter 14

Compatibility limit: what cannot be jointly sharp

The non-classical part of the theory appears when two otherwise legal questions do not compose into one common sharp question. Commutators, uncertainty relations, contextuality, Bell tests, and entanglement live here.

14.1 Why This Branch Exists

The non-classical core appears as failure of joint sharpness. Entanglement, Bell phenomena, and uncertainty are different faces of this compatibility structure.

14.2 Page Map

Page
Bell's theorem
Quantum entanglement
Commutator
Uncertainty principle
Einstein–Podolsky–Rosen paradox
Quantum nonlocality

14.3 Mechanism Pages

Each page below is read as a specialization of the compact quantum constructor. Topic-specific pages use explicit equations or overrides; core-derived pages use the branch-level equation form associated with their mechanism role.

14.4 Bell's theorem

Central Claim

Bell's theorem is a compatibility/locality stress test: quantum correlations violate bounds satisfied by local hidden-variable assignments.

Formal Role

Bell's theorem is not a page about a mysterious object. It is a falsifier for a classical joint-assignment model of measurement outcomes. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its mathematical presentation emphasizes local notation, information profile, spectral profile.

Quantum Mechanism Frame

- **Role**. Bell's theorem contributes a compatibility or joint-readout role to the quantum construction.
- **Placement**. This page is read first as a compatibility move: it asks which otherwise legal questions cannot share one sharp answer set.
- **Carrier or domain**. One state space or a multipartite state space on which several questions can be asked.
- **Operator or map**. Two or more observables, contexts, correlation operators, or hidden-variable assignments being compared.
- **Admissibility**. Commutators, uncertainty bounds, contextuality constraints, or Bell-type inequalities decide which joint assignments are possible.
- **Readout**. Joint spectra, correlations, inequality violations, uncertainty products, or incompatible outcome statistics.
- **Check**. The non-classical content appears only if the incompatible questions cannot be replaced by one common sharp classical assignment.

Topic Equations

Standard constructor skeleton: CHSH inequality and quantum violation bound.

$$|E(a,b) + E(a,b') + E(a',b) - E(a',b')| \leq 2$$
$$\text{quantum prediction can reach } 2\sqrt{2}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:1604.06537
- arXiv:0912.2823
- arXiv:1801.03283
- arXiv:quant-ph0205159

14.5 Quantum entanglement

Central Claim

Quantum entanglement is the non-factorization constructor: a composite state can carry correlations not reducible to independent subsystem states.

Formal Role

Entanglement belongs to composition and compatibility. It appears when the tensor-product state cannot be written as a product or mixture of local states, producing correlations that stress classical separability. The linked equation set is concentrated in state evolution, operator-to-spectrum readout, normalization or admissibility; its mathematical presentation emphasizes local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum entanglement contributes a compatibility or joint-readout role to the quantum construction.
- **Placement**. This page is read first as a compatibility move: it asks which otherwise legal questions cannot share one sharp answer set.
- **Carrier or domain**. One state space or a multipartite state space on which several questions can be asked.
- **Operator or map**. Two or more observables, contexts, correlation operators, or hidden-variable assignments being compared.
- **Admissibility**. Commutators, uncertainty bounds, contextuality constraints, or Bell-type inequalities decide which joint assignments are possible.
- **Readout**. Joint spectra, correlations, inequality violations, uncertainty products, or incompatible outcome statistics.

- **Check**. The non-classical content appears only if the incompatible questions cannot be replaced by one common sharp classical assignment.

Topic Equations

Standard constructor skeleton: tensor composition, non-factorization, and reduced state.

$$\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B$$
$$|\Psi\rangle_{AB} \neq |\psi\rangle_A \otimes |\phi\rangle_B$$
$$\rho_A = \mathrm{Tr}_B |\Psi\rangle\langle\Psi|$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:0912.2823
- arXiv:1604.06537
- arXiv:1801.03283
- arXiv:quant-ph0205159

14.6 Commutator

Central Claim

Commutator is the incompatibility constructor: it measures the failure of two transformations or questions to be freely exchanged.

Formal Role

The commutator is the algebraic source of many non-classical restrictions. If two observables do not commute, they generally cannot be resolved in one common sharp basis. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, non-commuting compatibility limits; its mathematical presentation emphasizes local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Commutator contributes a compatibility or joint-readout role to the quantum construction.
- **Placement**. This page is read first as a compatibility move: it asks which otherwise legal questions cannot share one sharp answer set.
- **Carrier or domain**. A common state space on which two transformations, observables, or questions are both defined.
- **Operator or map**. The ordered products AB and BA, compared through the obstruction [A,B]=AB-BA.
- **Admissibility**. A nonzero commutator marks an order-dependence or compatibility limit; a zero commutator permits a common sharp refinement only when the remaining spectral conditions hold.
- **Readout**. Compatibility tests, uncertainty bounds, common eigenspaces, or canonical commutation relations.
- **Check**. The mechanism is supported only when changing operator order changes the algebraic or statistical prediction.

Topic Equations

Standard constructor skeleton: order obstruction and canonical commutation.

$$[A, B] = AB - BA$$

$$[A, B] = 0 \quad \Rightarrow \quad \text{common spectral refinement may exist}$$

$$[x, p] = i\hbar$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.06537
- arXiv:2111.12617
- arXiv:1612.00682
- arXiv:2108.07838
- arXiv:0805.4565

14.7 Uncertainty principle

Central Claim

Uncertainty principle is a compatibility-limit theorem: non-commuting observables impose lower bounds on joint sharpness.

Formal Role

Uncertainty is not detector imperfection. It is a structural consequence of state variance and non-commuting observables. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its mathematical presentation emphasizes local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Uncertainty principle contributes a compatibility or joint-readout role to the quantum construction.
- **Placement**. This page is read first as a compatibility move: it asks which otherwise legal questions cannot share one sharp answer set.
- **Carrier or domain**. One state space or a multipartite state space on which several questions can be asked.

- **Operator or map**. Two or more observables, contexts, correlation operators, or hidden-variable assignments being compared.
- **Admissibility**. Commutators, uncertainty bounds, contextuality constraints, or Bell-type inequalities decide which joint assignments are possible.
- **Readout**. Joint spectra, correlations, inequality violations, uncertainty products, or incompatible outcome statistics.
- **Check**. The non-classical content appears only if the incompatible questions cannot be replaced by one common sharp classical assignment.

Topic Equations

Standard constructor skeleton: variance bound from commutator structure.

$$\Delta_\rho A \, \Delta_\rho B \geq \frac{1}{2} \left|\mathrm{Tr}(\rho[A, B])\right|$$
$$\Delta x \, \Delta p \geq \frac{\hbar}{2}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:1604.06537
- arXiv:2111.12617
- arXiv:0912.2823
- arXiv:1612.00682

14.8 Einstein–Podolsky–Rosen paradox

Central Claim

the Einstein–Podolsky–Rosen paradox modifies the interpretation of the probability/readout layer while preserving the formal quantum dynamics.

Formal Role

the Einstein–Podolsky–Rosen paradox acts on the readout layer of the quantum constructor. The formal ingredients remain the state assignment, the operator or measurement being applied, and the Born-rule map from projectors to probabilities. What changes is the status assigned to those ingredients: for this topic, the state or probability is treated through the agent, measurement context, or interpretive stance attached to the formalism. The page should therefore be read as a statement about the interpretation of state, probability, update, or recorded outcome while the Hamiltonian, spectral resolution, and commutator structure remain the formal reference layer. The linked equation set is concentrated in state evolution, normalization or admissibility, non-commuting compatibility limits; its mathematical presentation emphasizes local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Einstein–Podolsky–Rosen paradox contributes a topic-native constructor role to the quantum construction.
- **Placement**. This page is read first as a compatibility move: it asks which otherwise legal questions cannot share one sharp answer set.
- **Carrier or domain**. State terms: wave function, quantum state, or superposition. Context/domain terms: preparation condition, measurement setup, or potential or domain.
- **Operator or map**. Operator terms: Hamiltonian, observable operator, or generator. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: incompatible and uncertainty. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:0912.2823
- arXiv:2501.07524
- arXiv:1706.03846
- arXiv:1604.06537

14.9 Quantum nonlocality

Central Claim

a quantum nonlocality modifies the interpretation of the probability/readout layer while preserving the formal quantum dynamics.

Formal Role

a quantum nonlocality acts on the readout layer of the quantum constructor. The formal ingredients remain the state assignment, the operator or measurement being applied, and the Born-rule map from projectors to probabilities. What changes is the status assigned to those ingredients: for this topic, the state or probability is treated through the agent, measurement context, or interpretive stance attached to the formalism. The page should therefore be read as a statement about the interpretation of state, probability, update, or recorded outcome while the Hamiltonian, spectral resolution, and commutator structure remain the formal reference layer. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, controlled update protocol; its mathematical presentation emphasizes local notation, information profile, spectral profile.

Quantum Mechanism Frame

- **Role**. Quantum nonlocality contributes a compatibility or joint-readout role to the quantum construction.
- **Placement**. This page is read first as a compatibility move: it asks which otherwise legal questions cannot share one sharp answer set.
- **Carrier or domain**. State terms: wavefunction, quantum state, or wave function. Context/domain terms: basis.
- **Operator or map**. Operator terms: Hamiltonian, observable operator, or generator. Protocol or update terms: collapse.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:1612.00682
- arXiv:1801.03283
- arXiv:1506.05598
- arXiv:quant-ph0205159

Chapter 15

Boundary realization: how effects appear

Many named quantum effects are boundary realizations of the same construction. A potential, barrier, box, cavity, detector, or medium changes the allowed spectral channels without changing the basic prediction problem.

15.1 Why This Branch Exists

Named effects such as tunnelling and particle-in-a-box are boundary realizations of the state-operator-readout construction.

15.2 Page Map

Page
Potential well
Particle in a box
Scattering
Wave interference
Quantum optics
Spectral line
S-matrix
Quantum tunnelling
Quantum imaging
Quantum harmonic oscillator
Macroscopic quantum phenomena
Quantum metamaterial

15.3 Mechanism Pages

Each page below is read as a specialization of the compact quantum constructor. Topic-specific pages use explicit equations or overrides; core-derived pages use the branch-level equation form associated with their mechanism role.

15.4 Potential well

Central Claim

A potential well can be read as a quantum construction: the measurement basis and experimental arrangement fixes the admissible state space; the self-adjoint observable being asked of that state defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a potential well is described by a prepared quantum state before the measurement. The physical question is represented by the self-adjoint observable being asked of that state; the experimental or mathematical setting is the measurement basis and experimental arrangement. The observable content is obtained from the observable's spectral projectors and the Born probabilities assigned to them. In the local terminology of this topic, the same construction appears through wave function or state vector, Hamiltonian or observable operator, and spectrum or energy level. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its mathematical presentation emphasizes local

Quantum Mechanism Frame

- **Role**. Potential well contributes a boundary-shaped spectrum role to the quantum construction.
- **Placement**. This page is read first as a realization move: it changes the domain, boundary, geometry, or interface in which the operator acts.
- **Carrier or domain**. State terms: wave function. Context/domain terms: Potential.
- **Operator or map**. Operator terms: Hamiltonian, observable operator, or generator. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: spectrum, energy level, or mode. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$

$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$

$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$

$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:0912.2823
- arXiv:1506.05598
- arXiv:1612.00682
- arXiv:2108.07838

15.5 Particle in a box

Central Claim

Particle in a box is a boundary-spectrum constructor: a spatial domain and boundary condition discretize the allowed energy spectrum.

Formal Role

The page shows how a boundary condition changes the domain of the Hamiltonian and therefore the allowed spectra. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its mathematical presentation emphasizes local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Particle in a box contributes a many-mode field or particle-realization role to the quantum construction.
- **Placement**. This page is read first as a realization move: it changes the domain, boundary, geometry, or interface in which the operator acts.
- **Carrier or domain**. Fock space, field configuration space, or a sector selected by charge, spin, momentum, statistics, or gauge data.
- **Operator or map**. Creation, annihilation, field, charge, spin, Hamiltonian, or scattering operators acting on the admissible sector.
- **Admissibility**. Statistics, gauge constraints, commutation or anticommutation rules, domain conditions, and sector labels decide which states are legal.
- **Readout**. Occupation number, charge, spin, momentum, energy, correlation function, cross-section, or scattering amplitude.
- **Check**. The field description must preserve the relevant observables under changes of representation and reduce to the expected particle or quasiparticle limit when that limit exists.

Topic Equations

Standard constructor skeleton: boundary condition and discrete spectrum.

$$\psi(0) = \psi(L) = 0$$
$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L}$$
$$E_n = \frac{\hbar^2 \pi^2 n^2}{2mL^2}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels

- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:0912.2823
- arXiv:1801.03283
- arXiv:1612.00682
- arXiv:1506.05598
- arXiv:quant-ph0205159

15.6 Scattering

Central Claim

A scattering can be read as a quantum construction: the potential, domain, initial condition, or boundary condition fixes the admissible state space; the Hamiltonian, whose exponential gives unitary time evolution defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a scattering is described by a wave function or density operator defined on the Hilbert space allowed by the system's domain. The physical question is represented by the Hamiltonian, whose exponential gives unitary time evolution; the experimental or mathematical setting is the potential, domain, initial condition, or boundary condition. The observable content is obtained from the eigenvalues and eigenfunctions of the relevant observable. In the local terminology of this topic, the same construction appears through quantum state or wave function, Matrix or Hamiltonian, and eigenvalue or energy level. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in state evolution, normalization or admissibility, operator-to-spectrum readout; its mathematical

Quantum Mechanism Frame

- **Role**. Scattering contributes a boundary-shaped spectrum role to the quantum construction.
- **Placement**. This page is read first as a realization move: it changes the domain, boundary, geometry, or interface in which the operator acts.
- **Carrier or domain**. State terms: quantum state, wave function, or density operator. Context/domain terms: boundary condition and potential.
- **Operator or map**. Operator terms: Matrix. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$
\begin{aligned}
B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)}\\
\rho_t = U_t \rho_B U_t^\dagger \quad &\text{(unitary evolution from preparation to readout)}\\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)}\\
[O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:0912.2823
- arXiv:1604.06537
- arXiv:1708.03640
- arXiv:1801.03283

15.7 Wave interference

Central Claim

A wave interference can be read as a quantum construction: the measurement basis and experimental arrangement fixes the admissible state space; the self-adjoint observable being asked of that state defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a wave interference is described by a prepared quantum state before the measurement. The physical question is represented by the self-adjoint observable being asked of that state; the experimental or mathematical setting is the measurement basis and experimental arrangement. The observable content is obtained from the observable's spectral projectors and the Born probabilities assigned to them. In the local terminology of this topic, the same construction appears through superposition or wave function, observable operator or Hamiltonian, and eigenvalue or energy level. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its mathematical presentation emphasizes local

Quantum Mechanism Frame

- **Role**. Wave interference contributes a topic-native constructor role to the quantum construction.
- **Placement**. This page is read first as a realization move: it changes the domain, boundary, geometry, or interface in which the operator acts.
- **Carrier or domain**. State terms: superposition. Context/domain terms: preparation condition, measurement setup, or potential or domain.
- **Operator or map**. Operator terms: observable. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$
\begin{aligned}
B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)} \\
\rho_t = U_t \rho_B U_t^\dagger \quad &\text{(unitary evolution from preparation to readout)} \\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)} \\
[O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:1506.05598
- arXiv:1801.03283
- arXiv:1612.00682
- arXiv:0908.0752

15.8 Quantum optics

Central Claim

Quantum optics can be read as a quantum construction: the chosen basis, pulse sequence, or measurement axis fixes the admissible state space; a Hamiltonian or unitary matrix rotating that state between preparation and measurement defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, quantum optics is described by a two-dimensional Hilbert space, usually written as a qubit state or a density matrix. The physical question is represented by a Hamiltonian or unitary matrix rotating that state between preparation and measurement; the experimental or mathematical setting is the chosen basis, pulse sequence, or measurement axis. The observable content is obtained from projectors onto the two eigenstates of the measured observable. In the local terminology of this topic, the same construction appears through wavefunction or wave function, Hamiltonian or observable operator, and spectrum or eigenvalue. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility;

Quantum Mechanism Frame

- **Role**. Quantum optics contributes a boundary-shaped spectrum role to the quantum construction.
- **Placement**. This page is read first as a realization move: it changes the domain, boundary, geometry, or interface in which the operator acts.

- **Carrier or domain**. State terms: wavefunction. Context/domain terms: preparation condition, measurement setup, or potential or domain.
- **Operator or map**. Operator terms: Hamiltonian, observable operator, or generator. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: spectrum. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:0912.2823
- arXiv:0805.4565
- arXiv:1708.03640
- arXiv:quant-ph0205159

15.9 Spectral line

Central Claim

A spectral line can be read as a quantum construction: the measurement basis and experimental arrangement fixes the admissible state space; the self-adjoint observable being asked of that state defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a spectral line is described by a prepared quantum state before the measurement. The physical question is represented by the self-adjoint observable being asked of that state; the experimental or mathematical setting is the measurement basis and experimental arrangement. The observable content is obtained from the observable's spectral projectors and the Born probabilities assigned to them. In the local terminology of this topic, the same construction appears through quantum state or wave function, Hamiltonian or observable operator, and spectrum or eigenvalue. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through uncertainty relation or commutator. The linked equation set is concentrated in operator-to-spectrum readout, preparation, basis, or boundary context, state evolution; its mathematical presentation emphasizes local

Quantum Mechanism Frame

- **Role**. Spectral line contributes a boundary-shaped spectrum role to the quantum construction.
- **Placement**. This page is read first as a realization move: it changes the domain, boundary, geometry, or interface in which the operator acts.
- **Carrier or domain**. State terms: quantum state, wave function, or density operator. Context/domain terms: potential.
- **Operator or map**. Operator terms: Hamiltonian, observable operator, or generator. Protocol or update terms: projection.
- **Admissibility**. Compatibility or closure terms: uncertainty. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: spectrum. These name the outcome labels, projectors, amplitudes, or records used for testing.

- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:0912.2823
- arXiv:1604.06537
- arXiv:1604.05385
- arXiv:1801.03283
- arXiv:quant-ph0205159

15.10 S-matrix

Central Claim

The s-matrix can be read as a quantum construction: the potential, domain, initial condition, or boundary condition fixes the admissible state space; the Hamiltonian, whose exponential gives unitary time evolution defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, the S-matrix is described by a wave function or density operator defined on the Hilbert space allowed by the system's domain. The physical question is represented by the Hamiltonian, whose exponential gives unitary time evolution; the experimental or mathematical setting is the potential, domain, initial condition, or boundary condition. The observable content is obtained from the eigenvalues and eigenfunctions of the relevant observable. In the local terminology of this topic, the same construction appears through wave function or probability amplitude, matrix or Hamiltonian, and eigenvalue or energy level. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its

Quantum Mechanism Frame

- **Role**. S-matrix contributes a boundary-shaped spectrum role to the quantum construction.
- **Placement**. This page is read first as a realization move: it changes the domain, boundary, geometry, or interface in which the operator acts.
- **Carrier or domain**. State terms: wave function and probability amplitude. Context/-domain terms: potential, context, or basis.
- **Operator or map**. Operator terms: matrix, Hamiltonian, or unitary. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:0912.2823
- arXiv:1604.06537
- arXiv:1801.03283
- arXiv:1612.00682
- arXiv:1506.05598

15.11 Quantum tunnelling

Central Claim

Quantum tunnelling is a boundary-realization constructor: a state has nonzero transmission through a classically forbidden region.

Formal Role

Tunnelling shows that the realization layer matters. A potential barrier changes the admissible wave solutions and produces transmission even where classical kinetic energy would be negative. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its mathematical presentation emphasizes local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum tunnelling contributes a boundary-shaped spectrum role to the quantum construction.
- **Placement**. This page is read first as a realization move: it changes the domain, boundary, geometry, or interface in which the operator acts.
- **Carrier or domain**. A Hilbert space with a selected domain, potential, interface, asymptotic channel, cavity, well, or boundary condition.
- **Operator or map**. A Hamiltonian, wave operator, transfer operator, or scattering map whose domain is changed by the boundary.
- **Admissibility**. Boundary conditions and matching conditions determine allowed states, resonances, transmission amplitudes, and spectra.
- **Readout**. Eigenvalues, resonances, tunnelling probabilities, phase shifts, reflection/transmission amplitudes, or scattering data.
- **Check**. Changing the boundary should change the spectrum or amplitude in the predicted way, while the free or asymptotic limit is recovered when the boundary is removed.

Topic Equations

Standard constructor skeleton: barrier-domain Schrödinger equation and WKB transmission.

$$T \sim \exp\left(-2\int_{x_1}^{x_2}\sqrt{\frac{2m(V(x)-E)}{\hbar^2}}\,dx\right)$$

$$-\frac{\hbar^2}{2m}\psi''(x)+V(x)\psi(x)=E\psi(x)$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:0912.2823
- arXiv:1612.00682
- arXiv:quant-ph0205159
- arXiv:1801.03283

15.12 Quantum imaging

Central Claim

A quantum imaging can be read as a quantum construction: the measurement basis and experimental arrangement fixes the admissible state space; the self-adjoint observable being asked of that state defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a quantum imaging is described by a prepared quantum state before the measurement. The physical question is represented by the self-adjoint observable being asked of that state; the experimental or mathematical setting is the measurement basis and experimental arrangement. The observable content is obtained from the observable's spectral projectors and the Born probabilities assigned to them. In the local terminology of this topic, the same construction appears through quantum state or wave function, Hamiltonian or observable operator, and mode or eigenvalue. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in preparation, basis, or boundary context, operator-to-spectrum readout, state evolution; its mathematical presentation emphasizes local

Quantum Mechanism Frame

- **Role**. Quantum imaging contributes an instrument-mediated readout role to the quantum construction.
- **Placement**. This page is read first as a realization move: it changes the domain, boundary, geometry, or interface in which the operator acts.
- **Carrier or domain**. State terms: quantum state, wave function, or density operator. Context/domain terms: potential and detector.
- **Operator or map**. Operator terms: Hamiltonian, observable operator, or generator. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: mode. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$
\begin{aligned}
B \longmapsto \rho_B \quad & \text{(context specifies an admissible state)} \\
\rho_t = U_t \rho_B U_t^\dagger \quad & \text{(unitary evolution from preparation to readout)} \\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad & \text{(spectral probability measure)} \\
[O_1, O_2] \neq 0 \quad & \text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:0912.2823
- arXiv:1604.05385
- arXiv:1612.00682
- arXiv:quant-ph0205159
- arXiv:gr-qc0411110

15.13 Quantum harmonic oscillator

Central Claim

A quantum harmonic oscillator can be read as a quantum construction: the potential, domain, initial condition, or boundary condition fixes the admissible state space; the Hamiltonian, whose exponential gives unitary time evolution defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a quantum harmonic oscillator is described by a wave function or density operator defined on the Hilbert space allowed by the system's domain. The physical question is represented by the Hamiltonian, whose exponential gives unitary time evolution; the experimental or mathematical setting is the potential, domain, initial condition, or boundary condition. The observable content is obtained from the eigenvalues and eigenfunctions of the relevant observable. In the local terminology of this topic, the same construction appears through wave function or wavefunction, Hamiltonian or observable operator, and energy level or eigenvalue. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through uncertainty relation or commutator. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or

Quantum Mechanism Frame

- **Role**. Quantum harmonic oscillator contributes a topic-native constructor role to the quantum construction.
- **Placement**. This page is read first as a realization move: it changes the domain, boundary, geometry, or interface in which the operator acts.

- **Carrier or domain**. State terms: wave function and wavefunction. Context/domain terms: potential and basis.
- **Operator or map**. Operator terms: Hamiltonian. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: uncertainty. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: energy level, eigenvalue, eigenstate, or spectrum. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$
\begin{aligned}
B &\longmapsto \rho_B \quad \text{(context specifies an admissible state)} \\
\rho_t = U_t \rho_B U_t^\dagger &\quad \text{(unitary evolution from preparation to readout)} \\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) &\quad \text{(spectral probability measure)} \\
[O_1, O_2] \neq 0 &\quad \text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.06537
- arXiv:1801.03283
- arXiv:1612.00682
- arXiv:0912.2823
- arXiv:1506.05598

15.14 Macroscopic quantum phenomena

Central Claim

A macroscopic quantum phenomena can be read as a quantum construction: the potential, domain, initial condition, or boundary condition fixes the admissible state space; the Hamiltonian, whose exponential gives unitary time evolution defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a macroscopic quantum phenomena is described by a wave function or density operator defined on the Hilbert space allowed by the system's domain. The physical question is represented by the Hamiltonian, whose exponential gives unitary time evolution; the experimental or mathematical setting is the potential, domain, initial condition, or boundary condition. The observable content is obtained from the eigenvalues and eigenfunctions of the relevant observable. In the local terminology of this topic, the same construction appears through wave function or state vector, Hamiltonian or observable operator, and eigenvalue or energy level. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in state evolution, preparation, basis, or boundary context,

Quantum Mechanism Frame

- **Role**. Macroscopic quantum phenomena contributes an open-system transport and coherence role to the quantum construction.
- **Placement**. This page is read first as a realization move: it changes the domain, boundary, geometry, or interface in which the operator acts.
- **Carrier or domain**. State terms: wave function. Context/domain terms: potential.
- **Operator or map**. Operator terms: Hamiltonian, observable operator, or generator. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.

- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:0912.2823
- arXiv:1706.03846
- arXiv:hep-lat9608080
- arXiv:1708.03640
- arXiv:astro-ph0604157

15.15 Quantum metamaterial

Central Claim

A quantum metamaterial can be read as a quantum construction: the chosen basis, pulse sequence, or measurement axis fixes the admissible state space; a Hamiltonian or unitary matrix rotating that state between preparation and measurement defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a quantum metamaterial is described by a two-dimensional Hilbert space, usually written as a qubit state or a density matrix. The physical question is represented by a Hamiltonian or unitary matrix rotating that state between preparation and measurement; the experimental or mathematical setting is the chosen basis, pulse sequence, or measurement axis. The observable content is obtained from projectors onto the two eigenstates of the measured observable. In the local terminology of this topic, the same construction appears through quantum state or wave function, Hamiltonian or observable operator, and eigenvalue or energy level. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in state evolution, operator-to-spectrum readout, normalization or

Quantum Mechanism Frame

- **Role**. Quantum metamaterial contributes a boundary-shaped spectrum role to the quantum construction.
- **Placement**. This page is read first as a realization move: it changes the domain, boundary, geometry, or interface in which the operator acts.
- **Carrier or domain**. State terms: quantum state. Context/domain terms: preparation condition, measurement setup, or potential or domain.
- **Operator or map**. Operator terms: Hamiltonian, observable operator, or generator. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1706.03846
- arXiv:1708.03640
- arXiv:quant-ph0312187
- arXiv:0908.0752
- arXiv:0805.4565

Chapter 16

Many-mode extension: fields, particles, and scaling

Quantum field theory, gauge theory, renormalization, photons, fermions, and related topics extend the same state-operator-spectrum logic to variable particle number, local fields, and scale-dependent descriptions.

16.1 Why This Branch Exists

Field theory and gauge theory extend the same construction to many modes, local generators, and scale-dependent descriptions.

16.2 Page Map

Page
Fermi–Dirac statistics
Dirac equation
Quantum electrodynamics
Renormalization
Gauge theory
Photon
Quantum field theory
Fermion
Boson
Quantum chromodynamics
AdS/CFT correspondence
Klein–Gordon equation
Quantum gravity
Spin network
Electron
Loop quantum gravity
Spin foam
String theory
Quantum geometry
Fock space

Page
Quantum spacetime

16.3 Mechanism Pages

Each page below is read as a specialization of the compact quantum constructor. Topic-specific pages use explicit equations or overrides; core-derived pages use the branch-level equation form associated with their mechanism role.

16.4 Fermi–Dirac statistics

Central Claim

A fermi–dirac statistics can be read as a quantum construction: the chosen basis, pulse sequence, or measurement axis fixes the admissible state space; a Hamiltonian or unitary matrix rotating that state between preparation and measurement defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a fermi–Dirac statistics is described by a two-dimensional Hilbert space, usually written as a qubit state or a density matrix. The physical question is represented by a Hamiltonian or unitary matrix rotating that state between preparation and measurement; the experimental or mathematical setting is the chosen basis, pulse sequence, or measurement axis. The observable content is obtained from projectors onto the two eigenstates of the measured observable. In the local terminology of this topic, the same construction appears through quantum state or wave function, Hamiltonian or observable operator, and eigenvalue or energy level. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in state evolution, preparation, basis, or boundary context,

Quantum Mechanism Frame

- **Role**. Fermi–Dirac statistics contributes a lawful state-transport role to the quantum construction.
- **Placement**. This page is read first as a many-mode or field-realization move: it extends the state and operator construction beyond a single-particle carrier.
- **Carrier or domain**. State terms: quantum state, wave function, or density operator. Context/domain terms: potential.
- **Operator or map**. Operator terms: Hamiltonian, observable operator, or generator. Protocol or update terms: collapse.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.

- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$

$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$

$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$

$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:0912.2823
- arXiv:1604.05385
- arXiv:1801.03283
- arXiv:1612.00682
- arXiv:quant-ph0205159

16.5 Dirac equation

Central Claim

The dirac equation can be read as a quantum construction: the chosen basis, pulse sequence, or measurement axis fixes the admissible state space; a Hamiltonian or unitary matrix rotating that state between preparation and measurement defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, the Dirac equation is described by a two-dimensional Hilbert space, usually written as a qubit state or a density matrix. The physical question is represented by a Hamiltonian or unitary matrix rotating that state between preparation and measurement; the experimental or mathematical setting is the chosen basis, pulse sequence, or measurement axis. The observable content is obtained from projectors onto the two eigenstates of the measured observable. In the local terminology of this topic, the same construction appears through wavefunction or wave function, matrix or Hamiltonian, and spectrum or eigenvalue. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its

Quantum Mechanism Frame

- **Role**. Dirac equation contributes a lawful state-transport role to the quantum construction.
- **Placement**. This page is read first as a many-mode or field-realization move: it extends the state and operator construction beyond a single-particle carrier.
- **Carrier or domain**. State terms: wavefunction and wave function. Context/domain terms: context.
- **Operator or map**. Operator terms: matrix and Hamiltonian. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: spectrum. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:0912.2823
- arXiv:1604.06537
- arXiv:1801.03283
- arXiv:quant-ph0205159
- arXiv:1612.00682

16.6 Quantum electrodynamics

Central Claim

Quantum electrodynamics can be read as a quantum construction: the chosen basis, pulse sequence, or measurement axis fixes the admissible state space; a Hamiltonian or unitary matrix rotating that state between preparation and measurement defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, quantum electrodynamics is described by a two-dimensional Hilbert space, usually written as a qubit state or a density matrix. The physical question is represented by a Hamiltonian or unitary matrix rotating that state between preparation and measurement; the experimental or mathematical setting is the chosen basis, pulse sequence, or measurement axis. The observable content is obtained from projectors onto the two eigenstates of the measured observable. In the local terminology of this topic, the same construction appears through probability amplitude or wave function, observable operator or Hamiltonian, and eigenvalue or energy level. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through incompatible observables or commutator. The linked equation set is concentrated in operator-to-spectrum readout, state evolution,

Quantum Mechanism Frame

- **Role**. Quantum electrodynamics contributes a lawful state-transport role to the quantum construction.
- **Placement**. This page is read first as a many-mode or field-realization move: it extends the state and operator construction beyond a single-particle carrier.
- **Carrier or domain**. State terms: probability amplitude. Context/domain terms: experimental setup.
- **Operator or map**. Operator terms: observable. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: incompatible. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:0912.2823
- arXiv:1604.06537
- arXiv:2111.12617
- arXiv:1801.03283
- arXiv:2501.07524

16.7 Renormalization

Central Claim

Renormalization is the scale-flow constructor: the effective parameters of the theory change with resolution while predictions remain controlled.

Formal Role

Renormalization explains why a mechanism can preserve its operator role while changing its apparent parameters across scales. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its mathematical presentation emphasizes local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Renormalization contributes an unresolved constructor role to the quantum construction.
- **Placement**. This page is read first as a many-mode or field-realization move: it extends the state and operator construction beyond a single-particle carrier.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.
- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

Standard constructor skeleton: beta flow and effective operator expansion.

$$\mu \frac{dg}{d\mu} = \beta(g)$$

$$g = g(\mu)$$

$$\mathcal{L}_{\text{eff}}(\mu) = \sum_i c_i(\mu)\mathcal{O}_i$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:0912.2823
- arXiv:1604.06537
- arXiv:1612.00682
- arXiv:1801.03283
- arXiv:2111.12617

16.8 Gauge theory

Central Claim

Gauge theory is a redundancy-and-constraint constructor: different local presentations can represent the same physical state.

Formal Role

Gauge theory belongs at the field/geometry interface. It separates physical degrees of freedom from representational choices and imposes covariant transport through a connection. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its mathematical presentation emphasizes local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Gauge theory contributes an unresolved constructor role to the quantum construction.
- **Placement**. This page is read first as a many-mode or field-realization move: it extends the state and operator construction beyond a single-particle carrier.
- **Carrier or domain**. A context-selected state space or effective carrier for prediction.

- **Operator or map**. The relevant Hamiltonian, observable, channel, constraint, or update map.
- **Admissibility**. Domain, normalization, positivity, compatibility, boundary, or gauge requirements state what is legal.
- **Readout**. The outcome probabilities, spectra, correlations, amplitudes, or records used to test the mechanism.
- **Check**. A complete account must specify state carrier, operator or map, admissibility condition, readout, and at least one possible falsifier.

Topic Equations

Standard constructor skeleton: covariant derivative, curvature, and local gauge transformation.

$$\begin{aligned} D_\mu &= \partial_\mu + igA_\mu \\ F_{\mu\nu} &= \partial_\mu A_\nu - \partial_\nu A_\mu + ig[A_\mu, A_\nu] \\ \psi(x) &\mapsto U(x)\psi(x) \end{aligned}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:0912.2823
- arXiv:1604.05385
- arXiv:1612.00682
- arXiv:1801.03283
- arXiv:quant-ph0205159

16.9 Photon

Central Claim

Photon is a field-mode constructor: a one-quantum excitation of the electromagnetic field, constrained by massless dispersion and transverse polarization.

Formal Role

The native photon mechanism is field quantization. The electromagnetic field is decomposed into modes, creation and annihilation operators act on those modes, and a one-photon state is created from the vacuum. The relevant readouts are occupation number, energy, momentum, polarization, and detector clicks; the constraints are dispersion and gauge-compatible transversality. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its mathematical presentation emphasizes local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Photon contributes a many-mode field or particle-realization role to the quantum construction.
- **Placement**. This page is read first as a many-mode or field-realization move: it extends the state and operator construction beyond a single-particle carrier.
- **Carrier or domain**. Fock space, field configuration space, or a sector selected by charge, spin, momentum, statistics, or gauge data.
- **Operator or map**. Creation, annihilation, field, charge, spin, Hamiltonian, or scattering operators acting on the admissible sector.
- **Admissibility**. Statistics, gauge constraints, commutation or anticommutation rules, domain conditions, and sector labels decide which states are legal.
- **Readout**. Occupation number, charge, spin, momentum, energy, correlation function, cross-section, or scattering amplitude.
- **Check**. The field description must preserve the relevant observables under changes of representation and reduce to the expected particle or quasiparticle limit when that limit exists.

Topic Equations

Topic-specific constructor: the equations express massless dispersion, one-mode occupation, number readout, and transverse polarization.

$$E = \hbar\omega, \qquad \mathbf{p} = \hbar\mathbf{k}, \qquad \omega = c|\mathbf{k}|$$
$$|1_{\mathbf{k},\lambda}\rangle = a^\dagger_{\mathbf{k},\lambda}|0\rangle$$
$$\hat{N}_{\mathbf{k},\lambda} = a^\dagger_{\mathbf{k},\lambda}a_{\mathbf{k},\lambda}$$
$$\mathbf{k}\cdot\boldsymbol{\epsilon}_{\mathbf{k},\lambda} = 0$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:0912.2823
- arXiv:1612.00682
- arXiv:1801.03283
- arXiv:quant-ph0205159

16.10 Quantum field theory

Central Claim

Quantum field theory is the many-mode local-field extension of the quantum constructor.

Formal Role

Quantum field theory lifts the state-operator-spectrum construction to fields, local operators, creation and annihilation modes, and scattering amplitudes. Particles become stable excitation/readout roles of fields. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its mathematical presentation emphasizes local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum field theory contributes a many-mode field or particle-realization role to the quantum construction.
- **Placement**. This page is read first as a many-mode or field-realization move: it extends the state and operator construction beyond a single-particle carrier.
- **Carrier or domain**. Fock space, field configuration space, or a sector selected by charge, spin, momentum, statistics, or gauge data.
- **Operator or map**. Creation, annihilation, field, charge, spin, Hamiltonian, or scattering operators acting on the admissible sector.
- **Admissibility**. Statistics, gauge constraints, commutation or anticommutation rules, domain conditions, and sector labels decide which states are legal.
- **Readout**. Occupation number, charge, spin, momentum, energy, correlation function, cross-section, or scattering amplitude.
- **Check**. The field description must preserve the relevant observables under changes of representation and reduce to the expected particle or quasiparticle limit when that limit exists.

Topic Equations

Standard constructor skeleton: field expansion, mode algebra, and correlation readout.

$$\Phi(x) = \sum_k \left(a_k u_k(x) + a_k^\dagger u_k^*(x) \right)$$

$$[a_k, a_l^\dagger]_\mp = \delta_{kl}$$

$$\langle 0 | T\{\Phi(x)\Phi(y)\} | 0 \rangle$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:0912.2823
- arXiv:1604.06537
- arXiv:1708.03640
- arXiv:quant-ph0205159
- arXiv:1801.03283

16.11 Fermion

Central Claim

Fermion is an exchange-symmetry constructor: identical fermions live in antisymmetric sectors and obey Pauli exclusion.

Formal Role

The mechanism is an admissibility rule on many-body state space. Antisymmetry, anticommutation, and zero-or-one mode occupation define the portable role. The linked equation set is concentrated in state evolution, normalization or admissibility, preparation, basis, or boundary context; its mathematical presentation emphasizes local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Fermion contributes a many-mode field or particle-realization role to the quantum construction.
- **Placement**. This page is read first as a many-mode or field-realization move: it extends the state and operator construction beyond a single-particle carrier.
- **Carrier or domain**. Fock space, field configuration space, or a sector selected by charge, spin, momentum, statistics, or gauge data.

- **Operator or map**. Creation, annihilation, field, charge, spin, Hamiltonian, or scattering operators acting on the admissible sector.
- **Admissibility**. Statistics, gauge constraints, commutation or anticommutation rules, domain conditions, and sector labels decide which states are legal.
- **Readout**. Occupation number, charge, spin, momentum, energy, correlation function, cross-section, or scattering amplitude.
- **Check**. The field description must preserve the relevant observables under changes of representation and reduce to the expected particle or quasiparticle limit when that limit exists.

Topic Equations

Topic-specific constructor: the equations express antisymmetric sectors, anticommutation, and occupation restriction.

$$\mathcal{F}_-(\mathcal{H}) = \bigoplus_{n=0}^{\infty} \wedge^n \mathcal{H}$$
$$\{a_i, a_j^\dagger\} = \delta_{ij}, \qquad \{a_i, a_j\} = 0$$
$$n_i = a_i^\dagger a_i \in \{0, 1\}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1706.03846
- arXiv:0912.2823
- arXiv:1708.03640
- arXiv:0908.0752
- arXiv:astro-ph0604157

16.12 Boson

Central Claim

Boson is an exchange-symmetry constructor: identical bosons live in symmetric sectors and can share the same mode.

Formal Role

The mechanism is symmetric exchange. Commuting creation and annihilation operators allow arbitrary nonnegative occupation of a mode, which supports field modes, coherent states, condensates, and photon-like readouts. The linked equation set is concentrated in normalization or admissibility, state evolution, operator-to-spectrum readout; its mathematical presentation emphasizes local notation, information profile, spectral profile.

Quantum Mechanism Frame

- **Role**. Boson contributes a many-mode field or particle-realization role to the quantum construction.
- **Placement**. This page is read first as a many-mode or field-realization move: it extends the state and operator construction beyond a single-particle carrier.
- **Carrier or domain**. Fock space, field configuration space, or a sector selected by charge, spin, momentum, statistics, or gauge data.
- **Operator or map**. Creation, annihilation, field, charge, spin, Hamiltonian, or scattering operators acting on the admissible sector.
- **Admissibility**. Statistics, gauge constraints, commutation or anticommutation rules, domain conditions, and sector labels decide which states are legal.
- **Readout**. Occupation number, charge, spin, momentum, energy, correlation function, cross-section, or scattering amplitude.
- **Check**. The field description must preserve the relevant observables under changes of representation and reduce to the expected particle or quasiparticle limit when that limit exists.

Topic Equations

Topic-specific constructor: the equations express symmetric sectors, commutation, and unrestricted mode occupation.

$$\mathcal{F}_+(\mathcal{H}) = \bigoplus_{n=0}^{\infty} \mathrm{Sym}^n\, \mathcal{H}$$
$$[a_i, a_j^\dagger] = \delta_{ij}, \qquad [a_i, a_j] = 0$$
$$n_i = a_i^\dagger a_i \in \{0, 1, 2, \ldots\}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1706.03846
- arXiv:1708.03640
- arXiv:1801.08949
- arXiv:0908.0752
- arXiv:1801.03283

16.13 Quantum chromodynamics

Central Claim

Quantum chromodynamics can be read as a quantum construction: the chosen basis, pulse sequence, or measurement axis fixes the admissible state space; a Hamiltonian or unitary matrix rotating that state between preparation and measurement defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, quantum chromodynamics is described by a two-dimensional Hilbert space, usually written as a qubit state or a density matrix. The physical question is represented by a Hamiltonian or unitary matrix rotating that state between preparation and measurement; the experimental or mathematical setting is the chosen basis, pulse sequence, or measurement axis. The observable content is obtained from projectors onto the two eigenstates of the measured observable. In the local terminology of this topic, the same construction appears through quantum state or wave function, matrix or Hamiltonian, and eigenvalue or energy level. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility;

Quantum Mechanism Frame

- **Role**. Quantum chromodynamics contributes a lawful state-transport role to the quantum construction.
- **Placement**. This page is read first as a many-mode or field-realization move: it extends the state and operator construction beyond a single-particle carrier.
- **Carrier or domain**. State terms: quantum state, wave function, or density operator. Context/domain terms: context and basis.
- **Operator or map**. Operator terms: matrix. Protocol or update terms: projection.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$

$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$

$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$

$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:0912.2823
- arXiv:1604.05385
- arXiv:1604.06537
- arXiv:1801.03283
- arXiv:quant-ph0205159

16.14 AdS/CFT correspondence

Central Claim

AdS/CFT correspondence is a geometry-translation constructor: bulk gravitational data and boundary field data are treated as dual presentations of one operator structure.

Formal Role

AdS/CFT belongs at the interface where geometry becomes a representation of the quantum construction rather than the invariant root. The practical content is a dictionary between bulk fields and boundary operators. The linked equation set is concentrated in state evolution, operator-to-spectrum readout, preparation, basis, or boundary context; its mathematical presentation emphasizes local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. AdS/CFT correspondence contributes a geometry or holographic realization role to the quantum construction.
- **Placement**. This page is read first as a many-mode or field-realization move: it extends the state and operator construction beyond a single-particle carrier.
- **Carrier or domain**. A spacetime, boundary algebra, gauge orbit, spin network, bulk/boundary pair, or geometric representation of a quantum state space.
- **Operator or map**. Hamiltonian, action, constraint, boundary operator, correlation map, or dictionary between two representations.
- **Admissibility**. Gauge, boundary, metric, covariance, and constraint conditions decide which geometric descriptions represent the same physical content.
- **Readout**. Boundary correlators, spectra, entropies, scattering data, geometric invariants, or reconstructed bulk quantities.
- **Check**. A geometric reformulation is physical only to the extent that it preserves observables or correlation functions across the representation change.

Topic Equations

Standard constructor skeleton: boundary-source/bulk-field dictionary.

$$Z_{\mathrm{grav}}[\phi_0] \simeq Z_{\mathrm{CFT}}[J = \phi_0]$$

$$\left\langle \exp \int J\mathcal{O} \right\rangle_{\mathrm{CFT}} = Z_{\mathrm{bulk}}[\phi|_{\partial} = J]$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:0912.2823
- arXiv:quant-ph0205159
- arXiv:1801.03283
- arXiv:1708.03640
- arXiv:1506.05598

16.15 Klein–Gordon equation

Central Claim

The klein–gordon equation can be read as a quantum construction: the chosen basis, pulse sequence, or measurement axis fixes the admissible state space; a Hamiltonian or unitary matrix rotating that state between preparation and measurement defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, the Klein–Gordon equation is described by a two-dimensional Hilbert space, usually written as a qubit state or a density matrix. The physical question is represented by a Hamiltonian or unitary matrix rotating that state between preparation and measurement; the experimental or mathematical setting is the chosen basis, pulse sequence, or measurement axis. The observable content is obtained from projectors onto the two eigenstates of the measured observable. In the local terminology of this topic, the same construction appears through quantum state or wave function, matrix or Hamiltonian, and spectrum or eigenvalue. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through uncertainty relation or commutator. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, non-commuting compatibility limits;

Quantum Mechanism Frame

- **Role**. Klein–Gordon equation contributes a lawful state-transport role to the quantum construction.
- **Placement**. This page is read first as a many-mode or field-realization move: it extends the state and operator construction beyond a single-particle carrier.
- **Carrier or domain**. State terms: quantum state, wave function, or density operator. Context/domain terms: preparation condition, measurement setup, or potential or domain.
- **Operator or map**. Operator terms: matrix. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: uncertainty. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: spectrum. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$\begin{aligned}
B \longmapsto \rho_B \quad & \text{(context specifies an admissible state)} \\
\rho_t = U_t \rho_B U_t^\dagger \quad & \text{(unitary evolution from preparation to readout)} \\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad & \text{(spectral probability measure)} \\
[O_1, O_2] \neq 0 \quad & \text{(incompatible observables: no common sharp basis)}
\end{aligned}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.06537
- arXiv:0908.0752
- arXiv:1506.05598
- arXiv:1708.03640
- arXiv:quant-ph0205159

16.16 Quantum gravity

Central Claim

Quantum gravity asks whether geometry is a background realization, a constrained quantum carrier, or an emergent readout of a deeper quantum state.

Formal Role

The constructor cannot treat quantum gravity as an ordinary wave function on a fixed domain. A candidate theory must specify the state space of geometric and matter degrees of freedom, the constraint or evolution operators, the gauge-invariant or relational observables, and the limit in which classical spacetime is recovered. The linked equation set is concentrated in state evolution, normalization or admissibility, operator-to-spectrum readout; its mathematical presentation emphasizes local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum gravity contributes a geometry or holographic realization role to the quantum construction.
- **Placement**. This page is read first as a many-mode or field-realization move: it extends the state and operator construction beyond a single-particle carrier.
- **Carrier or domain**. A state space for geometric and matter degrees of freedom, or a deeper carrier from which geometry is reconstructed.
- **Operator or map**. Constraint, evolution, or amplitude operators that do not presuppose an unexamined fixed spacetime background.
- **Admissibility**. Gauge and diffeomorphism constraints determine the physical state space and which operators are observable.
- **Readout**. Relational observables, boundary amplitudes, geometric spectra, or semiclassical spacetime data.

- **Check**. The construction must recover controlled classical geometry and reproduce established low-energy quantum field predictions in its domain of validity.

Topic Equations

Schematic constructor shared by constrained approaches: quantum carrier, constraints, physical observables, and semiclassical recovery must all be specified.

$$\Psi \in \mathcal{H}_{\text{geom}\otimes\text{matter}}, \qquad \widehat{\mathcal{C}}_a \Psi = 0$$
$$[\widehat{O}_{\text{phys}}, \widehat{\mathcal{C}}_a]\Psi = 0$$
$$\langle \Psi | \widehat{g}_{\mu\nu} | \Psi \rangle \longrightarrow g^{\text{cl}}_{\mu\nu} \quad \text{in a controlled semiclassical regime}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:0912.2823
- arXiv:1708.03640
- arXiv:1801.03283
- arXiv:quant-ph0205159

16.17 Spin network

Central Claim

A spin network can be read as a quantum construction: the potential, domain, initial condition, or boundary condition fixes the admissible state space; the Hamiltonian, whose exponential gives unitary time evolution defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a spin network is described by a wave function or density operator defined on the Hilbert space allowed by the system's domain. The physical question is represented by the Hamiltonian, whose exponential gives unitary time evolution; the experimental or mathematical setting is the potential, domain, initial condition, or boundary condition. The observable content is obtained from the eigenvalues and eigenfunctions of the relevant observable. In the local terminology of this topic, the same construction appears through quantum state or wave function, matrix or Hamiltonian, and spectrum or eigenstate. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its mathematical

Quantum Mechanism Frame

- **Role**. Spin network contributes an engineered operation-sequence role to the quantum construction.
- **Placement**. This page is read first as a many-mode or field-realization move: it extends the state and operator construction beyond a single-particle carrier.
- **Carrier or domain**. State terms: quantum state. Context/domain terms: context and basis.
- **Operator or map**. Operator terms: matrix and Hamiltonian. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: spectrum, eigenstate, or eigenvalue. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:0912.2823
- arXiv:1801.03283
- arXiv:1612.00682
- arXiv:quant-ph0205159
- arXiv:1506.05598

16.18 Electron

Central Claim

Electron is a charged spinor constructor: its identity is fixed by mass, charge, spin-1/2 representation, fermionic statistics, and electromagnetic coupling.

Formal Role

The electron is not a generic object label in this tree. Its native mechanism combines a spinor state, a Schrödinger/Pauli/Dirac generator depending on regime, conserved charge, and fermionic anticommutation. The readouts are charge, spin, momentum, energy, and scattering response. The linked equation set is concentrated in operator-to-spectrum readout, preparation, basis, or boundary context, state evolution; its mathematical presentation emphasizes local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Electron contributes a many-mode field or particle-realization role to the quantum construction.
- **Placement**. This page is read first as a many-mode or field-realization move: it extends the state and operator construction beyond a single-particle carrier.
- **Carrier or domain**. Fock space, field configuration space, or a sector selected by charge, spin, momentum, statistics, or gauge data.
- **Operator or map**. Creation, annihilation, field, charge, spin, Hamiltonian, or scattering operators acting on the admissible sector.
- **Admissibility**. Statistics, gauge constraints, commutation or anticommutation rules, domain conditions, and sector labels decide which states are legal.
- **Readout**. Occupation number, charge, spin, momentum, energy, correlation function, cross-section, or scattering amplitude.
- **Check**. The field description must preserve the relevant observables under changes of representation and reduce to the expected particle or quasiparticle limit when that limit exists.

Topic Equations

Topic-specific constructor: the equations express relativistic spinor transport, electromagnetic coupling, and fermionic field statistics.

$$(i\hbar\gamma^\mu D_\mu - mc)\psi = 0$$

$$D_\mu = \partial_\mu + \frac{ie}{\hbar c}A_\mu$$

$$\{\psi_\alpha(x), \psi^\dagger_\beta(y)\} = \delta_{\alpha\beta}\delta(x-y)$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation

- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:0912.2823
- arXiv:1604.05385
- arXiv:1612.00682
- arXiv:quant-ph0205159
- arXiv:gr-qc0411110

16.19 Loop quantum gravity

Central Claim

A loop quantum gravity can be read as a quantum construction: the potential, domain, initial condition, or boundary condition fixes the admissible state space; the Hamiltonian, whose exponential gives unitary time evolution defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a loop quantum gravity is described by a wave function or density operator defined on the Hilbert space allowed by the system's domain. The physical question is represented by the Hamiltonian, whose exponential gives unitary time evolution; the experimental or mathematical setting is the potential, domain, initial condition, or boundary condition. The observable content is obtained from the eigenvalues and eigenfunctions of the relevant observable. In the local terminology of this topic, the same construction appears through quantum state or wave function, Hamiltonian or observable operator, and spectrum or eigenvalue. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or

Quantum Mechanism Frame

- **Role**. Loop quantum gravity contributes a geometry or holographic realization role to the quantum construction.
- **Placement**. This page is read first as a many-mode or field-realization move: it extends the state and operator construction beyond a single-particle carrier.
- **Carrier or domain**. State terms: quantum state, wave function, or density operator. Context/domain terms: basis.
- **Operator or map**. Operator terms: Hamiltonian. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: spectrum. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:0912.2823
- arXiv:quant-ph0205159
- arXiv:1612.00682
- arXiv:1801.03283
- arXiv:1506.05598

16.20 Spin foam

Central Claim

A spin foam can be read as a quantum construction: the boundary conditions that define which histories are included fixes the admissible state space; the action functional assigning phases to histories defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a spin foam is described by boundary states or field configurations at the endpoints of a process. The physical question is represented by the action functional assigning phases to histories; the experimental or mathematical setting is the boundary conditions that define which histories are included. The observable content is obtained from transition amplitudes and probabilities obtained by summing over histories. In the local terminology of this topic, the same construction appears through superposition or wave function, Hamiltonian or observable operator, and eigenvalue or energy level. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its mathematical presentation

Quantum Mechanism Frame

- **Role**. Spin foam contributes a geometry or holographic realization role to the quantum construction.
- **Placement**. This page is read first as a many-mode or field-realization move: it extends the state and operator construction beyond a single-particle carrier.
- **Carrier or domain**. State terms: superposition. Context/domain terms: basis.
- **Operator or map**. Operator terms: Hamiltonian, observable operator, or generator. Protocol or update terms: path integral.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:0912.2823
- arXiv:1604.06537
- arXiv:1612.00682
- arXiv:2108.07838
- arXiv:quant-ph0205159

16.21 String theory

Central Claim

String theory can be read as a quantum construction: the measurement basis and experimental arrangement fixes the admissible state space; the self-adjoint observable being asked of that state defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, string theory is described by a prepared quantum state before the measurement. The physical question is represented by the self-adjoint observable being asked of that state; the experimental or mathematical setting is the measurement basis and experimental arrangement. The observable content is obtained from the observable's spectral projectors and the Born probabilities assigned to them. In the local terminology of this topic, the same construction appears through quantum state or wave function, Hamiltonian or observable operator, and spectrum or eigenvalue. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its mathematical presentation emphasizes local notation,

Quantum Mechanism Frame

- **Role**. String theory contributes a geometry or holographic realization role to the quantum construction.
- **Placement**. This page is read first as a many-mode or field-realization move: it extends the state and operator construction beyond a single-particle carrier.

- **Carrier or domain**. State terms: quantum state, wave function, or density operator. Context/domain terms: basis.
- **Operator or map**. Operator terms: Hamiltonian, observable operator, or generator. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: spectrum. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:0912.2823
- arXiv:1612.00682
- arXiv:quant-ph0205159
- arXiv:1506.05598
- arXiv:1801.03283

16.22 Quantum geometry

Central Claim

Quantum geometry is a geometry-realization page: geometric quantities are promoted to quantum observables rather than assumed as a smooth background.

Formal Role

Quantum geometry uses a quantum state of geometry, often represented by graph or spin-network data. The operator-to-spectrum step asks for eigenvalues of geometric observables such as area or volume. This places the page near the geometry/boundary interface rather than inside a generic many-mode field layer. The linked equation set is concentrated in operator-to-spectrum readout, normalization or admissibility, state evolution; its mathematical presentation emphasizes local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum geometry contributes a geometry or holographic realization role to the quantum construction.
- **Placement**. This page is read first as a many-mode or field-realization move: it extends the state and operator construction beyond a single-particle carrier.
- **Carrier or domain**. A spacetime, boundary algebra, gauge orbit, spin network, bulk/boundary pair, or geometric representation of a quantum state space.
- **Operator or map**. Hamiltonian, action, constraint, boundary operator, correlation map, or dictionary between two representations.
- **Admissibility**. Gauge, boundary, metric, covariance, and constraint conditions decide which geometric descriptions represent the same physical content.
- **Readout**. Boundary correlators, spectra, entropies, scattering data, geometric invariants, or reconstructed bulk quantities.
- **Check**. A geometric reformulation is physical only to the extent that it preserves observables or correlation functions across the representation change.

Topic Equations

Topic-specific constructor: the equations express graph-based geometry states and spectral readout of geometric observables.

$$\mathcal{H}_\Gamma = L^2\big(SU(2)^E/SU(2)^V\big)\,, \qquad |\Gamma, j_e, \iota_v\rangle$$
$$\hat{A}(S)|\Gamma, j, \iota\rangle = 8\pi\gamma\ell_P^2 \sum_{e\cap S} \sqrt{j_e(j_e+1)}\,|\Gamma, j, \iota\rangle$$
$$\text{geometry readout :} \quad \hat{G}\,|g_i\rangle = g_i|g_i\rangle$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1708.03640
- arXiv:quant-ph0205159
- arXiv:1801.03283
- arXiv:1612.00682
- arXiv:1506.05598

16.23 Fock space

Central Claim

Fock space is the occupation-number state space: the construction that replaces a fixed-particle Hilbert space by a direct sum over particle number.

Formal Role

Fock space changes the carrier of the quantum state. Instead of describing one system in one Hilbert space, it builds sectors with zero, one, two, and more identical quanta, then imposes the bosonic or fermionic exchange rule. Creation and annihilation operators are the native coordinates of this page because they move the state between occupation sectors. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its mathematical presentation emphasizes local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Fock space contributes a many-mode field or particle-realization role to the quantum construction.
- **Placement**. This page is read first as a many-mode or field-realization move: it extends the state and operator construction beyond a single-particle carrier.
- **Carrier or domain**. Fock space, field configuration space, or a sector selected by charge, spin, momentum, statistics, or gauge data.
- **Operator or map**. Creation, annihilation, field, charge, spin, Hamiltonian, or scattering operators acting on the admissible sector.
- **Admissibility**. Statistics, gauge constraints, commutation or anticommutation rules, domain conditions, and sector labels decide which states are legal.
- **Readout**. Occupation number, charge, spin, momentum, energy, correlation function, cross-section, or scattering amplitude.
- **Check**. The field description must preserve the relevant observables under changes of representation and reduce to the expected particle or quasiparticle limit when that limit exists.

Topic Equations

Topic-specific constructor: the equations express variable particle number, exchange symmetry, and occupation-number readout.

$$\mathcal{F}_\pm(\mathcal{H}) = \bigoplus_{n=0}^{\infty} \mathcal{S}_\pm \mathcal{H}^{\otimes n}$$

$$[a_i, a_j^\dagger]_\mp = \delta_{ij}, \qquad [a_i, a_j]_\mp = 0$$

$$N = \sum_i a_i^\dagger a_i, \qquad N|n_1, n_2, \ldots\rangle = \left(\sum_i n_i\right)|n_1, n_2, \ldots\rangle$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:2308.15676
- arXiv:2108.07838
- arXiv:1612.00682
- arXiv:0809.5271
- arXiv:2105.11733

16.24 Quantum spacetime

Central Claim

A quantum spacetime can be read as a quantum construction: the chosen basis, pulse sequence, or measurement axis fixes the admissible state space; a Hamiltonian or unitary matrix rotating that state between preparation and measurement defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a quantum spacetime is described by a two-dimensional Hilbert space, usually written as a qubit state or a density matrix. The physical question is represented by a Hamiltonian or unitary matrix rotating that state between preparation and measurement; the experimental or mathematical setting is the chosen basis, pulse sequence, or measurement axis. The observable content is obtained from projectors onto the two eigenstates of the measured observable. In the local terminology of this topic, the same construction appears through quantum state or wave function, matrix or Hamiltonian, and eigenvalue or energy level. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through uncertainty relation or commutator. The linked equation set is concentrated in non-commuting compatibility limits, state evolution, normalization or admissibility;

Quantum Mechanism Frame

- **Role**. Quantum spacetime contributes a geometry or holographic realization role to the quantum construction.
- **Placement**. This page is read first as a many-mode or field-realization move: it extends the state and operator construction beyond a single-particle carrier.
- **Carrier or domain**. State terms: quantum state, wave function, or density operator. Context/domain terms: preparation condition, measurement setup, or potential or domain.
- **Operator or map**. Operator terms: matrix. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: uncertainty. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation

- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.06537
- arXiv:1706.03846
- arXiv:1604.05385
- arXiv:2111.12617
- arXiv:1708.03640

Chapter 17

Protocol layer: engineered transformations

Quantum computing, channels, circuits, algorithms, networks, sensors, and error correction turn the same formal machinery into controlled sequences of operations.

17.1 Why This Branch Exists

Quantum information is the engineering layer: the same state-operator-readout machinery becomes a controlled sequence of transformations.

17.2 Page Map

Page
Quantum information science
Quantum network
Quantum algorithm
Quantum error correction
Quantum channel
Quantum logic gate
Quantum neural network
Quantum circuit
Quantum information
Quantum computing
Quantum key distribution
Quantum sensor
Quantum teleportation
Quantum bus
Quantum image processing
Quantum finite automaton
Quantum cryptography
Quantum programming

17.3 Mechanism Pages

Each page below is read as a specialization of the compact quantum constructor. Topic-specific pages use explicit equations or overrides; core-derived pages use the branch-level equation form associated with their mechanism role.

17.4 Quantum information science

Central Claim

Quantum information science can be read as a quantum construction: the chosen basis, pulse sequence, or measurement axis fixes the admissible state space; a Hamiltonian or unitary matrix rotating that state between preparation and measurement defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, quantum information science is described by a two-dimensional Hilbert space, usually written as a qubit state or a density matrix. The physical question is represented by a Hamiltonian or unitary matrix rotating that state between preparation and measurement; the experimental or mathematical setting is the chosen basis, pulse sequence, or measurement axis. The observable content is obtained from projectors onto the two eigenstates of the measured observable. In the local terminology of this topic, the same construction appears through wave function or superposition, Hamiltonian or observable operator, and spectrum or eigenvalue. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, controlled update

Quantum Mechanism Frame

- **Role**. Quantum information science contributes a topic-native constructor role to the quantum construction.
- **Placement**. This page is read first as an operation-sequence move: it specifies an ordered composition of allowed maps.
- **Carrier or domain**. State terms: wave function and superposition. Context/domain terms: preparation condition, measurement setup, or potential or domain.
- **Operator or map**. Operator terms: Hamiltonian, observable operator, or generator. Protocol or update terms: collapse.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: spectrum. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:2308.15676
- arXiv:2105.11733
- arXiv:0809.5271
- arXiv:cond-mat0108470

17.5 Quantum network

Central Claim

A quantum network can be read as a quantum construction: the chosen basis, pulse sequence, or measurement axis fixes the admissible state space; a Hamiltonian or unitary matrix rotating that state between preparation and measurement defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a quantum network is described by a two-dimensional Hilbert space, usually written as a qubit state or a density matrix. The physical question is represented by a Hamiltonian or unitary matrix rotating that state between preparation and measurement; the experimental or mathematical setting is the chosen basis, pulse sequence, or measurement axis. The observable content is obtained from projectors onto the two eigenstates of the measured observable. In the local terminology of this topic, the same construction appears through quantum state or wave function, Hamiltonian or observable operator, and mode or eigenvalue. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility;

Quantum Mechanism Frame

- **Role**. Quantum network contributes an engineered operation-sequence role to the quantum construction.
- **Placement**. This page is read first as an operation-sequence move: it specifies an ordered composition of allowed maps.
- **Carrier or domain**. State terms: quantum state. Context/domain terms: potential.
- **Operator or map**. Operator terms: Hamiltonian, observable operator, or generator. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: mode. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:0912.2823
- arXiv:0805.4565
- arXiv:1708.03640
- arXiv:quant-ph0205159

17.6 Quantum algorithm

Central Claim

A quantum algorithm can be read as a quantum construction: the chosen basis, pulse sequence, or measurement axis fixes the admissible state space; a Hamiltonian or unitary matrix rotating that state between preparation and measurement defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a quantum algorithm is described by a two-dimensional Hilbert space, usually written as a qubit state or a density matrix. The physical question is represented by a Hamiltonian or unitary matrix rotating that state between preparation and measurement; the experimental or mathematical setting is the chosen basis, pulse sequence, or measurement axis. The observable content is obtained from projectors onto the two eigenstates of the measured observable. In the local terminology of this topic, the same construction appears through quantum state or superposition, Hamiltonian or matrix, and eigenvalue or energy level. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its

Quantum Mechanism Frame

- **Role**. Quantum algorithm contributes an engineered operation-sequence role to the quantum construction.
- **Placement**. This page is read first as an operation-sequence move: it specifies an ordered composition of allowed maps.
- **Carrier or domain**. State terms: quantum state and superposition. Context/domain terms: domain.
- **Operator or map**. Operator terms: Hamiltonian, matrix, or unitary. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:0912.2823
- arXiv:1604.05385
- arXiv:1708.03640
- arXiv:quant-ph0205159
- arXiv:1801.03283

17.7 Quantum error correction

Central Claim

A quantum error correction can be read as a quantum construction: the chosen basis, pulse sequence, or measurement axis fixes the admissible state space; a Hamiltonian or unitary matrix rotating that state between preparation and measurement defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a quantum error correction is described by a two-dimensional Hilbert space, usually written as a qubit state or a density matrix. The physical question is represented by a Hamiltonian or unitary matrix rotating that state between preparation and measurement; the experimental or mathematical setting is the chosen basis, pulse sequence, or measurement axis. The observable content is obtained from projectors onto the two eigenstates of the measured observable. In the local terminology of this topic, the same construction appears through superposition or wave function, Hamiltonian or unitary operator, and eigenvalue or energy level. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or

Quantum Mechanism Frame

- **Role**. Quantum error correction contributes an engineered operation-sequence role to the quantum construction.
- **Placement**. This page is read first as an operation-sequence move: it specifies an ordered composition of allowed maps.
- **Carrier or domain**. State terms: superposition. Context/domain terms: preparation.
- **Operator or map**. Operator terms: Hamiltonian and unitary. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$
\begin{aligned}
B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)}\\
\rho_t = U_t \rho_B U_t^\dagger \quad &\text{(unitary evolution from preparation to readout)}\\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)}\\
[O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:1604.06537
- arXiv:0912.2823
- arXiv:1612.00682
- arXiv:quant-ph0205159

17.8 Quantum channel

Central Claim

Quantum channel is the open-system protocol constructor: it maps input states to output states while preserving complete positivity and trace.

Formal Role

A channel is the mechanism for noisy transformations, measurements with forgotten outcomes, and subsystem evolution. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its mathematical presentation emphasizes local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum channel contributes an engineered operation-sequence role to the quantum construction.
- **Placement**. This page is read first as an operation-sequence move: it specifies an ordered composition of allowed maps.
- **Carrier or domain**. Input and output density operators, possibly on different Hilbert spaces or subsystem carriers.
- **Operator or map**. A completely positive trace-preserving map, often represented by Kraus operators or by a Stinespring dilation.
- **Admissibility**. Complete positivity and trace preservation are the legal conditions; non-trace-preserving maps require an explicitly conditioned outcome.
- **Readout**. Output state, final POVM probabilities, fidelity, capacity, error rate, or recovered subsystem statistics.
- **Check**. The channel claim requires a map that stays positive under extension by an untouched reference system and preserves total probability.

Topic Equations

Standard constructor skeleton: completely positive trace-preserving map and readout.

$$\mathcal{E}(\rho) = \sum_a K_a \rho K_a^\dagger$$

$$\sum_a K_a^\dagger K_a = I$$

$$p(y) = \mathrm{Tr}(M_y \mathcal{E}(\rho))$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:1604.06537
- arXiv:1506.05598
- arXiv:0805.4565
- arXiv:1708.03640

17.9 Quantum logic gate

Central Claim

A quantum logic gate can be read as a quantum construction: the chosen basis, pulse sequence, or measurement axis fixes the admissible state space; a Hamiltonian or unitary matrix rotating that state between preparation and measurement defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a quantum logic gate is described by a two-dimensional Hilbert space, usually written as a qubit state or a density matrix. The physical question is represented by a Hamiltonian or unitary matrix rotating that state between preparation and measurement; the experimental or mathematical setting is the chosen basis, pulse sequence, or measurement axis. The observable content is obtained from projectors onto the two eigenstates of the measured observable. In the local terminology of this topic, the same construction appears through quantum state or wave function, matrix or Hamiltonian, and eigenvalue or energy level. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its

Quantum Mechanism Frame

- **Role**. Quantum logic gate contributes an engineered operation-sequence role to the quantum construction.
- **Placement**. This page is read first as an operation-sequence move: it specifies an ordered composition of allowed maps.
- **Carrier or domain**. State terms: quantum state. Context/domain terms: basis.
- **Operator or map**. Operator terms: matrix, Hamiltonian, or unitary. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$
\begin{aligned}
B \longmapsto \rho_B \quad & \text{(context specifies an admissible state)} \\
\rho_t = U_t \rho_B U_t^\dagger \quad & \text{(unitary evolution from preparation to readout)} \\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad & \text{(spectral probability measure)} \\
[O_1, O_2] \neq 0 \quad & \text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:1801.03283
- arXiv:1612.00682
- arXiv:quant-ph0205159
- arXiv:1506.05598

17.10 Quantum neural network

Central Claim

a quantum neural network modifies the interpretation of the probability/readout layer while preserving the formal quantum dynamics.

Formal Role

a quantum neural network acts on the readout layer of the quantum constructor. The formal ingredients remain the state assignment, the operator or measurement being applied, and the Born-rule map from projectors to probabilities. What changes is the status assigned to those ingredients: for this topic, the state or probability is treated through the agent, measurement context, or interpretive stance attached to the formalism. The page should therefore be read as a statement about the interpretation of state, probability, update, or recorded outcome while the Hamiltonian, spectral resolution, and commutator structure remain the formal reference layer. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its mathematical presentation emphasizes local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum neural network contributes an engineered operation-sequence role to the quantum construction.
- **Placement**. This page is read first as an operation-sequence move: it specifies an ordered composition of allowed maps.
- **Carrier or domain**. State terms: superposition. Context/domain terms: basis.
- **Operator or map**. Operator terms: unitary. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.

- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$\begin{aligned}
B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)} \\
\rho_t = U_t \rho_B U_t^{\dagger} \quad &\text{(unitary evolution from preparation to readout)} \\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)} \\
[O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)}
\end{aligned}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:2108.07838
- arXiv:0805.4565
- arXiv:1612.00682
- arXiv:quant-ph0205159

17.11 Quantum circuit

Central Claim

Quantum circuit is the engineered-composition constructor: a finite sequence of admissible maps prepares, transforms, and measures a register.

Formal Role

A circuit is the protocol layer of the same state-operator-readout machinery. Gates are controlled unitary or channel maps; measurement converts final states into output probabilities. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its mathematical presentation emphasizes local notation, information profile, formula structure.

Quantum Mechanism Frame

- **Role**. Quantum circuit contributes an engineered operation-sequence role to the quantum construction.
- **Placement**. This page is read first as an operation-sequence move: it specifies an ordered composition of allowed maps.
- **Carrier or domain**. An input state, register, channel state, error syndrome, key, or controlled experimental configuration.
- **Operator or map**. An ordered sequence of gates, channels, measurements, corrections, encodings, or conditional maps.
- **Admissibility**. Each step must belong to the claimed map class: unitary, completely positive, trace-preserving, projective, conditional, or corrective.
- **Readout**. Output state, key, error rate, fidelity, channel capacity, algorithmic success probability, or sensor estimate.
- **Check**. Changing operation order, inserting classical controls, or replacing a quantum channel should identify which step carries the effect.

Topic Equations

Standard constructor skeleton: composed gates and final measurement.

$$\begin{aligned} \rho_{\text{out}} &= U_m \cdots U_2 U_1 \, \rho_{\text{in}} \, U_1^\dagger U_2^\dagger \cdots U_m^\dagger \\ p(y) &= \mathrm{Tr}(M_y \rho_{\text{out}}) \end{aligned}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:0912.2823
- arXiv:1612.00682
- arXiv:2108.07838
- arXiv:quant-ph0205159

17.12 Quantum information

Central Claim

A quantum information can be read as a quantum construction: the chosen basis, pulse sequence, or measurement axis fixes the admissible state space; a Hamiltonian or unitary matrix rotating that state between preparation and measurement defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a quantum information is described by a two-dimensional Hilbert space, usually written as a qubit state or a density matrix. The physical question is represented by a Hamiltonian or unitary matrix rotating that state between preparation and measurement; the experimental or mathematical setting is the chosen basis, pulse sequence, or measurement axis. The observable content is obtained from projectors onto the two eigenstates of the measured observable. In the local terminology of this topic, the same construction appears through quantum state or density matrix, observable operator or matrix, and eigenstate or eigenvalue. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through uncertainty relation or non-commuting observables. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization

Quantum Mechanism Frame

- **Role**. Quantum information contributes a topic-native constructor role to the quantum construction.
- **Placement**. This page is read first as an operation-sequence move: it specifies an ordered composition of allowed maps.
- **Carrier or domain**. State terms: quantum state and density matrix. Context/domain terms: preparation and basis.
- **Operator or map**. Operator terms: observable, matrix, or unitary. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: uncertainty and non-commuting. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenstate and eigenvalue. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation

- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.06537
- arXiv:0912.2823
- arXiv:1604.05385
- arXiv:1612.00682
- arXiv:1801.03283

17.13 Quantum computing

Central Claim

Quantum computing can be read as a quantum construction: the chosen basis, pulse sequence, or measurement axis fixes the admissible state space; a Hamiltonian or unitary matrix rotating that state between preparation and measurement defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, quantum computing is described by a two-dimensional Hilbert space, usually written as a qubit state or a density matrix. The physical question is represented by a Hamiltonian or unitary matrix rotating that state between preparation and measurement; the experimental or mathematical setting is the chosen basis, pulse sequence, or measurement axis. The observable content is obtained from projectors onto the two eigenstates of the measured observable. In the local terminology of this topic, the same construction appears through quantum state or superposition, Hamiltonian or observable operator, and eigenvalue or energy level. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, controlled update

Quantum Mechanism Frame

- **Role**. Quantum computing contributes an engineered operation-sequence role to the quantum construction.
- **Placement**. This page is read first as an operation-sequence move: it specifies an ordered composition of allowed maps.
- **Carrier or domain**. State terms: quantum state and superposition. Context/domain terms: basis.
- **Operator or map**. Operator terms: Hamiltonian, observable operator, or generator. Protocol or update terms: Born rule.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:1801.03283
- arXiv:quant-ph0205159
- arXiv:1708.03640
- arXiv:1612.00682

17.14 Quantum key distribution

Central Claim

A quantum key distribution can be read as a quantum construction: the chosen basis, pulse sequence, or measurement axis fixes the admissible state space; a Hamiltonian or unitary matrix rotating that state between preparation and measurement defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a quantum key distribution is described by a two-dimensional Hilbert space, usually written as a qubit state or a density matrix. The physical question is represented by a Hamiltonian or unitary matrix rotating that state between preparation and measurement; the experimental or mathematical setting is the chosen basis, pulse sequence, or measurement axis. The observable content is obtained from projectors onto the two eigenstates of the measured observable. In the local terminology of this topic, the same construction appears through quantum state or wave function, Hamiltonian or observable operator, and Mode or eigenstate. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or

Quantum Mechanism Frame

- **Role**. Quantum key distribution contributes a topic-native constructor role to the quantum construction.
- **Placement**. This page is read first as an operation-sequence move: it specifies an ordered composition of allowed maps.
- **Carrier or domain**. State terms: quantum state. Context/domain terms: basis.
- **Operator or map**. Operator terms: Hamiltonian, observable operator, or generator. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: Mode and eigenstate. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$
\begin{aligned}
B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)} \\
\rho_t = U_t \rho_B U_t^\dagger \quad &\text{(unitary evolution from preparation to readout)} \\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)} \\
[O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:1612.00682
- arXiv:quant-ph0205159
- arXiv:1801.03283
- arXiv:1506.05598

17.15 Quantum sensor

Central Claim

A quantum sensor can be read as a quantum construction: the chosen basis, pulse sequence, or measurement axis fixes the admissible state space; a Hamiltonian or unitary matrix rotating that state between preparation and measurement defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a quantum sensor is described by a two-dimensional Hilbert space, usually written as a qubit state or a density matrix. The physical question is represented by a Hamiltonian or unitary matrix rotating that state between preparation and measurement; the experimental or mathematical setting is the chosen basis, pulse sequence, or measurement axis. The observable content is obtained from projectors onto the two eigenstates of the measured observable. In the local terminology of this topic, the same construction appears through superposition or wave function, Hamiltonian or observable operator, and mode or eigenvalue. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through uncertainty relation or commutator. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its

Quantum Mechanism Frame

- **Role**. Quantum sensor contributes an instrument-mediated readout role to the quantum construction.
- **Placement**. This page is read first as an operation-sequence move: it specifies an ordered composition of allowed maps.

- **Carrier or domain**. State terms: superposition. Context/domain terms: potential.
- **Operator or map**. Operator terms: Hamiltonian, observable operator, or generator. Protocol or update terms: projection.
- **Admissibility**. Compatibility or closure terms: uncertainty. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: mode. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:0912.2823
- arXiv:1604.06537
- arXiv:quant-ph0205159
- arXiv:1612.00682

17.16 Quantum teleportation

Central Claim

A quantum teleportation can be read as a quantum construction: the chosen basis, pulse sequence, or measurement axis fixes the admissible state space; a Hamiltonian or unitary matrix rotating that state between preparation and measurement defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a quantum teleportation is described by a two-dimensional Hilbert space, usually written as a qubit state or a density matrix. The physical question is represented by a Hamiltonian or unitary matrix rotating that state between preparation and measurement; the experimental or mathematical setting is the chosen basis, pulse sequence, or measurement axis. The observable content is obtained from projectors onto the two eigenstates of the measured observable. In the local terminology of this topic, the same construction appears through quantum state or superposition, Hamiltonian or observable operator, and eigenvalue or energy level. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, controlled update

Quantum Mechanism Frame

- **Role**. Quantum teleportation contributes a topic-native constructor role to the quantum construction.
- **Placement**. This page is read first as an operation-sequence move: it specifies an ordered composition of allowed maps.
- **Carrier or domain**. State terms: quantum state and superposition. Context/domain terms: basis.
- **Operator or map**. Operator terms: Hamiltonian, observable operator, or generator. Protocol or update terms: collapse and projection.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.

- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:0912.2823
- arXiv:2308.15676
- arXiv:1612.00682
- arXiv:quant-ph0205159

17.17 Quantum bus

Central Claim

A quantum bus can be read as a quantum construction: the chosen basis, pulse sequence, or measurement axis fixes the admissible state space; a Hamiltonian or unitary matrix rotating that state between preparation and measurement defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a quantum bus is described by a two-dimensional Hilbert space, usually written as a qubit state or a density matrix. The physical question is represented by a Hamiltonian or unitary matrix rotating that state between preparation and measurement; the experimental or mathematical setting is the chosen basis, pulse sequence, or measurement axis. The observable content is obtained from projectors onto the two eigenstates of the measured observable. In the local terminology of this topic, the same construction appears through superposition or wave function, hamiltonian or observable operator, and eigenvalue or energy level. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, preparation, basis, or

Quantum Mechanism Frame

- **Role**. Quantum bus contributes a topic-native constructor role to the quantum construction.
- **Placement**. This page is read first as an operation-sequence move: it specifies an ordered composition of allowed maps.
- **Carrier or domain**. State terms: superposition. Context/domain terms: preparation condition, measurement setup, or potential or domain.
- **Operator or map**. Operator terms: hamiltonian. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$

$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$

$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$

$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:0912.2823
- arXiv:2108.07838
- arXiv:0805.4565
- arXiv:0908.0752
- arXiv:1612.00682

17.18 Quantum image processing

Central Claim

A quantum image processing can be read as a quantum construction: the chosen basis, pulse sequence, or measurement axis fixes the admissible state space; a Hamiltonian or unitary matrix rotating that state between preparation and measurement defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a quantum image processing is described by a two-dimensional Hilbert space, usually written as a qubit state or a density matrix. The physical question is represented by a Hamiltonian or unitary matrix rotating that state between preparation and measurement; the experimental or mathematical setting is the chosen basis, pulse sequence, or measurement axis. The observable content is obtained from projectors onto the two eigenstates of the measured observable. In the local terminology of this topic, the same construction appears through quantum state or wave function, Hamiltonian or observable operator, and eigenvalue or energy level. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, non-commuting

Quantum Mechanism Frame

- **Role**. Quantum image processing contributes a topic-native constructor role to the quantum construction.
- **Placement**. This page is read first as an operation-sequence move: it specifies an ordered composition of allowed maps.
- **Carrier or domain**. State terms: quantum state, wave function, or density operator. Context/domain terms: basis.
- **Operator or map**. Operator terms: Hamiltonian, observable operator, or generator. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:2111.12617
- arXiv:1604.06537
- arXiv:2308.15676
- arXiv:1506.05598
- arXiv:1612.00682

17.19 Quantum finite automaton

Central Claim

A quantum finite automaton can be read as a quantum construction: the potential, domain, initial condition, or boundary condition fixes the admissible state space; the Hamiltonian, whose exponential gives unitary time evolution defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a quantum finite automaton is described by a wave function or density operator defined on the Hilbert space allowed by the system's domain. The physical question is represented by the Hamiltonian, whose exponential gives unitary time evolution; the experimental or mathematical setting is the potential, domain, initial condition, or boundary condition. The observable content is obtained from the eigenvalues and eigenfunctions of the relevant observable. In the local terminology of this topic, the same construction appears through quantum state or state vector, matrix or unitary operator, and eigenvalue or energy level. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in state evolution, normalization or admissibility, operator-to-spectrum readout;

Quantum Mechanism Frame

- **Role**. Quantum finite automaton contributes a topic-native constructor role to the quantum construction.
- **Placement**. This page is read first as an operation-sequence move: it specifies an ordered composition of allowed maps.
- **Carrier or domain**. State terms: quantum state and state vector. Context/domain terms: preparation condition, measurement setup, or potential or domain.
- **Operator or map**. Operator terms: matrix and unitary. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$

$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$

$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$

$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:1706.03846
- arXiv:1708.03640
- arXiv:0908.0752
- arXiv:1801.03283

17.20 Quantum cryptography

Central Claim

A quantum cryptography can be read as a quantum construction: the chosen basis, pulse sequence, or measurement axis fixes the admissible state space; a Hamiltonian or unitary matrix rotating that state between preparation and measurement defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a quantum cryptography is described by a two-dimensional Hilbert space, usually written as a qubit state or a density matrix. The physical question is represented by a Hamiltonian or unitary matrix rotating that state between preparation and measurement; the experimental or mathematical setting is the chosen basis, pulse sequence, or measurement axis. The observable content is obtained from projectors onto the two eigenstates of the measured observable. In the local terminology of this topic, the same construction appears through quantum state or wave function, Hamiltonian or observable operator, and eigenvalue or energy level. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or

Quantum Mechanism Frame

- **Role**. Quantum cryptography contributes an engineered operation-sequence role to the quantum construction.
- **Placement**. This page is read first as an operation-sequence move: it specifies an ordered composition of allowed maps.
- **Carrier or domain**. State terms: quantum state, wave function, or superposition. Context/domain terms: potential and basis.
- **Operator or map**. Operator terms: Hamiltonian, observable operator, or generator. Protocol or update terms: collapse.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$

$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$

$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$

$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:0912.2823
- arXiv:1604.05385
- arXiv:1801.03283
- arXiv:quant-ph0205159
- arXiv:1612.00682

17.21 Quantum programming

Central Claim

Quantum programming can be read as a quantum construction: the chosen basis, pulse sequence, or measurement axis fixes the admissible state space; a Hamiltonian or unitary matrix rotating that state between preparation and measurement defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, quantum programming is described by a two-dimensional Hilbert space, usually written as a qubit state or a density matrix. The physical question is represented by a Hamiltonian or unitary matrix rotating that state between preparation and measurement; the experimental or mathematical setting is the chosen basis, pulse sequence, or measurement axis. The observable content is obtained from projectors onto the two eigenstates of the measured observable. In the local terminology of this topic, the same construction appears through quantum state or wave function, unitary operator or Hamiltonian, and eigenvalue or energy level. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or

Quantum Mechanism Frame

- **Role**. Quantum programming contributes an engineered operation-sequence role to the quantum construction.
- **Placement**. This page is read first as an operation-sequence move: it specifies an ordered composition of allowed maps.
- **Carrier or domain**. State terms: quantum state, wave function, or density operator. Context/domain terms: basis.
- **Operator or map**. Operator terms: unitary. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:0805.4565
- arXiv:1612.00682
- arXiv:2108.07838
- arXiv:quant-ph0205159

Chapter 18

Annotations: history, interpretations, and popular frames

Some pages help readers navigate the subject but do not form steps in the mechanism. They are kept as annotations so books, historical figures, interpretations, and popular frames do not distort the constructive tree.

18.1 Why This Branch Exists

These pages remain useful for orientation and are placed downstream of the construction steps. This prevents biographies, books, and interpretations from becoming false roots of the mechanism.

18.2 Page Map

Page
Introduction to quantum mechanics
Old quantum theory
History of quantum mechanics *(annotation)*
Introduction to Quantum Mechanics (book) *(annotation)*
Quantum mind *(annotation)*
History of quantum field theory *(annotation)*
Erwin Schrödinger *(annotation)*
Interpretations of quantum mechanics *(annotation)*
Quantum mysticism *(annotation)*
David Hilbert *(annotation)*
Quantum Computing: A Gentle Introduction *(annotation)*
QBism *(annotation)*
Relational quantum mechanics *(annotation)*
Modern Quantum Mechanics *(annotation)*
The Quantum Universe *(annotation)*
Quantum Theory: Concepts and Methods *(annotation)*

18.3 Mechanism Pages

Each page below is read as a specialization of the compact quantum constructor. Topic-specific pages use explicit equations or overrides; core-derived pages use the branch-level equation form associated with their mechanism role.

18.4 Introduction to quantum mechanics

Central Claim

An introduction to quantum mechanics can be read as a quantum construction: the chosen basis, pulse sequence, or measurement axis fixes the admissible state space; a Hamiltonian or unitary matrix rotating that state between preparation and measurement defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, an introduction to quantum mechanics is described by a two-dimensional Hilbert space, usually written as a qubit state or a density matrix. The physical question is represented by a Hamiltonian or unitary matrix rotating that state between preparation and measurement; the experimental or mathematical setting is the chosen basis, pulse sequence, or measurement axis. The observable content is obtained from projectors onto the two eigenstates of the measured observable. In the local terminology of this topic, the same construction appears through quantum state or wave function, Hamiltonian or observable operator, and eigenvalue or energy level. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through uncertainty relation or complementarity. The linked equation set is concentrated in operator-to-spectrum readout, state evolution,

Quantum Mechanism Frame

- **Role**. Introduction to quantum mechanics contributes a topic-native constructor role to the quantum construction.
- **Placement**. This page is read first as an interpretive or historical move: it clarifies which formal layer is being discussed.
- **Carrier or domain**. State terms: quantum state, wave function, or density operator. Context/domain terms: basis.
- **Operator or map**. Operator terms: Hamiltonian, observable operator, or generator. Protocol or update terms: Born rule.
- **Admissibility**. Compatibility or closure terms: uncertainty and complementarity. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:1604.06537
- arXiv:0912.2823
- arXiv:1708.03640
- arXiv:quant-ph0205159

18.5 Old quantum theory

Central Claim

An old quantum theory can be read as a quantum construction: the chosen basis, pulse sequence, or measurement axis fixes the admissible state space; a Hamiltonian or unitary matrix rotating that state between preparation and measurement defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, an old quantum theory is described by a two-dimensional Hilbert space, usually written as a qubit state or a density matrix. The physical question is represented by a Hamiltonian or unitary matrix rotating that state between preparation and measurement; the experimental or mathematical setting is the chosen basis, pulse sequence, or measurement axis. The observable content is obtained from projectors onto the two eigenstates of the measured observable. In the local terminology of this topic, the same construction appears through quantum state or wave function, matrix or Hamiltonian, and spectrum or eigenvalue. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its

Quantum Mechanism Frame

- **Role**. Old quantum theory contributes a topic-native constructor role to the quantum construction.
- **Placement**. This page is read first as an interpretive or historical move: it clarifies which formal layer is being discussed.
- **Carrier or domain**. State terms: quantum state, wave function, or density operator. Context/domain terms: preparation condition, measurement setup, or potential or domain.
- **Operator or map**. Operator terms: matrix and Hamiltonian. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: spectrum. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$
\begin{aligned}
B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)}\\
\rho_t = U_t \rho_B U_t^\dagger \quad &\text{(unitary evolution from preparation to readout)}\\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)}\\
[O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.06537
- arXiv:0912.2823
- arXiv:1604.05385
- arXiv:1708.03640
- arXiv:1801.03283

18.6 History of quantum mechanics

This page is treated as historical, interpretive, or popular context rather than as a conceptual root.

Central Claim

A history of quantum mechanics can be read as a quantum construction: the chosen basis, pulse sequence, or measurement axis fixes the admissible state space; a Hamiltonian or unitary matrix rotating that state between preparation and measurement defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a history of quantum mechanics is described by a two-dimensional Hilbert space, usually written as a qubit state or a density matrix. The physical question is represented by a Hamiltonian or unitary matrix rotating that state between preparation and measurement; the experimental or mathematical setting is the chosen basis, pulse sequence, or measurement axis. The observable content is obtained from projectors onto the two eigenstates of the measured observable. In the local terminology of this topic, the same construction appears through quantum state or wave function, Hamiltonian or observable operator, and spectrum or eigenvalue. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or

Quantum Mechanism Frame

- **Role**. History of quantum mechanics contributes a topic-native constructor role to the quantum construction.
- **Placement**. This page is read first as an interpretive or historical move: it clarifies which formal layer is being discussed.
- **Carrier or domain**. State terms: quantum state, wave function, or density operator. Context/domain terms: basis.
- **Operator or map**. Operator terms: Hamiltonian, observable operator, or generator. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: spectrum. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:1604.06537
- arXiv:1612.00682
- arXiv:quant-ph0205159
- arXiv:1801.03283

18.7 Introduction to Quantum Mechanics (book)

This page is treated as historical, interpretive, or popular context rather than as a conceptual root.

Central Claim

An introduction to quantum mechanics (book) can be read as a quantum construction: the potential, domain, initial condition, or boundary condition fixes the admissible state space; the Hamiltonian, whose exponential gives unitary time evolution defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, an introduction to Quantum Mechanics (book) is described by a wave function or density operator defined on the Hilbert space allowed by the system's domain. The physical question is represented by the Hamiltonian, whose exponential gives unitary time evolution; the experimental or mathematical setting is the potential, domain, initial condition, or boundary condition. The observable content is obtained from the eigenvalues and eigenfunctions of the relevant observable. In the local terminology of this topic, the same construction appears through Wave Function or state vector, Hamiltonian or observable operator, and eigenvalue or energy level. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in state evolution, normalization or admissibility,

Quantum Mechanism Frame

- **Role**. Introduction to Quantum Mechanics (book) contributes a topic-native constructor role to the quantum construction.
- **Placement**. This page is read first as an interpretive or historical move: it clarifies which formal layer is being discussed.
- **Carrier or domain**. State terms: Wave Function. Context/domain terms: preparation condition, measurement setup, or potential or domain.
- **Operator or map**. Operator terms: Hamiltonian, observable operator, or generator. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1706.03846
- arXiv:1708.03640
- arXiv:0806.4515
- arXiv:0908.0752
- arXiv:1801.03283

18.8 Quantum mind

This page is treated as historical, interpretive, or popular context rather than as a conceptual root.

Central Claim

a quantum mind modifies the interpretation of the probability/readout layer while preserving the formal quantum dynamics.

Formal Role

a quantum mind acts on the readout layer of the quantum constructor. The formal ingredients remain the state assignment, the operator or measurement being applied, and the Born-rule map from projectors to probabilities. What changes is the status assigned to those ingredients: for this topic, the state or probability is treated through the agent, measurement context, or interpretive stance attached to the formalism. The page should therefore be read as a statement about the interpretation of state, probability, update, or recorded outcome while the Hamiltonian, spectral resolution, and commutator structure remain the formal reference layer. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, controlled update protocol; its mathematical presentation emphasizes local notation, information profile, formula structure. Its constructive use is to identify which formal layer is being interpreted: state assignment, probability, update, readout, or ontology.

Quantum Mechanism Frame

- **Role**. Quantum mind contributes a topic-native constructor role to the quantum construction.
- **Placement**. This page is read first as an interpretive or historical move: it clarifies which formal layer is being discussed.
- **Carrier or domain**. State terms: wave function and superposition. Context/domain terms: basis.
- **Operator or map**. Operator terms: Hamiltonian, observable operator, or generator. Protocol or update terms: collapse.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:0912.2823
- arXiv:quant-ph0205159
- arXiv:1612.00682
- arXiv:0805.4565

18.9 History of quantum field theory

This page is treated as historical, interpretive, or popular context rather than as a conceptual root.

Central Claim

A history of quantum field theory can be read as a quantum construction: the chosen basis, pulse sequence, or measurement axis fixes the admissible state space; a Hamiltonian or unitary matrix rotating that state between preparation and measurement defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a history of quantum field theory is described by a two-dimensional Hilbert space, usually written as a qubit state or a density matrix. The physical question is represented by a Hamiltonian or unitary matrix rotating that state between preparation and measurement; the experimental or mathematical setting is the chosen basis, pulse sequence, or measurement axis. The observable content is obtained from projectors onto the two eigenstates of the measured observable. In the local terminology of this topic, the same construction appears through quantum state or wave function, Hamiltonian or observable operator, and spectrum or eigenvalue. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through uncertainty relation or commutator. The linked equation set is concentrated in state evolution, operator-to-spectrum readout, non-commuting

Quantum Mechanism Frame

- **Role**. History of quantum field theory contributes a many-mode field or particle-realization role to the quantum construction.
- **Placement**. This page is read first as an interpretive or historical move: it clarifies which formal layer is being discussed.
- **Carrier or domain**. State terms: quantum state. Context/domain terms: preparation condition, measurement setup, or potential or domain.
- **Operator or map**. Operator terms: Hamiltonian, observable operator, or generator. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: Uncertainty. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: spectrum. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$
\begin{aligned}
B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)}\\
\rho_t = U_t \rho_B U_t^\dagger \quad &\text{(unitary evolution from preparation to readout)}\\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)}\\
[O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.06537
- arXiv:1604.05385
- arXiv:0912.2823
- arXiv:1708.03640
- arXiv:1801.03283

18.10 Erwin Schrödinger

This page is treated as historical, interpretive, or popular context rather than as a conceptual root.

Central Claim

an erwin Schrödinger modifies the interpretation of the probability/readout layer while preserving the formal quantum dynamics.

Formal Role

an erwin Schrödinger acts on the readout layer of the quantum constructor. The formal ingredients remain the state assignment, the operator or measurement being applied, and the Born-rule map from projectors to probabilities. What changes is the status assigned to those ingredients: for this topic, the state or probability is treated through the agent, measurement context, or interpretive stance attached to the formalism. The page should therefore be read as a statement about the interpretation of state, probability, update, or recorded outcome while the Hamiltonian, spectral resolution, and commutator structure remain the formal reference layer. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its mathematical presentation emphasizes local notation, information profile, formula structure. Its constructive use is to identify which formal layer is being interpreted: state assignment, probability, update, readout, or ontology.

Quantum Mechanism Frame

- **Role**. Erwin Schrödinger contributes a lawful state-transport role to the quantum construction.
- **Placement**. This page is read first as an interpretive or historical move: it clarifies which formal layer is being discussed.
- **Carrier or domain**. State terms: wave function. Context/domain terms: preparation condition, measurement setup, or potential or domain.
- **Operator or map**. Operator terms: Hamiltonian, observable operator, or generator. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: Eigenvalue. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$\begin{aligned} B &\longmapsto \rho_B \quad \text{(context specifies an admissible state)} \\ \rho_t &= U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)} \\ O &= \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)} \\ [O_1, O_2] &\neq 0 \quad \text{(incompatible observables: no common sharp basis)} \end{aligned}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation

- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:0912.2823
- arXiv:1604.06537
- arXiv:1801.03283
- arXiv:1708.03640

18.11 Interpretations of quantum mechanics

This page is treated as historical, interpretive, or popular context rather than as a conceptual root.

Central Claim

an interpretations of quantum mechanics modifies the interpretation of the probability/readout layer while preserving the formal quantum dynamics.

Formal Role

an interpretations of quantum mechanics acts on the readout layer of the quantum constructor. The formal ingredients remain the state assignment, the operator or measurement being applied, and the Born-rule map from projectors to probabilities. What changes is the status assigned to those ingredients: for this topic, the state or probability is treated through the agent, measurement context, or interpretive stance attached to the formalism. The page should therefore be read as a statement about the interpretation of state, probability, update, or recorded outcome while the Hamiltonian, spectral resolution, and commutator structure remain the formal reference layer. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its mathematical presentation emphasizes local notation, information profile, formula structure. Its constructive use is to identify which formal layer is being interpreted: state assignment, probability, update, readout, or ontology.

Quantum Mechanism Frame

- **Role**. Interpretations of quantum mechanics contributes a probability/readout role to the quantum construction.
- **Placement**. This page is read first as an interpretive or historical move: it clarifies which formal layer is being discussed.
- **Carrier or domain**. State terms: wave function, wavefunction, state vector, quantum state, or superposition. Context/domain terms: domain.
- **Operator or map**. Operator terms: matrix. Protocol or update terms: Born rule and collapse.
- **Admissibility**. Compatibility or closure terms: Complementarity. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenstate. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$\begin{aligned} B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)} \\ \rho_t = U_t \rho_B U_t^\dagger \quad &\text{(unitary evolution from preparation to readout)} \\ O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)} \\ [O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)} \end{aligned}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels

- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:0912.2823
- arXiv:1604.06537
- arXiv:1708.03640
- arXiv:quant-ph0205159

18.12 Quantum mysticism

This page is treated as historical, interpretive, or popular context rather than as a conceptual root.

Central Claim

a quantum mysticism modifies the interpretation of the probability/readout layer while preserving the formal quantum dynamics.

Formal Role

a quantum mysticism acts on the readout layer of the quantum constructor. The formal ingredients remain the state assignment, the operator or measurement being applied, and the Born-rule map from projectors to probabilities. What changes is the status assigned to those ingredients: for this topic, the state or probability is treated through the agent, measurement context, or interpretive stance attached to the formalism. The page should therefore be read as a statement about the interpretation of state, probability, update, or recorded outcome while the Hamiltonian, spectral resolution, and commutator structure remain the formal reference layer. The linked equation set is concentrated in non-commuting compatibility limits, state evolution, normalization or admissibility; its mathematical presentation emphasizes local notation, information profile, formula structure. Its constructive use is to identify which formal layer is being interpreted: state assignment, probability, update, readout, or ontology.

Quantum Mechanism Frame

- **Role**. Quantum mysticism contributes a topic-native constructor role to the quantum construction.
- **Placement**. This page is read first as an interpretive or historical move: it clarifies which formal layer is being discussed.
- **Carrier or domain**. State terms: quantum state, wave function, or density operator. Context/domain terms: preparation condition, measurement setup, or potential or domain.
- **Operator or map**. Operator terms: Hamiltonian, observable operator, or generator. Protocol or update terms: collapse.
- **Admissibility**. Compatibility or closure terms: uncertainty. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:2111.12617
- arXiv:1604.06537
- arXiv:1604.05385
- arXiv:1706.03846
- arXiv:2410.23449

18.13 David Hilbert

This page is treated as historical, interpretive, or popular context rather than as a conceptual root.

Central Claim

A david hilbert can be read as a quantum construction: the measurement basis and experimental arrangement fixes the admissible state space; the self-adjoint observable being asked of that state defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a david Hilbert is described by a prepared quantum state before the measurement. The physical question is represented by the self-adjoint observable being asked of that state; the experimental or mathematical setting is the measurement basis and experimental arrangement. The observable content is obtained from the observable's spectral projectors and the Born probabilities assigned to them. In the local terminology of this topic, the same construction appears through quantum state or wave function, Hamiltonian or observable operator, and eigenvalue or energy level. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its mathematical presentation emphasizes local

Quantum Mechanism Frame

- **Role**. David Hilbert contributes a state-carrier role to the quantum construction.
- **Placement**. This page is read first as an interpretive or historical move: it clarifies which formal layer is being discussed.
- **Carrier or domain**. State terms: quantum state, wave function, or density operator. Context/domain terms: basis.
- **Operator or map**. Operator terms: Hamiltonian, observable operator, or generator. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:0912.2823
- arXiv:1604.06537
- arXiv:1612.00682
- arXiv:quant-ph0205159
- arXiv:1801.03283

18.14 Quantum Computing: A Gentle Introduction

This page is treated as historical, interpretive, or popular context rather than as a conceptual root.

Central Claim

A quantum computing: a gentle introduction can be read as a quantum construction: the chosen basis, pulse sequence, or measurement axis fixes the admissible state space; a Hamiltonian or unitary matrix rotating that state between preparation and measurement defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a quantum Computing: A Gentle Introduction is described by a two-dimensional Hilbert space, usually written as a qubit state or a density matrix. The physical question is represented by a Hamiltonian or unitary matrix rotating that state between preparation and measurement; the experimental or mathematical setting is the chosen basis, pulse sequence, or measurement axis. The observable content is obtained from projectors onto the two eigenstates of the measured observable. In the local terminology of this topic, the same construction appears through superposition or wave function, unitary operator or Hamiltonian, and eigenvalue or energy level. Probabilities enter only after this spectral decomposition: the

Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in state evolution, normalization or admissibility,

Quantum Mechanism Frame

- **Role**. Quantum Computing: A Gentle Introduction contributes an engineered operation-sequence role to the quantum construction.
- **Placement**. This page is read first as an interpretive or historical move: it clarifies which formal layer is being discussed.
- **Carrier or domain**. State terms: superposition. Context/domain terms: preparation condition, measurement setup, or potential or domain.
- **Operator or map**. Operator terms: unitary. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$\begin{aligned} B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)} \\ \rho_t = U_t \rho_B U_t^\dagger \quad &\text{(unitary evolution from preparation to readout)} \\ O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)} \\ [O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)} \end{aligned}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1706.03846
- arXiv:1708.03640
- arXiv:0908.0752
- arXiv:1801.03283
- arXiv:quant-ph0205159

18.15 QBism

This page is treated as historical, interpretive, or popular context rather than as a conceptual root.

Central Claim

QBism modifies the interpretation of the probability/readout layer while preserving the formal quantum dynamics.

Formal Role

QBism acts on the readout layer of the quantum constructor. The formal ingredients remain the state assignment, the operator or measurement being applied, and the Born-rule map from projectors to probabilities. What changes is the status assigned to those ingredients: for this topic, the state or probability is treated through the agent, measurement context, or interpretive stance attached to the formalism. The page should therefore be read as a statement about the interpretation of state, probability, update, or recorded outcome while the Hamiltonian, spectral resolution, and commutator structure remain the formal reference layer. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its mathematical presentation emphasizes local notation, information profile, formula structure. Its constructive use is to identify which formal layer is being interpreted: state assignment, probability, update, readout, or ontology.

Quantum Mechanism Frame

- **Role**. QBism contributes a probability/readout role to the quantum construction.
- **Placement**. This page is read first as an interpretive or historical move: it clarifies which formal layer is being discussed.
- **Carrier or domain**. State terms: wavefunction, quantum state, wave function, or superposition. Context/domain terms: measurement apparatus.
- **Operator or map**. Operator terms: Hamiltonian, observable operator, or generator. Protocol or update terms: Born rule and collapse.
- **Admissibility**. Compatibility or closure terms: uncertainty. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:1604.06537
- arXiv:0912.2823
- arXiv:quant-ph0205159
- arXiv:1612.00682

18.16 Relational quantum mechanics

This page is treated as historical, interpretive, or popular context rather than as a conceptual root.

Central Claim

a relational quantum mechanics modifies the interpretation of the probability/readout layer while preserving the formal quantum dynamics.

Formal Role

a relational quantum mechanics acts on the readout layer of the quantum constructor. The formal ingredients remain the state assignment, the operator or measurement being applied, and the Born-rule map from projectors to probabilities. What changes is the status assigned to those ingredients: for this topic, the state or probability is treated through the agent, measurement context, or interpretive stance attached to the formalism. The page should therefore be read as a statement about the interpretation of state, probability, update, or recorded outcome while the Hamiltonian, spectral resolution, and commutator structure remain the formal reference layer. The linked equation set is concentrated in operator-to-spectrum readout, controlled update protocol, state evolution; its mathematical presentation emphasizes local notation, information profile, formula structure. Its constructive use is to identify which formal layer is being interpreted: state assignment, probability, update, readout, or ontology.

Quantum Mechanism Frame

- **Role**. Relational quantum mechanics contributes a probability/readout role to the quantum construction.
- **Placement**. This page is read first as an interpretive or historical move: it clarifies which formal layer is being discussed.

- **Carrier or domain**. State terms: state vector, Wavefunction, or superposition. Context/domain terms: preparation condition, measurement setup, or potential or domain.
- **Operator or map**. Operator terms: matrix. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.
- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenstate. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$\begin{aligned} B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)} \\ \rho_t = U_t \rho_B U_t^\dagger \quad &\text{(unitary evolution from preparation to readout)} \\ O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)} \\ [O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)} \end{aligned}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:0912.2823
- arXiv:quant-ph0607206
- arXiv:2306.13129
- arXiv:1612.00682

18.17 Modern Quantum Mechanics

This page is treated as historical, interpretive, or popular context rather than as a conceptual root.

Central Claim

A modern quantum mechanics can be read as a quantum construction: the potential, domain, initial condition, or boundary condition fixes the admissible state space; the Hamiltonian, whose exponential gives unitary time evolution defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, a modern Quantum Mechanics is described by a wave function or density operator defined on the Hilbert space allowed by the system's domain. The physical question is represented by the Hamiltonian, whose exponential gives unitary time evolution; the experimental or mathematical setting is the potential, domain, initial condition, or boundary condition. The observable content is obtained from the eigenvalues and eigenfunctions of the relevant observable. In the local terminology of this topic, the same construction appears through quantum state or wave function, Hamiltonian or observable operator, and eigenvalue or energy level. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through commutator or uncertainty relation. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or

Quantum Mechanism Frame

- **Role**. Modern Quantum Mechanics contributes a topic-native constructor role to the quantum construction.
- **Placement**. This page is read first as an interpretive or historical move: it clarifies which formal layer is being discussed.
- **Carrier or domain**. State terms: quantum state, wave function, or density operator. Context/domain terms: preparation condition, measurement setup, or potential or domain.
- **Operator or map**. Operator terms: Hamiltonian. Protocol or update terms: unitary evolution, projection or measurement update, or path integral weighting.

- **Admissibility**. Compatibility or closure terms: commutator, uncertainty relation, or non-commuting transformations. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$
\begin{aligned}
B \longmapsto \rho_B \quad &\text{(context specifies an admissible state)} \\
\rho_t = U_t \rho_B U_t^\dagger \quad &\text{(unitary evolution from preparation to readout)} \\
O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad &\text{(spectral probability measure)} \\
[O_1, O_2] \neq 0 \quad &\text{(incompatible observables: no common sharp basis)}
\end{aligned}
$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1612.00682
- arXiv:1801.03283
- arXiv:quant-ph0205159
- arXiv:1708.03640
- arXiv:1506.05598

18.18 The Quantum Universe

This page is treated as historical, interpretive, or popular context rather than as a conceptual root.

Central Claim

The quantum universe can be read as a quantum construction: the chosen basis, pulse sequence, or measurement axis fixes the admissible state space; a Hamiltonian or unitary matrix rotating that state between preparation and measurement defines the transformation or question; and spectral projectors with the Born rule determine the recorded probability distribution.

Formal Role

In quantum-mechanical terms, The Quantum Universe is described by a two-dimensional Hilbert space, usually written as a qubit state or a density matrix. The physical question is represented by a Hamiltonian or unitary matrix rotating that state between preparation and measurement; the experimental or mathematical setting is the chosen basis, pulse sequence, or measurement axis. The observable content is obtained from projectors onto the two eigenstates of the measured observable. In the local terminology of this topic, the same construction appears through quantum state or wave function, Hamiltonian or observable operator, and eigenvalue or energy level. Probabilities enter only after this spectral decomposition: the Born rule assigns weights to projectors, not to informal object names. When two observables have a non-zero commutator, no single basis diagonalizes both; the limitation is therefore a statement about jointly available spectra, not about detector imperfection. In this page the compatibility condition is expressed through uncertainty relation or commutator. The linked equation set is concentrated in state evolution, non-commuting compatibility limits, normalization or

Quantum Mechanism Frame

- **Role**. The Quantum Universe contributes a topic-native constructor role to the quantum construction.
- **Placement**. This page is read first as an interpretive or historical move: it clarifies which formal layer is being discussed.
- **Carrier or domain**. State terms: quantum state, wave function, or density operator. Context/domain terms: preparation condition, measurement setup, or potential or domain.
- **Operator or map**. Operator terms: Hamiltonian, observable operator, or generator. Protocol or update terms: path integral.

- **Admissibility**. Compatibility or closure terms: uncertainty. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.
- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
- the operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels
- the dependence of admissible readout on measurement context or boundary condition
- the non-commuting compatibility structure, which survives changes of representation

What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
- where time dependence is represented: on the state, on the operator, or in a path weight
- the coordinate system, basis, or geometric picture used to display the same relation
- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1706.03846
- arXiv:1604.06537
- arXiv:2111.12617
- arXiv:1708.03640
- arXiv:0908.0752

18.19 Quantum Theory: Concepts and Methods

This page is treated as historical, interpretive, or popular context rather than as a conceptual root.

Central Claim

a quantum Theory: Concepts and Methods modifies the interpretation of the probability/readout layer while preserving the formal quantum dynamics.

Formal Role

a quantum Theory: Concepts and Methods acts on the readout layer of the quantum constructor. The formal ingredients remain the state assignment, the operator or measurement being applied, and the Born-rule map from projectors to probabilities. What changes is the status assigned to those ingredients: for this topic, the state or probability is treated through the agent, measurement context, or interpretive stance attached to the formalism. The page should therefore be read as a statement about the interpretation of state, probability, update, or recorded outcome while the Hamiltonian, spectral resolution, and commutator structure remain the formal reference layer. The linked equation set is concentrated in operator-to-spectrum readout, state evolution, normalization or admissibility; its mathematical presentation emphasizes local notation, information profile, formula structure. Its constructive use is to identify which formal layer is being interpreted: state assignment, probability, update, readout, or ontology.

Quantum Mechanism Frame

- **Role**. Quantum Theory: Concepts and Methods contributes a topic-native constructor role to the quantum construction.
- **Placement**. This page is read first as an interpretive or historical move: it clarifies which formal layer is being discussed.
- **Carrier or domain**. State terms: wavefunction and quantum state. Context/domain terms: preparation condition, measurement setup, or potential or domain.
- **Operator or map**. Operator terms: Hamiltonian, observable operator, or generator. Protocol or update terms: collapse.
- **Admissibility**. Compatibility or closure terms: incompatible and uncertainty. These determine which questions, states, or updates are legal.
- **Readout**. Readout terms: eigenvalue, energy level, or measurement outcome. These name the outcome labels, projectors, amplitudes, or records used for testing.

- **Check**. A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.

Topic Equations

$$B \longmapsto \rho_B \quad \text{(context specifies an admissible state)}$$
$$\rho_t = U_t \rho_B U_t^\dagger \quad \text{(unitary evolution from preparation to readout)}$$
$$O = \sum_i \lambda_i P_i, \quad p_i = \mathrm{Tr}(P_i \rho_t) \quad \text{(spectral probability measure)}$$
$$[O_1, O_2] \neq 0 \quad \text{(incompatible observables: no common sharp basis)}$$

What Remains Stable

- the rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation
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What Changes With Realization

- the name of the carrier: particle, wave, field, qubit, or excitation
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- the physical implementation of detector, boundary, preparation, or readout

Checks

- A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical readout, but without a tested incompatibility relation.
- The validation criterion is that varying the context changes the admissible readout while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

Source Pointers

- arXiv:1604.05385
- arXiv:1604.06537
- arXiv:1612.00682
- arXiv:quant-ph0205159
- arXiv:1801.03283

Source Boundary

The generated page archive and tree record the source equations used for this synthesis. ArXiv links are evidence pointers, not claims that any individual paper proves the whole branch.